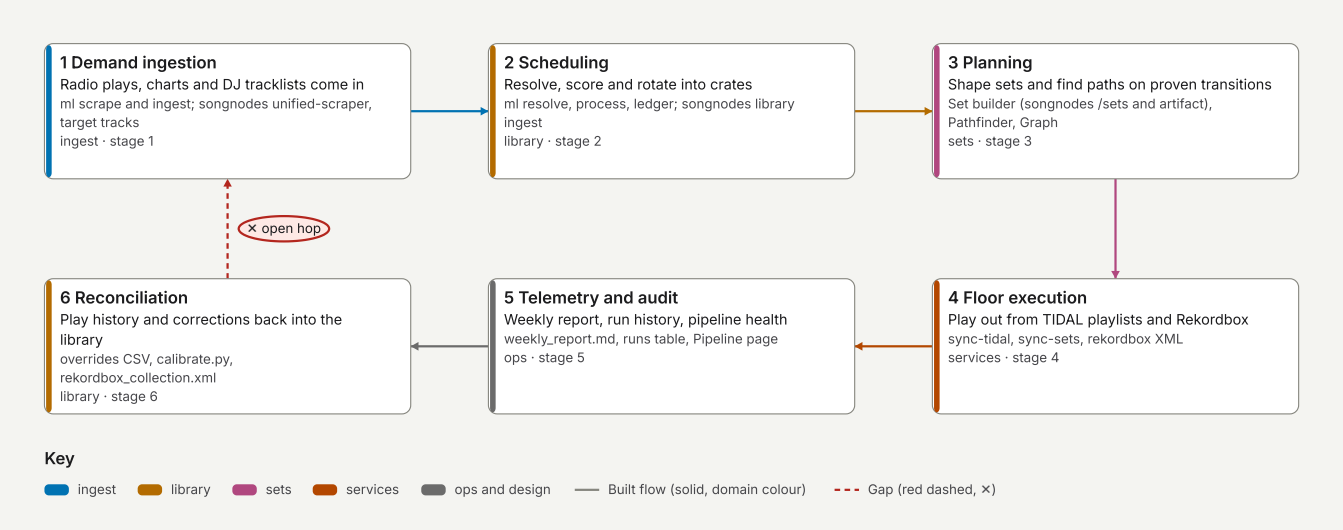
Data flow report: **7 Nov 2026**

The estate serves one loop: radio plays and DJ tracklists come in, become a curated club library, feed set building and the graph of proven transitions, play out in TIDAL and Rekordbox, and should come back as play history and corrections. Today the loop is open at the last step.

115	87	28	8
Pages and entry points	Numbered data flows	Critical data gaps	Personas

D1: **Closed loop**

The loop is open at one hop: reconciliation does not feed demand ingestion automatically.



D1 · n = 6 · Unit: stages · ● Built · Source: loop[] (name, mapping, systems, domain, gap) · Method: hand-placed ring of 6 cards, orthogonal hops; 0 crossings

Built 7 Nov 2026, 10:16 AEDT by Claude (Opus 5.5) for Markus, session 01NvWNfTGjhUeGYwDZx2H948 · songnodes 7429821 on docs/data-flow-investigation · musicLibrary snapshot 7 Nov 2026 · design system 1.10.0 · findings.json sha256 dc92fbc287f9

Contents

1 · Executive summary 3
2 · Architecture and numbered flows 6
3 · Page catalogue and standing rules 24
4 · Stores, signals and contracts 33
5 · Synchronisation protocols, timings and limits 54
6 · Personas and journeys 76
7 · Technical debt, risk and roadmap 93
8 · Gaps, proposed modules and storage backbone 100
9 · Built and Projected 116
10 · Sister-project tasks 119
11 · Verification register 121
12 · Methods, assumptions and open decisions 125
13 · Handover 129

Figures

D1 · Closed loop 5
D2a · System architecture – ingest, musiclibrary 10
D2b · System architecture – ingest, songnodes 11
D2c · System architecture – library, musiclibrary 12
D2d · System architecture – library, songnodes 13
D2e · System architecture – graph and sets 14
D2f · System architecture – services 15
D2g · System architecture – ops and design 16
D3a · Storage topology – musicLibrary stores 17
D3b · Storage topology – external services 18
D3c · Storage topology – songnodes medallion tables 19
D3d · Storage topology – songnodes operational tables 20
D3e · Storage topology – songnodes services and queues 21
D3f · Storage topology – songnodes browser stores 22
D4a · Page link graph – songnodes frontend 30
D4b · Page link graph – songnodes backend 31
D4c · Page link graph – musicLibrary 32
D5 · Persona overview 83
D5-P1 · Swimlane – P1 DJ performing 85
D5-P2 · Swimlane – P2 Set planner 86
D5-P3 · Swimlane – P3 Library curator 87
D5-P4 · Swimlane – P4 Pipeline operator 88
D5-P5 · Swimlane – P5 Developer 89
D5-P6 · Swimlane – P6 Scraper (system actor) 90
D5-P7 · Swimlane – P7 Weekly-run agent (system actor) 91
D5-P8 · Swimlane – P8 Beatport sync (assisted) 92
D6a · Sequence – Scrape to bronze, silver and projections 69
D6b · Sequence – Weekly musicLibrary run to TIDAL and songnodes 70
D6c · Sequence – TIDAL and Spotify OAuth connect 71
D7a · State machine – Scrape and ingestion job 72
D7b · State machine – Track rotation 73
D7c · State machine – OAuth token 74
D8a · Gap map – musicLibrary and cross-project 112
D8b · Gap map – songnodes backend 113
D8c · Gap map – songnodes frontend 114
D9 · Built to Projected change map 118
D10 · Storage backbone before and after 115
D11 · Roadmap timeline 99
D12 · Sister-project dependency graph 23
C1 · Pages by domain and status 24
C2 · Page and store access matrix 33

C3 · Signal primitives, built and projected 35
C4 · Store fan-in and fan-out 34
C5 · Timing constants on a log axis 75
C6 · Polling load by target store (Estimate) 61
C7 · Sizes against limits 29
C8 · Standing-rule conformance 29
C9 · Personas by page, with step numbers 84
C10 · Risk matrix, severity by effort 93
C11 · Verification outcome by area 121
C12 · Sister tasks by project, type and status 119

Section 1: **Executive summary**

Read as: one estate of two projects. songnodes (git, main @7429821) is the source of truth for SongNodes; E:\musicLibrary (no git) is read from a snapshot taken 7 Nov 2026 without its 58 GB data folder. Queued = open PRs #576–#580, musicLibrary plan/jules_payloads and unmerged briefs.

115
Pages and entry points

87
Numbered data flows

28
Critical data gaps
▲ Fix first

8
Personas

More figures 6

48
Stores and external systems

404
Signals extracted

17
Standing-rule violations
× To fix

34
Doc claims corrected or refuted
▲ Review

14
Queued items

16
Sister-project tasks

What holds

- songnodes is Git-first with one integration branch, a scope guard and design-system CI; the frontend copy of the design system is in sync and its checkers report zero findings.
- The scraping core is one unified-scraper container with complete-playlist rules, idempotency keys on ingest jobs and FOR UPDATE SKIP LOCKED job claiming.
- musicLibrary's weekly run is a clear eight-step file pipeline with an append-only ledger, a resumable resolve step and a full SQLite rebuild every run.
- Both projects already share one design system, one TIDAL developer app and one saved-set idea, so the join points are few and nameable.

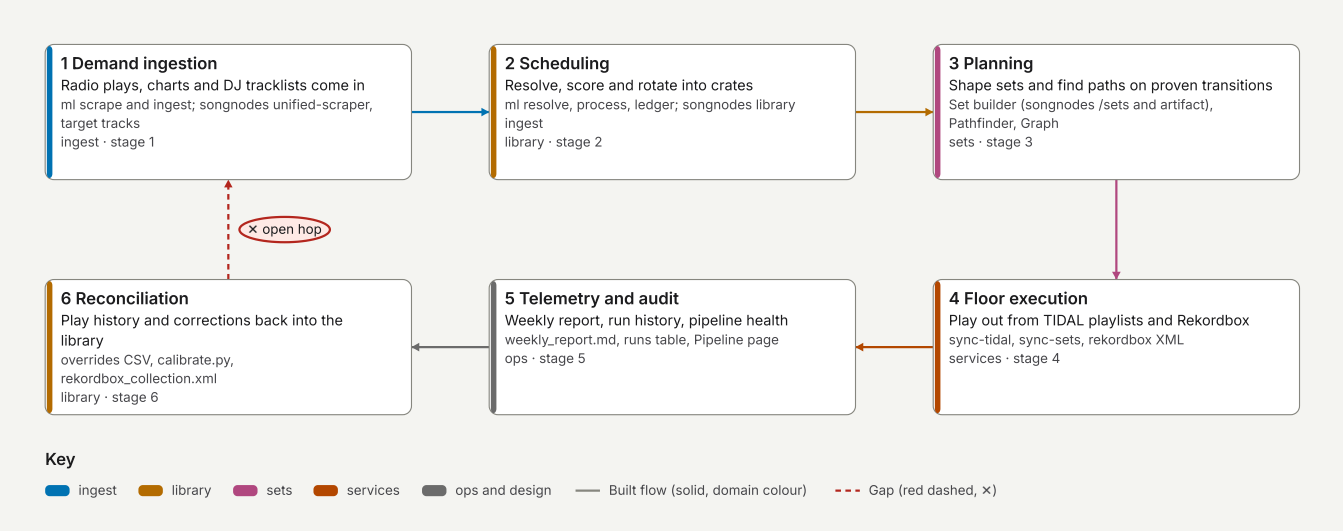
What strains

- Contracts between the two projects have drifted: the documented sync script and file do not exist, and the two Set Builders no longer share a saved-set shape.
- The medallion model in CLAUDE.md is not what runs: the scraper writes silver directly and no gold or operational job runs in compose.
- Feedback loops are missing: play history, overrides and proven transitions never flow back into curation or set building, and no track has a BPM or key yet.
- Security posture is not ready for the queued tunnel: write and admin routes are open and the queued login falls back to a default account.

The estate serves one loop: radio plays and DJ tracklists come in, become a curated club library, feed set building and the graph of proven transitions, play out in TIDAL and Rekordbox, and should come back as play history and corrections. Today the loop is open at the last step.

D1: **Closed loop**

The loop is open at one hop: reconciliation does not feed demand ingestion automatically.



D1 · n = 6 · Unit: stages · ● Built · Source: loop[] (name, mapping, systems, domain, gap) · Method: hand-placed ring of 6 cards, orthogonal hops; 0 crossings

Section 2: **Architecture and numbered flows** 87 flows

Every flow in the table is drawn in D2 with the same number.

Table: **Numbered flows** 87 flows

87 rows

FLOW	FROM	TO	DATA CONTRACT	CHANNEL	DOMAIN	STATE	EVIDENCE
1	ext-3rrr	ml-cli-scrape	3RRR episode tracks	–	ingest	Built	–
2	ext-abc-plays	ml-cli-scrape	triple j Doof plays	–	ingest	Built	–
3	ext-aria	ml-cli-scrape	ARIA Club rank and weeks	–	ingest	Built	–
4	ext-balance-wp	ml-cli-scrape	Balance tracklists	–	ingest	Built	–
5	ext-deezer	ml-cli-resolve	Deezer versions, BPM, ISRC, genre, label	–	ingest	Built	–
6	ext-metabrainz	ml-store-data	MusicBrainz dumps	–	ingest	Built	–
7	ext-pbs-airnet	ml-cli-scrape	PBS playlists	–	ingest	Built	–
8	ext-tidal	ml-cli-resolve	TIDAL versions	–	ingest	Built	–
9	ml-cli-ingest	ml-cli-resolve	to_resolve list	–	ingest	Built	–
10	ml-cli-scrape	ml-cli-ingest	raw rows	–	ingest	Built	–
11	ml-store-inbox	ml-cli-scrape	manual adds	–	ingest	Built	–
12	sn-be-svc-rest-api	sn-be-tbl-bronze	manual tracklist import	–	ingest	Built	–
13	sn-be-svc-unified-scraper	sn-be-store-redis	daily quota + setlist dedupe	–	ingest	Built	–
14	sn-be-svc-unified-scraper	sn-be-tbl-bronze	bronze playlists/tracks	–	ingest	Built	–
15	sn-be-svc-unified-scraper	sn-be-tbl-silver	layer-skipping silver writes	–	ingest	Built	–
16	sn-be-svc-worker	sn-be-svc-unified-scraper	scrape request	–	ingest	Built	–
17	sn-be-tbl-bronze	sn-be-tbl-silver	ingest_pending bronze→silver + transitions	–	ingest	Built	–
18	sn-be-tbl-silver	sn-be-tbl-outbox	track/artist/setlist events	–	ingest	Built	–
19	sn-be-tbl-target-tracks	sn-be-tbl-ingestion-jobs	scrape_targets hourly fan-out per source	–	ingest	Built	–
20	sn-fe-library	sn-be-svc-rest-api	mark library track as target (+ local sn-target-tracks-v1)	–	ingest	Built	–
21	sn-fe-library	sn-be-svc-rest-api	musicLibrary library.json export → ingest job	–	ingest	Built	–
22	sn-fe-pipeline	sn-be-svc-rest-api	scrape trigger (source, search_query, Idempotency-Key)	–	ingest	Built	–
23	sn-fe-sources-import	sn-be-svc-rest-api	AI-found setlists → import-tracklist	–	ingest	Built	–

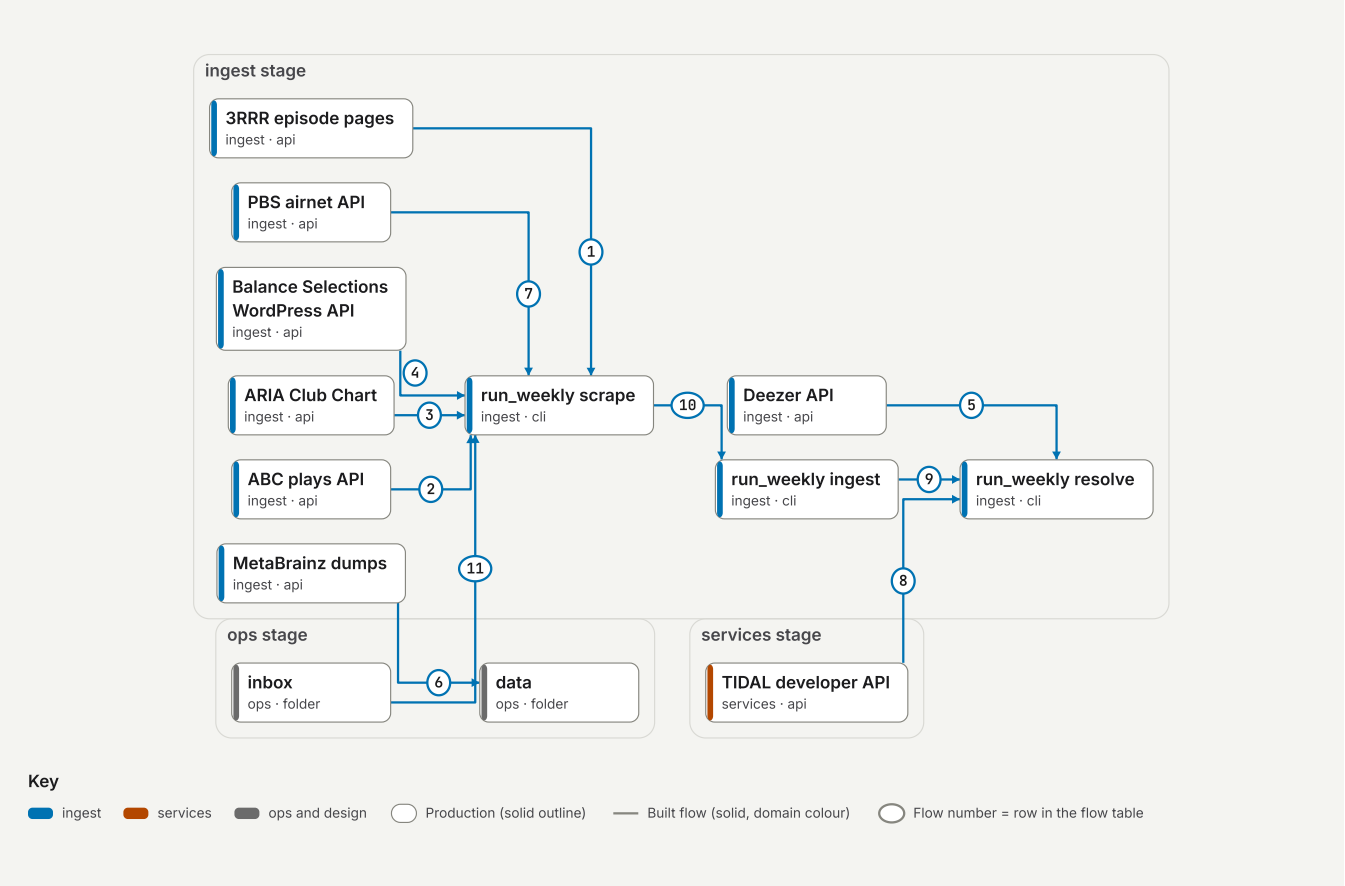
24	sn-fe-sources-targets	sn-be-svc-rest-api	target track CRUD	–	ingest	● Built	–
25	sn-fe-sources-targets	sn-fe-sources-import	Import a tracklist with AI link	–	ingest	● Built	–
26	ml-agent-beatport-sync	ml-cli-confirm	Beatport sync result	–	library	● Built	–
27	ml-agent-weekly-run	ml-store-artifact-files	republish library.json	–	library	● Built	–
28	ml-cli-build-db	ml-store-outputs	library.json	–	library	● Built	–
29	ml-cli-build-db	ml-store-sqlite	SQLite rebuild	–	library	● Built	–
30	ml-cli-db	sn-be-svc-rest-api	library.json ingest	–	library	● Built	–
31	ml-cli-process	ext-rekordbox	Rekordbox XML	–	library	● Built	–
32	ml-cli-process	ml-agent-beatport-sync	sync plan and buy list	–	library	● Built	–
33	ml-cli-process	ml-store-ledger	tracks, pending, rejected, rotation	–	library	● Built	–
34	ml-cli-resolve	ml-cli-process	resolved versions	–	library	● Built	–
35	ml-cli-sync-songnodes	sn-be-svc-rest-api	library.json bulk ingest	–	library	● Built	–
36	ml-cli-sync-tidal	ext-tidal	crate playlists	–	library	● Built	–
37	ml-cli-sync-tidal	ml-cli-confirm	sync result	–	library	● Built	–
38	ml-store-artifact-files	ml-page-set-builder	LibraryFile	–	library	● Built	–
39	ml-store-ledger	ml-cli-build-db	full ledger	–	library	● Built	–
40	ml-store-ledger	ml-cli-sync-tidal	crates	–	library	● Built	–
41	ml-store-overrides	ml-cli-process	hand corrections	–	library	● Built	–
42	sn-be-cli-import-music-library	sn-be-svc-rest-api	library.json via import-music-library.sh	–	library	● Built	–
43	sn-be-svc-rest-api	sn-fe-library	library tracks (uuid ids)	–	library	● Built	–
44	sn-be-svc-rest-api	sn-fe-library	library.tracks rows (compact keys)	–	library	● Built	–
45	sn-be-svc-worker	sn-be-tbl-library	library_ingest	–	library	● Built	–
46	sn-be-tbl-library	sn-be-tbl-target-tracks	auto-promote new library tracks	–	library	● Built	–
47	sn-be-tbl-outbox	sn-be-tbl-projection-tracks	projection_consume	–	library	● Built	–

48	sn-be-tbl-projection-tracks	sn-be-svc-rest-api	library search	–	library	● Built	–
49	sn-fe-home	sn-fe-library	AppHeader search / Ctrl K → /library?focus=search	–	library	● Built	–
50	ml-cli-sync-songnodes	sn-be-svc-rest-api	setlists.v1 (shape mismatch)	–	graph	● Built	–
51	sn-be-svc-graph-api	sn-fe-graph	graph nodes/edges for PIXI	–	graph	● Built	–
52	sn-be-svc-graph-api	sn-fe-graph	neighbourhood view loaded from API (view=neighbourhood&center&steps&max) instead of full graph	–	graph	▲ Queued	songnodes/STATUS.md (GV2 Next) +2
53	sn-be-tbl-outbox	sn-be-store-memgraph	graph_projection_consume (disabled)	–	graph	● Built	–
54	sn-be-tbl-silver	sn-be-svc-graph-api	graph read (silver direct)	–	graph	● Built	–
55	sn-fe-paths	sn-fe-graph	Open Pathfinder in Graph	–	graph	● Built	–
56	ml-agent-weekly-run	ml-cli-build-db	artifact_sets.json	–	sets	● Built	–
57	ml-cli-build-db	ml-cli-sync-sets	sets marked tidal	–	sets	● Built	–
58	ml-cli-sync-sets	ext-tidal	SET · <name> playlists	–	sets	● Built	–
59	ml-page-set-builder	ml-store-artifact-db	SavedSet documents	–	sets	● Built	–
60	ml-store-artifact-db	ml-agent-weekly-run	saved sets rows	–	sets	● Built	–
61	sn-be-svc-rest-api	sn-be-tbl-library	saved sets	–	sets	● Built	–
62	sn-fe-library	ml-store-inbox	artifact_sets.json download (manual copy)	–	sets	● Built	–
63	sn-fe-library	sn-be-tbl-library	saved sets	–	sets	● Built	–
64	sn-fe-library	sn-fe-sets	crate filters + pool=crate in URL; draft slots in sn-set-draft	–	sets	● Built	–
65	sn-fe-sets	ml-store-inbox	artifact_sets.json download (manual copy to musicLibrary inbox)	–	sets	● Built	–
66	sn-fe-sets	sn-be-svc-rest-api	SavedSet upsert {name,description,tracks[{track_id}]}	–	sets	● Built	–
67	ml-svc-ref-mb	sn-be-store-refmb	MusicBrainz reference lookups	–	services	● Built	–
68	sn-be-svc-rest-api	sn-be-legacy-metadata-enrichment	music search proxy (dead target)	–	services	● Built	–
69	sn-be-svc-rest-api	sn-be-store-redis	OAuth state	–	services	● Built	–
70	sn-be-svc-rest-api	sn-be-tbl-oauth	OAuth tokens and saved keys	–	services	● Built	–
71	sn-be-svc-worker	sn-be-svc-audio-analysis	analyse Deezer preview	–	services	● Built	–
72	sn-be-svc-worker	sn-be-tbl-silver	enrichment evidence/consensus	–	services	● Built	–
73	sn-be-tbl-oauth	sn-be-svc-worker	saved provider keys override env	–	services	● Built	–

74	sn-fe-oauth-callback	sn-be-svc-rest-api	tokens stored server-side for enrichment	–	services	● Built	–
75	sn-fe-oauth-callback	sn-fe-settings	oauth-success tokens (pop-up only)	–	services	● Built	–
76	sn-fe-sets	sn-be-svc-rest-api	send set to TIDAL	–	services	● Built	–
77	sn-fe-settings	sn-fe-oauth-callback	OAuth redirect with state in sessionStorage	–	services	● Built	–
78	ml-plan-jules	ext-jules	work-package sessions	–	ops	● Built	–
79	sn-be-q-store-cloudflare-tunnel	sn-be-svc-rest-api	public HTTPS via frontend nginx /api proxy	–	ops	▲ Queued	–
80	sn-be-svc-rest-api	sn-be-svc-unified-scraper	live scrape progress poll	–	ops	● Built	–
81	sn-be-svc-rest-api	sn-be-tbl-ingestion-jobs	scrape/library_ingest job enqueue (idempotency key)	–	ops	● Built	–
82	sn-be-svc-rest-api	sn-fe-pipeline	jobs, outbox status, stages (polled)	–	ops	● Built	–
83	sn-be-svc-worker	sn-be-store-rabbitmq	Celery beat ticks	–	ops	● Built	–
84	sn-fe-home	sn-fe-sets	LauncherCard links to every module via MODULE_ROUTES	–	ops	● Built	–
85	sn-fe-work-queue-tracker	ml-store-artifact-db	window.claude db collections meta, tasks (seeded from seed.json)	–	ops	● Built	–
86	sn-fe-library	ml-page-set-builder	compiled pages via build.mjs	–	design	● Built	–
87	sn-fe-settings	sn-fe-index-html	display prefs read before first paint	–	design	● Built	–

D2a: **System architecture – ingest, musiclibrary**

Every ingest, musiclibrary flow, numbered as in the flow table; all of them are built.

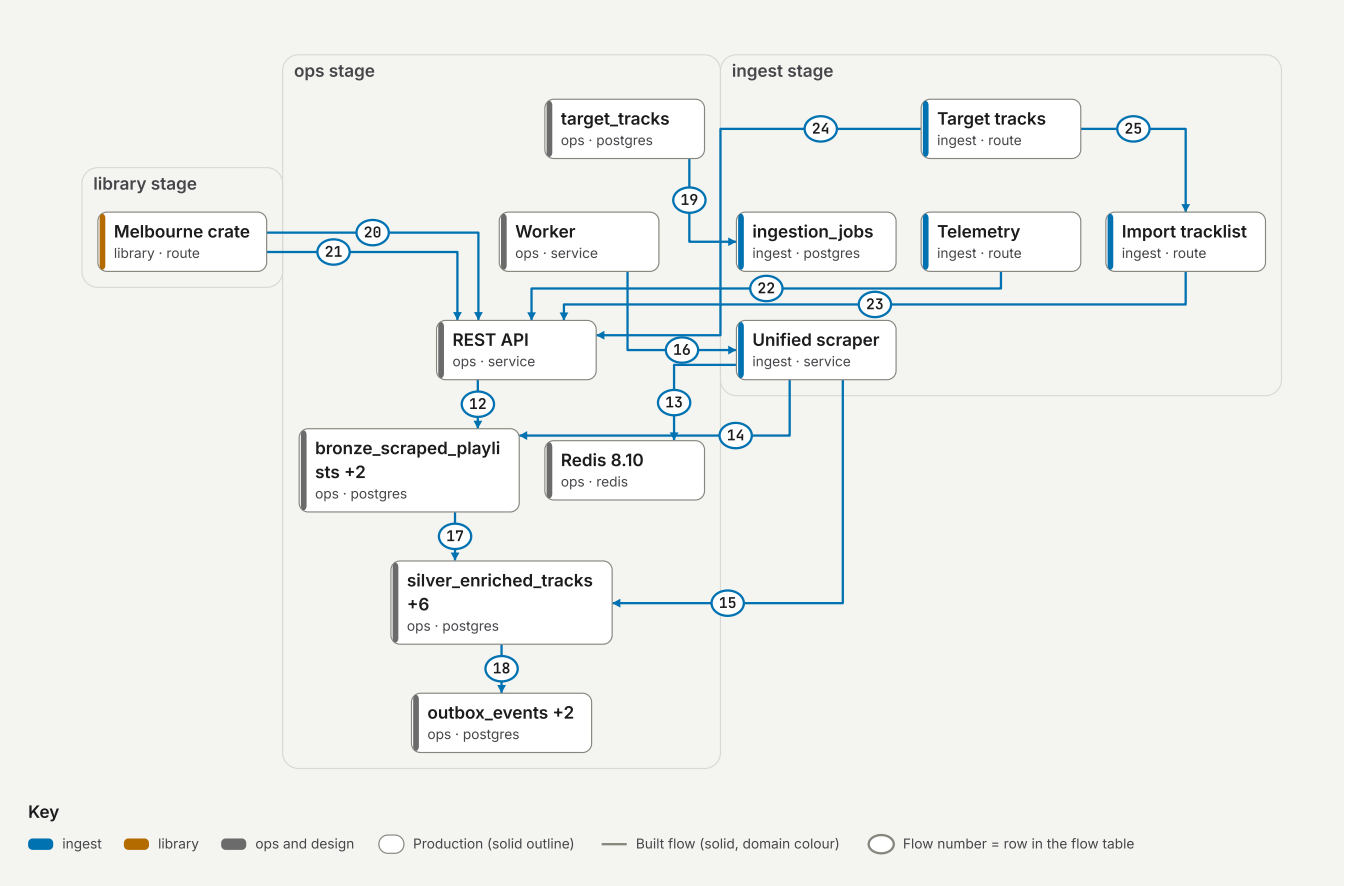


D2a · n = 11 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 1 crossings, 0 edges through nodes

D2b: **System architecture – ingest, songnodes**

Every ingest, songnodes flow, numbered as in the flow table; all of them are built.

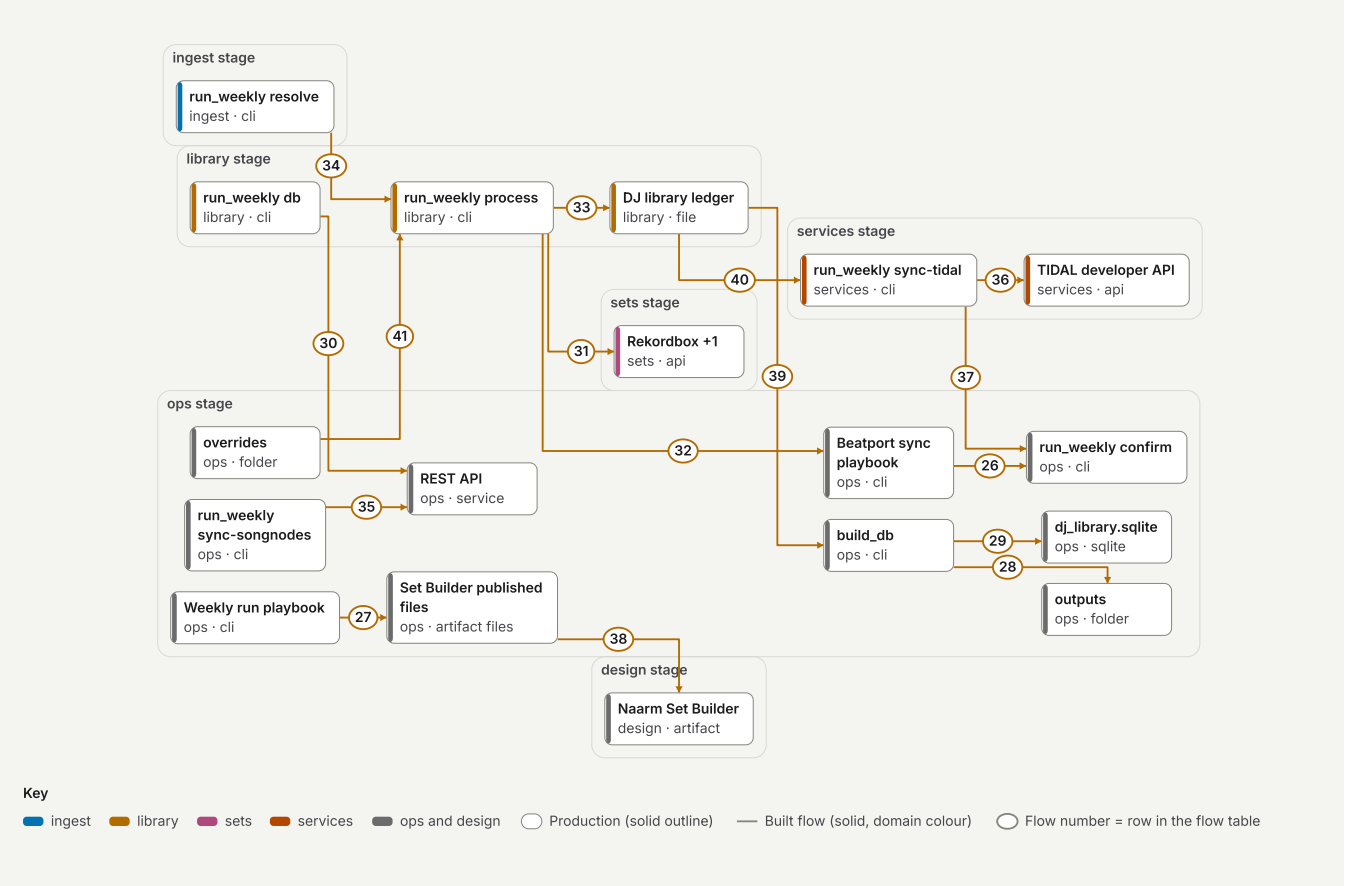


D2b · n = 14 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 7 crossings, 0 edges through nodes

D2c: **System architecture – library, musiclibrary**

Every library, musiclibrary flow, numbered as in the flow table; all of them are built.

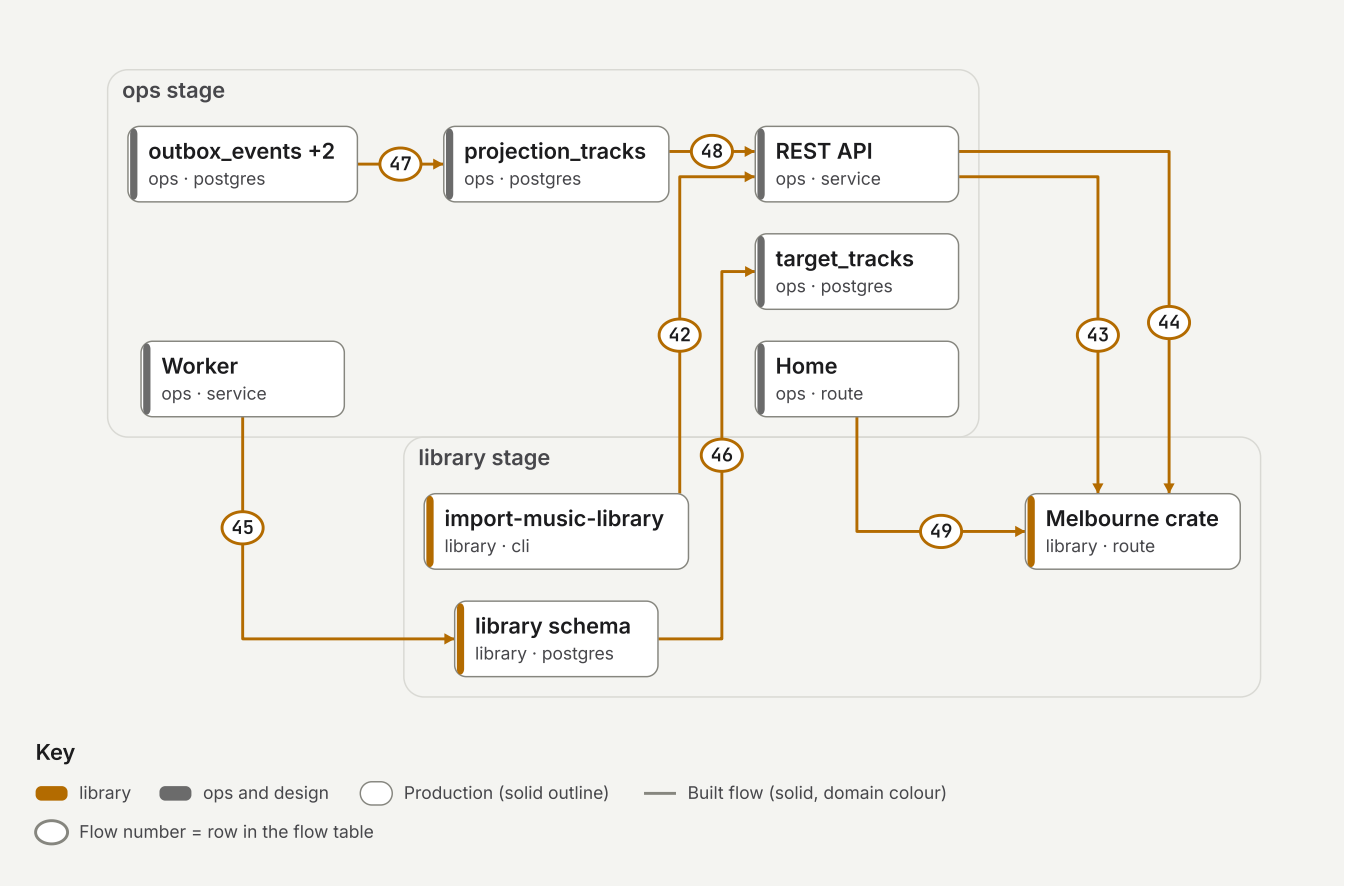


D2c · n = 16 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 3 crossings, 0 edges through nodes

D2d: **System architecture – library, songnodes**

Every library, songnodes flow, numbered as in the flow table; all of them are built.

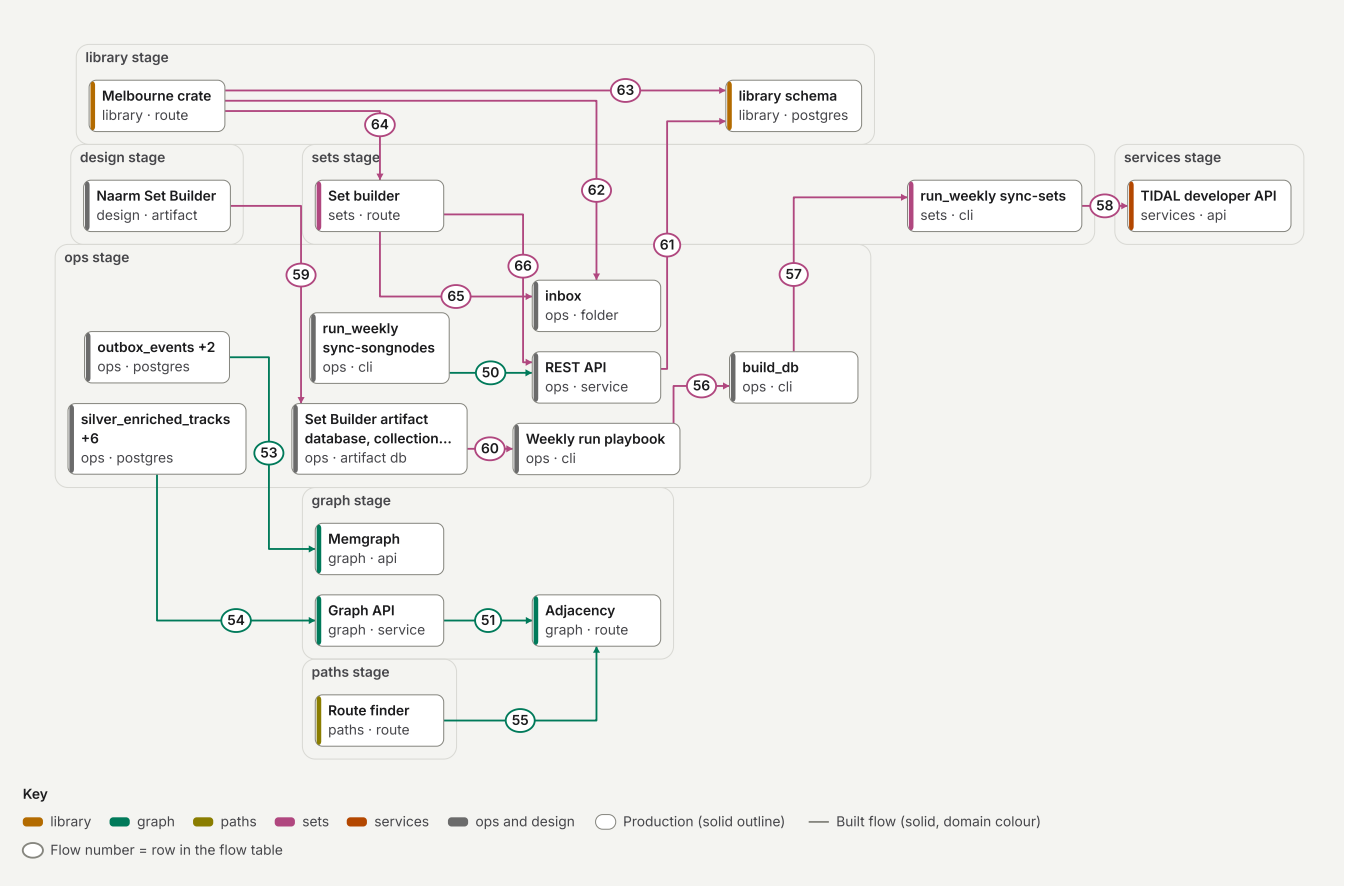


D2d · n = 8 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D2e: **System architecture – graph and sets**

Every graph and sets flow, numbered as in the flow table; 1 queued flow appears only in the Projected view.

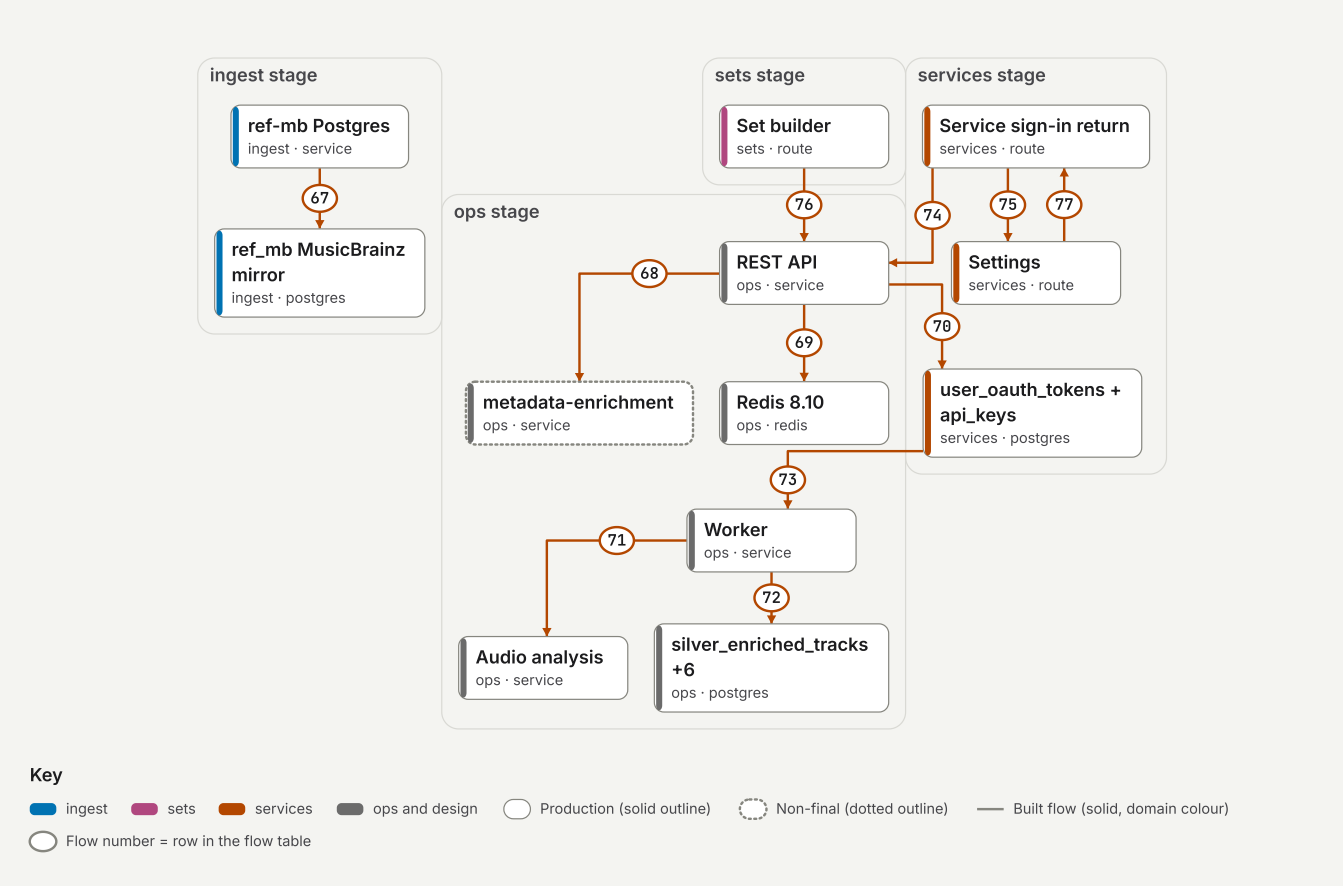


D2e · n = 17 · Unit: flows · **i built+queued**

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 1 crossings, 0 edges through nodes

D2f: **System architecture – services**

Every services flow, numbered as in the flow table; all of them are built.

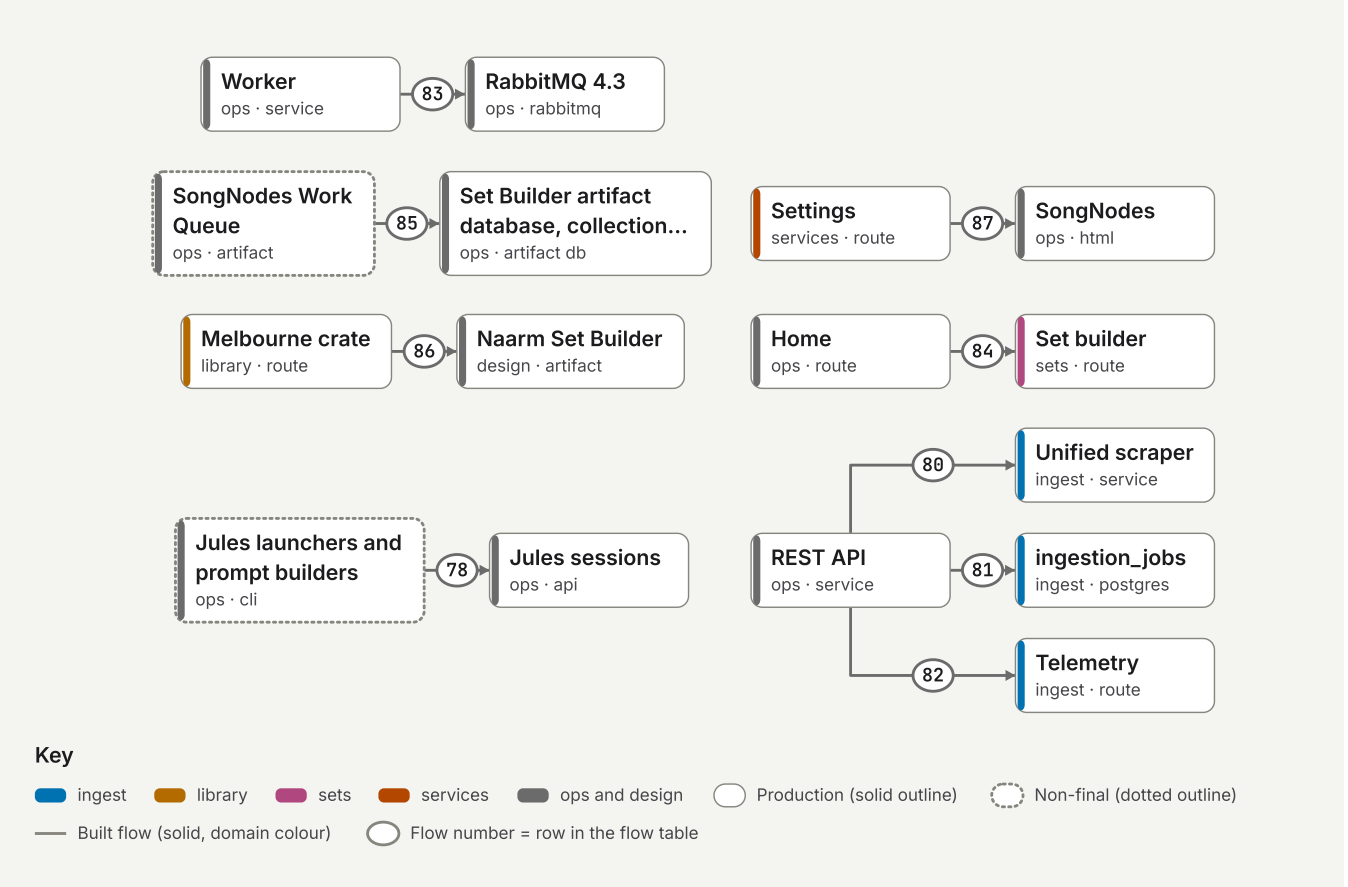


D2f · n = 11 · Unit: flows · ● Built

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D2g: **System architecture – ops and design**

Every ops and design flow, numbered as in the flow table; 1 queued flow appears only in the Projected view.

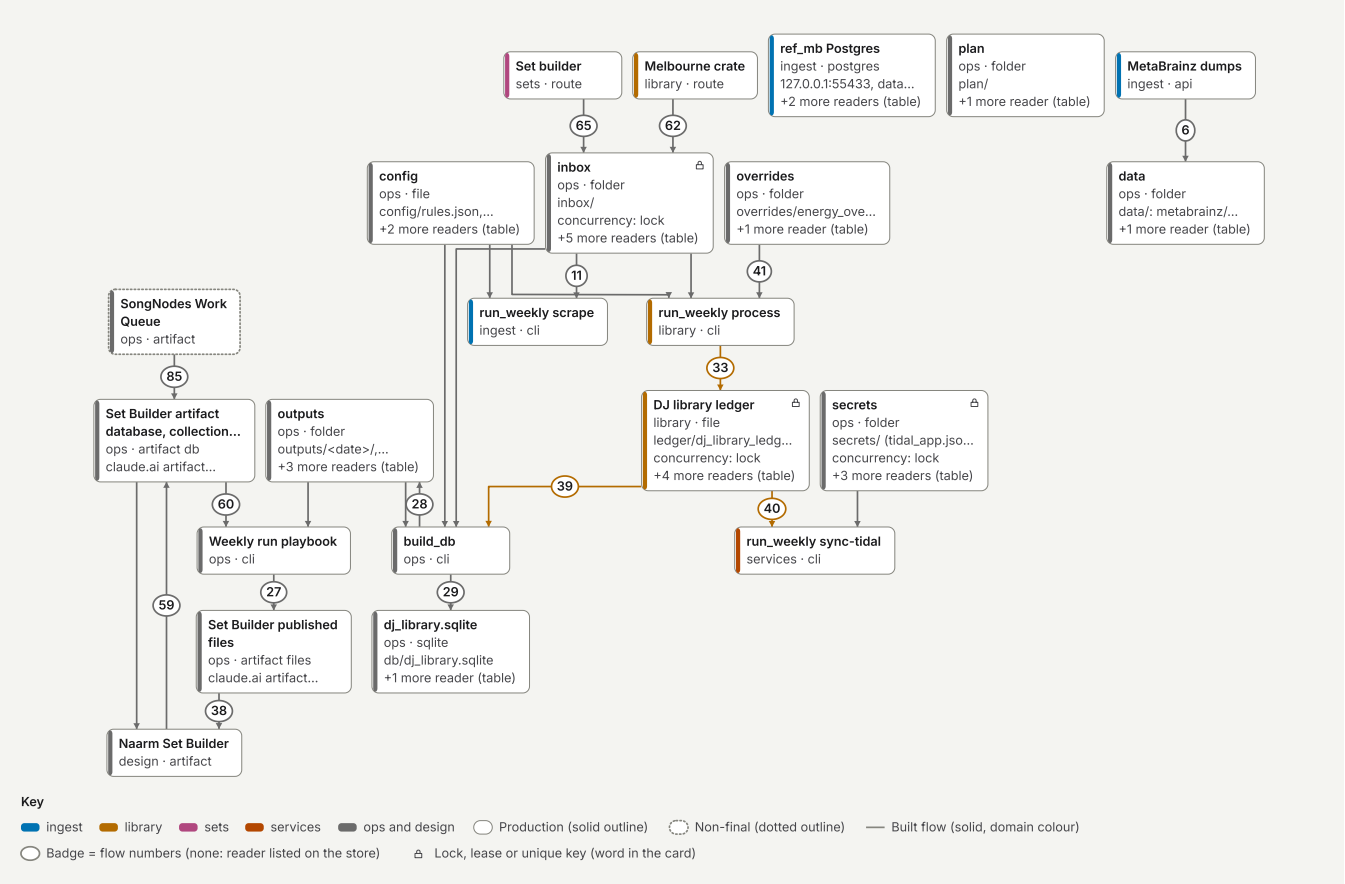


D2g · n = 10 · Unit: flows · **i built+queued**

· Source: flows[] (n, id, from, to, label, domain, state); pages[]/stores[] (title, name, domain, kind, classification, state); domain_order; colour_map · Method: graphviz dot layered layout (LR, order given, components packed 4 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D3a: **Storage topology – musicLibrary stores**

3 of 12 musicLibrary stores carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

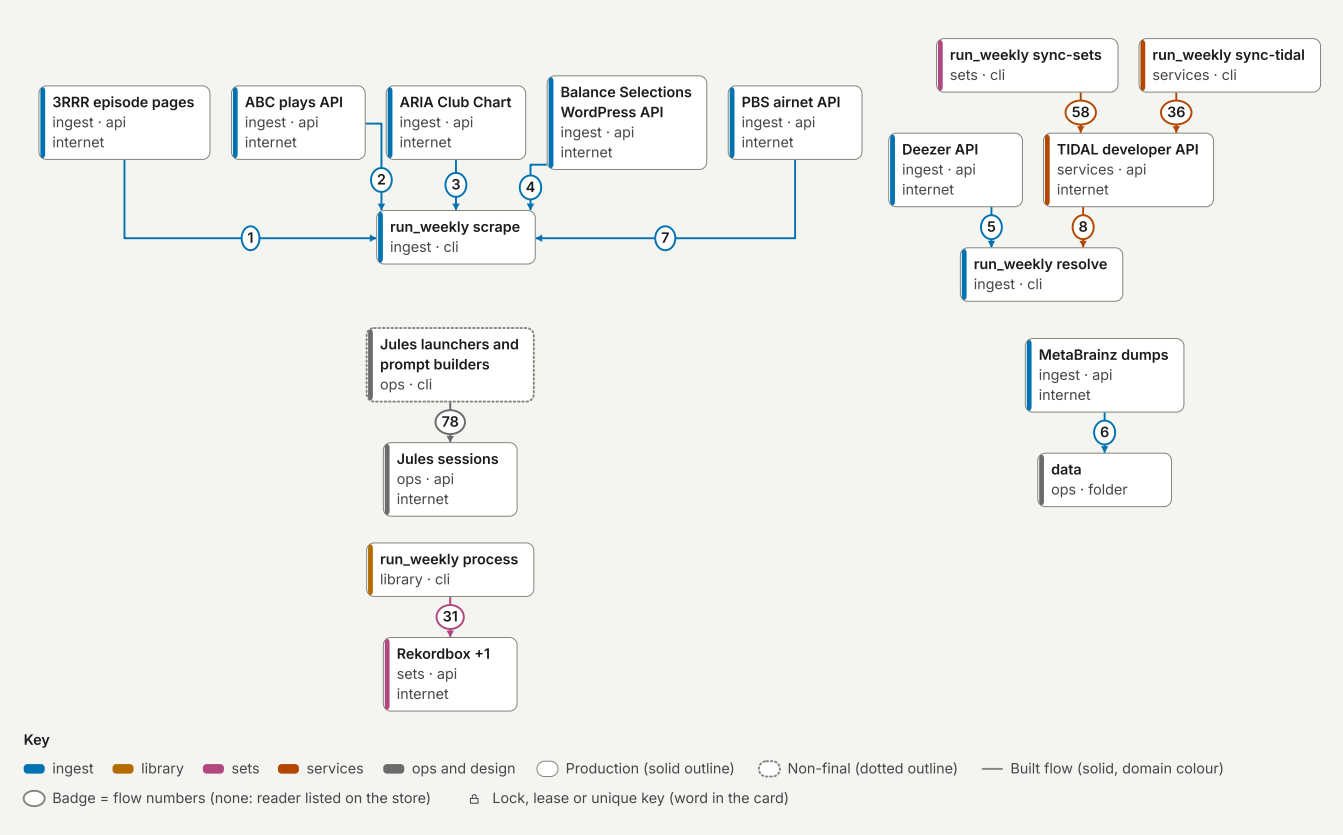


D3a · n = 12 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order reverse, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 2 crossings, 0 edges through nodes

D3b: **Storage topology – external services**

0 of 10 external services carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

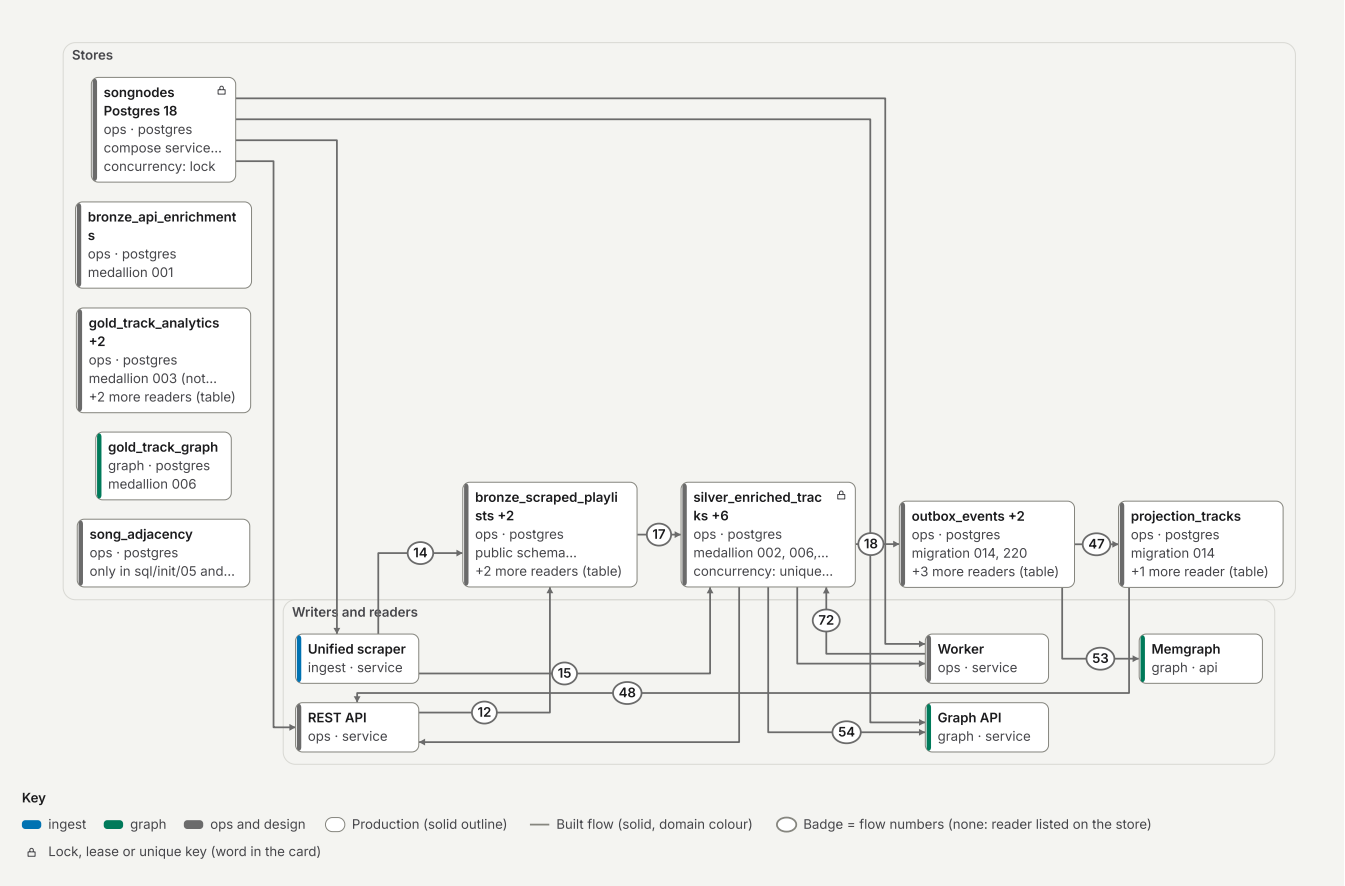


D3b · n = 10 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order given, components packed 2 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 0 crossings, 0 edges through nodes

D3c: **Storage topology – songnodes medallion tables**

2 of 9 songnodes medallion tables carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

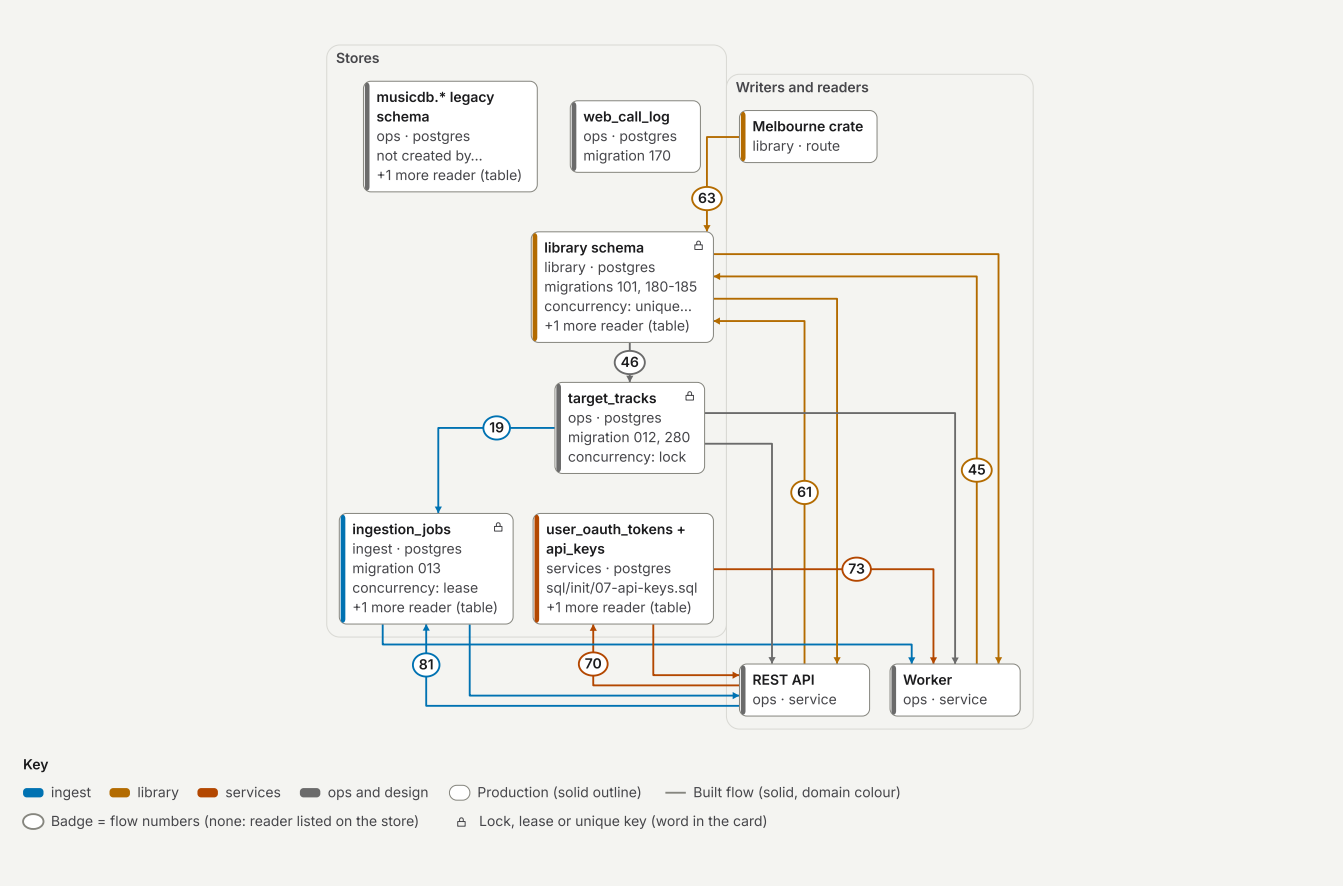


D3c · n = 9 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 10 crossings, 0 edges through nodes

D3d: **Storage topology – songnodes operational tables**

3 of 6 songnodes operational tables carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

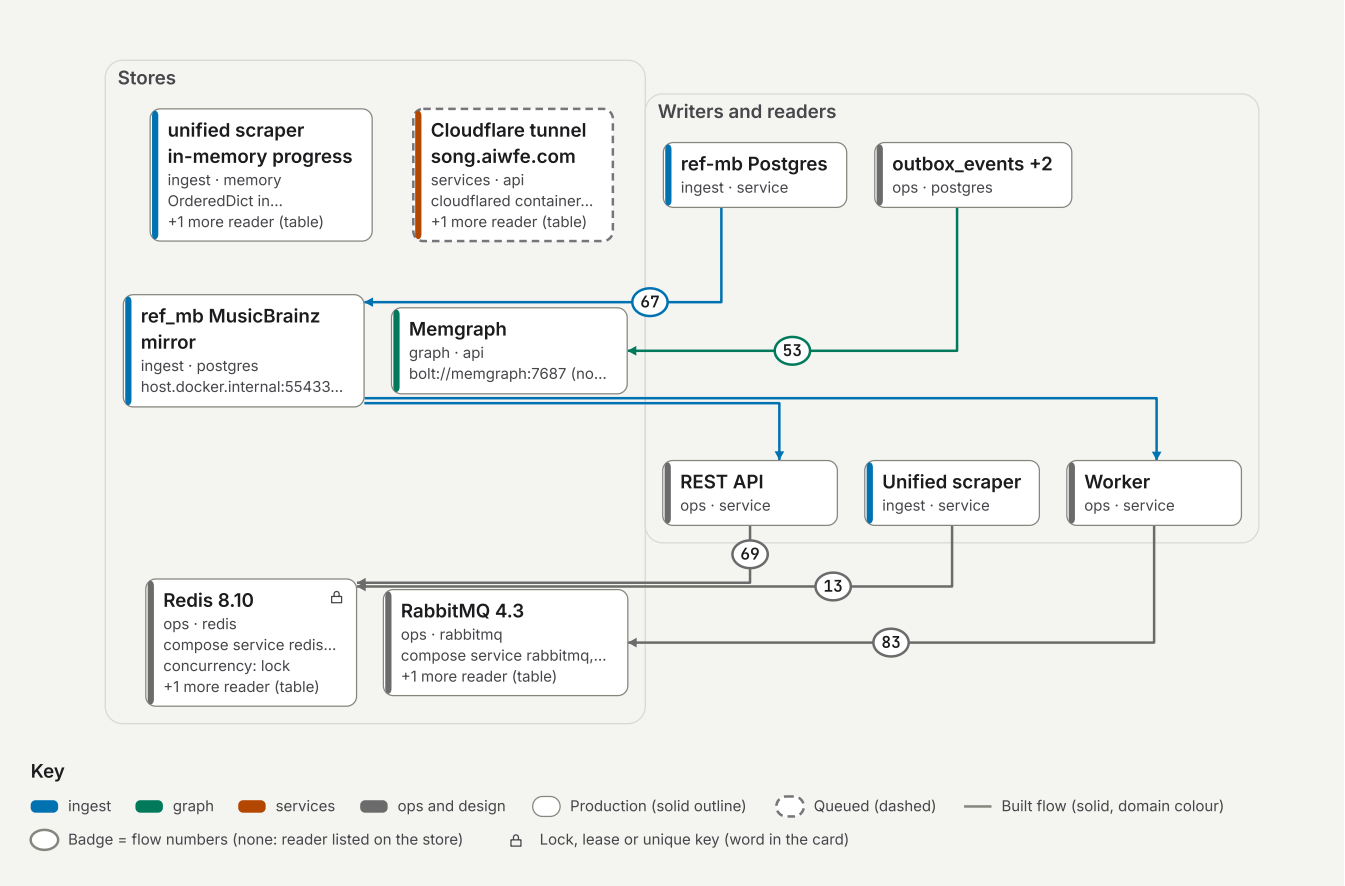


D3d · n = 6 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 12 crossings, 0 edges through nodes

D3e: **Storage topology – songnodes services and queues**

1 of 6 songnodes services and queues carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

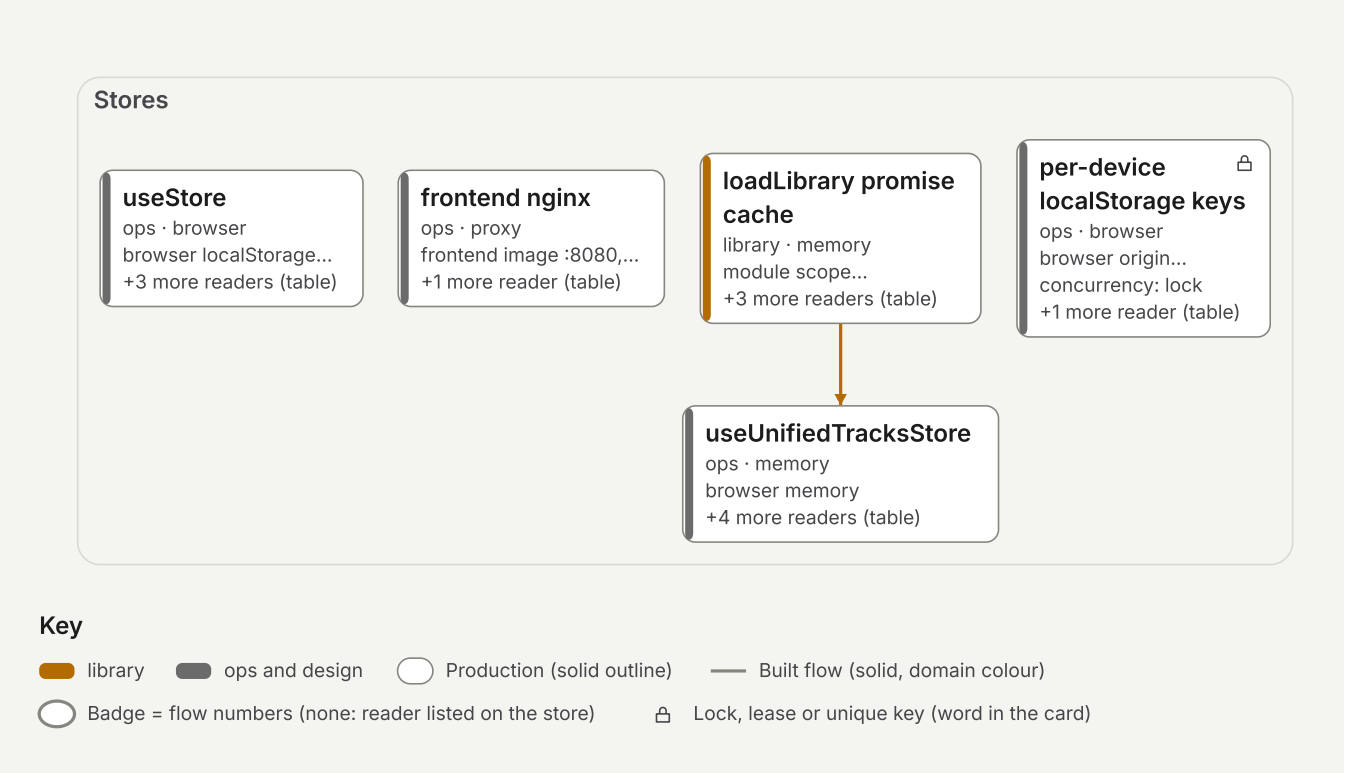


D3e · n = 6 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D3f: **Storage topology – songnodes browser stores**

1 of 5 songnodes browser stores carry a lock, lease or unique-key rule; the rest rely on whole-file overwrite or last writer wins.

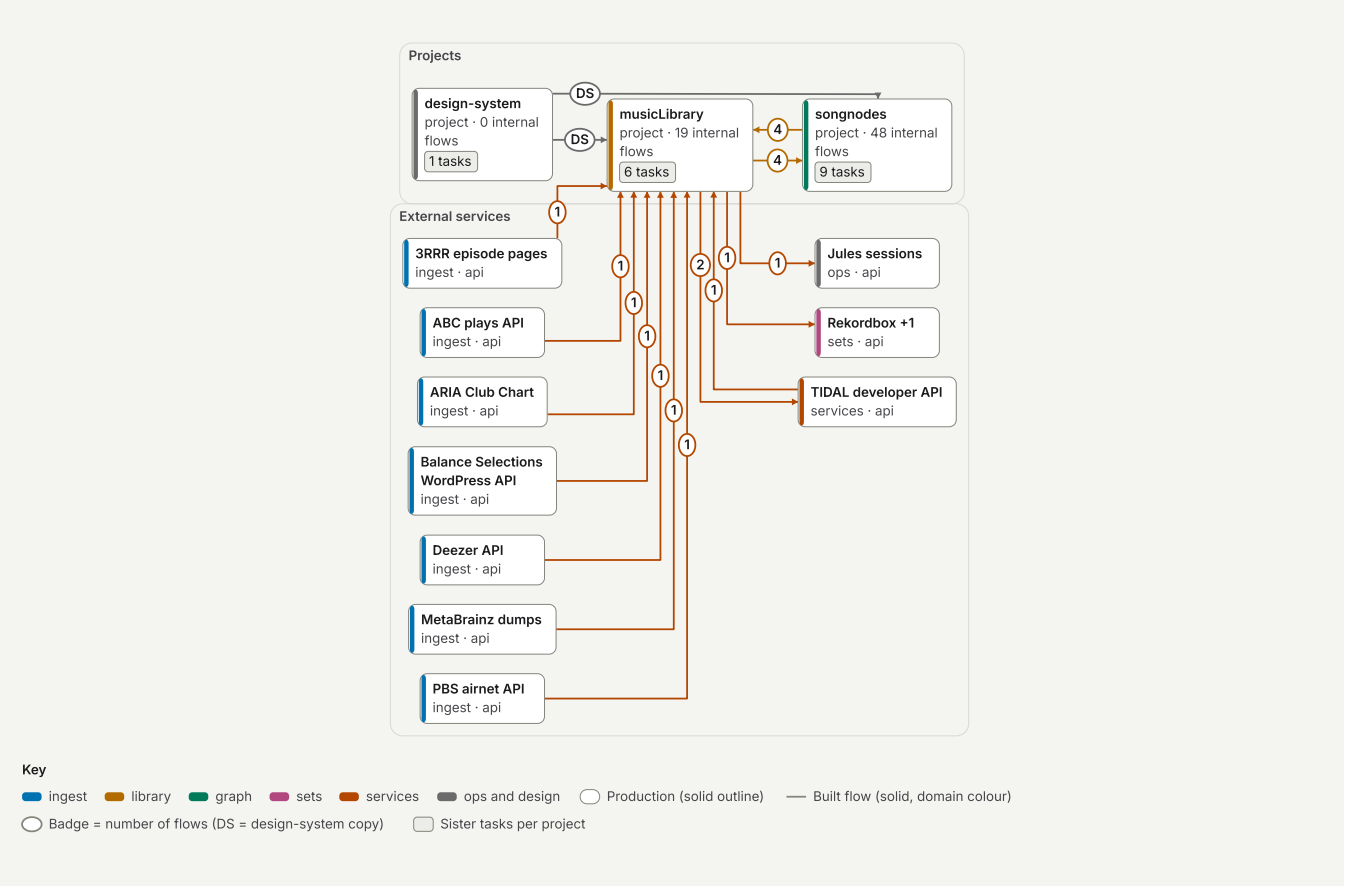


D3f · n = 5 · Unit: stores · ● Built

· Source: stores[] (kind, location, owner, readers, concurrency); flows[] touching stores · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D12: **Sister-project dependency graph**

Only 8 of 87 flows cross between songnodes and musicLibrary; the rest stay inside one project or reach an external service.



D12 · n = 15 · Unit: project pairs · ● Built

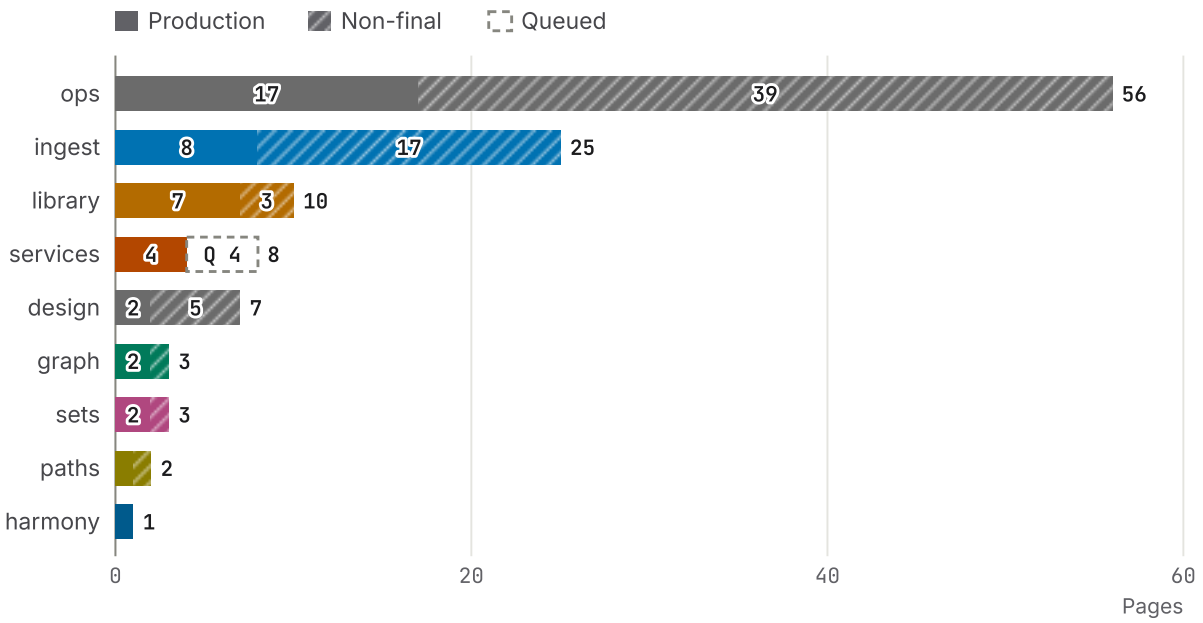
· Source: flows[] aggregated by repo pair (pages[]/stores[].repo); tasks[].project; design-system pages · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 0 crossings, 0 edges through nodes

Section 3: **Page catalogue and standing rules** 115 pages

44 production, 71 non-final, 17 rule violations.

C1: **Pages by domain and status**

ops holds the most pages (56 of 115); 67 pages are non-final and 4 queued.



C1 · n = 115 · Unit: pages · i Projected

· Source: pages[].domain, pages[].classification, pages[].state · Method: Count of pages per domain; built pages split by classification, queued pages counted separately; sorted by total.

Rules: **Standing rules audited** 8 rules

8 rows

RULE	RULE TEXT	SOURCE
R1	The NavDrawer is the only site navigation.	songnodes/CLAUDE.md §4 Rules: Interaction
R2	One primary button per view.	songnodes/CLAUDE.md §4 Rules: Interaction
R3	Never alert, confirm or prompt; use ConfirmDialog and Toast.	songnodes/CLAUDE.md §4 Rules: Interaction
R4	Tokens and sn-* classes only: no hex, rgb, named colours, px font sizes, ad-hoc radii, shadows or z-index numbers.	songnodes/CLAUDE.md §4 Rules: Colour and values
R5	No emoji in the interface; en-AU sentence case; missing values are an en dash.	songnodes/CLAUDE.md §4 Rules: Writing
R6	Pages register their route (G2) and REST routers register themselves (G3); do not wire App.tsx or main.py.	songnodes/AGENTS.md rule 12
R7	unified-scraper is the only scraper container.	songnodes/CLAUDE.md Critical requirement 3
R8	Valid artists on both endpoints of every transition; no NULL or Unknown Artist in the graph.	songnodes/CLAUDE.md Critical requirement 1

Table: **Page catalogue** 115 pages

115 rows

PAGE	REPO	DOMAIN	CATEGORY	STATE	BYTES	R1
Naarm Set Builder (design-system build) artifact/build/index.html	musicLibrary	design	● Production	● Built	134	●
Naarm Set Builder (original hand-written page) artifact/index.html	musicLibrary	sets	i Non-final superseded-prototype	● Built	42,185	×
Naarm Set Builder (Vite entry) artifact/app/index.html	musicLibrary	ops	i Non-final dev-tool	● Built	179	–

run_weekly weekly `pipeline/run_weekly.py weekly`	musicLibrary	ops	Production	Built	32,519	–
run_weekly scrape `pipeline/run_weekly.py scrape`	musicLibrary	ingest	Production	Built	32,519	–
run_weekly ingest `pipeline/run_weekly.py ingest`	musicLibrary	ingest	Production	Built	32,519	–
run_weekly resolve `pipeline/run_weekly.py resolve`	musicLibrary	ingest	Production	Built	32,519	–
run_weekly process `pipeline/run_weekly.py process`	musicLibrary	library	Production	Built	32,519	–
run_weekly sync-tidal `pipeline/run_weekly.py sync-tidal`	musicLibrary	services	Production	Built	32,519	–
run_weekly confirm `pipeline/run_weekly.py confirm`	musicLibrary	ops	Production	Built	32,519	–
run_weekly db `pipeline/run_weekly.py db`	musicLibrary	library	Production	Built	32,519	–
run_weekly sync-sets `pipeline/run_weekly.py sync-sets`	musicLibrary	sets	Production	Built	32,519	–
run_weekly sync-songnodes `pipeline/run_weekly.py sync-songnodes`	musicLibrary	ops	Production	Built	32,519	–
run_weekly status `pipeline/run_weekly.py status`	musicLibrary	library	Production	Built	32,519	–
build_db `pipeline/build_db.py`	musicLibrary	ops	Production	Built	13,928	–
calibrate `pipeline/calibrate.py`	musicLibrary	library	Production	Built	4,409	–
tidal_cli `pipeline/tidal_cli.py`	musicLibrary	services	Production	Built	2,166	–
test_pipeline `pipeline/test_pipeline.py`	musicLibrary	ops	Non-final test-harness	Built	4,525	–
fetch_metabrainz (py and sh) `pipeline/fetch_metabrainz.py`	musicLibrary	ingest	Non-final dev-tool	Built	5,097	–
ref-mb Postgres `ref_mb/compose.yaml`	musicLibrary	ingest	Production	Built	2,179	–
ref_mb load and maintenance scripts `ref_mb/scripts/load_ref_mb.ps1`	musicLibrary	ingest	Non-final dev-tool	Built	2,968	–
ref_mb accuracy evaluation `ref_mb/scripts/run_eval.ps1`	musicLibrary	ingest	Non-final test-harness	Built	1,327	–
Weekly run playbook `agent/weekly_run.md`	musicLibrary	ops	Production	Built	5,567	–
Beatport sync playbook `agent/beatport_sync.md`	musicLibrary	ops	Production	Built	1,744	–
Jules launchers and prompt builders `plan/launch_jules.py`	musicLibrary	ops	Non-final dev-tool	Built	14,261	–
songNodes weekend progress run prompt `plan/scheduled_run_prompt.md`	musicLibrary	ops	Non-final doc-reference	Built	3,548	–
songnodes pytest report `tests/reports/junit.xml`	musicLibrary	ops	Non-final doc-reference	Built	1,734	–
REST API `services/rest_api`	songnodes	ops	Production	Built	30,385	–
Graph API `services/graph-visualization-api`	songnodes	graph	Production	Built	40,500	–
Worker `services/worker/celery_app.py`	songnodes	ops	Production	Built	5,008	–
Unified scraper `scrapers/unified_scraper_api.py`	songnodes	ingest	Production	Built	24,173	–

Audio analysis `services/audio-analysis/main.py`	songnodes	ops	● Production	● Built	25,523	–
Postgres 18 `compose.dev.yaml#postgres`	songnodes	ops	● Production	● Built	8,505	–
api-gateway `services/api-gateway`	songnodes	ops	i Non-final superseded-prototype	● Built	-	–
api-gateway-internal `services/api-gateway-internal`	songnodes	ops	i Non-final superseded-prototype	● Built	17,030	–
browser-collector `services/browser-collector`	songnodes	ops	i Non-final superseded-prototype	● Built	26,662	–
data-transformer `services/data-transformer`	songnodes	ops	i Non-final superseded-prototype	● Built	52,495	–
data-validator `services/data-validator`	songnodes	ops	i Non-final superseded-prototype	● Built	57,259	–
db-connection-pool `services/db-connection-pool`	songnodes	ops	i Non-final superseded-prototype	● Built	16,452	–
dlq-manager `services/dlq-manager`	songnodes	ops	i Non-final superseded-prototype	● Built	22,041	–
enhanced-visualization-service `services/enhanced-visualization-service`	songnodes	ops	i Non-final superseded-prototype	● Built	-	–
gold-to-operational-etl `services/gold-to-operational-etl`	songnodes	ops	i Non-final superseded-prototype	● Built	23,029	–
gold-layer-processor `services/gold_layer_processor.py`	songnodes	library	i Non-final superseded-prototype	● Built	14,800	–
graph-projection `services/graph-projection`	songnodes	graph	i Non-final dev-tool	● Built	1,405	–
health-monitor `services/health-monitor`	songnodes	ops	i Non-final superseded-prototype	● Built	18,384	–
metadata-enrichment `services/metadata-enrichment`	songnodes	ops	i Non-final superseded-prototype	● Built	69,816	–
nlp-processor `services/nlp-processor`	songnodes	library	i Non-final superseded-prototype	● Built	25,410	–
reference-data `services/reference-data`	songnodes	ops	i Non-final superseded-prototype	● Built	654	–
serato-integration `services/serato-integration`	songnodes	ops	i Non-final migration-repair	● Built	16,312	–
websocket-api `services/websocket-api`	songnodes	ops	i Non-final superseded-prototype	● Built	35,590	–
database `services/database`	songnodes	ops	i Non-final doc-reference	● Built	-	–
scraper-orchestrator `scraper-orchestrator`	songnodes	ingest	i Non-final migration-repair	● Built	9,991	–
scraper_service.py `scrapers/scraper_service.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	7,957	–
scraper_api_real.py `scrapers/scraper_api_real.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	9,484	–
real_data_scraper.py `scrapers/real_data_scraper.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	19,724	–
raw_data_processor.py `scrapers/raw_data_processor.py`	songnodes	ingest	i Non-final migration-repair	● Built	23,683	–
recover_raw_data.py `scrapers/recover_raw_data.py`	songnodes	ingest	i Non-final migration-repair	● Built	35,415	–
enrich_existing_tracks.py `scrapers/enrich_existing_tracks.py`	songnodes	ingest	i Non-final migration-repair	● Built	10,550	–

backfill_artist_names.py `scrapers/backfill_artist_names.py`	songnodes	ingest	i Non-final migration-repair	● Built	12,228	–
batch_scrape_all_playlists.py `scrapers/batch_scrape_all_playlists.py`	songnodes	ingest	i Non-final dev-tool	● Built	3,754	–
import_nodes_data.py `scrapers/import_nodes_data.py`	songnodes	ingest	i Non-final migration-repair	● Built	7,983	–
import_setlist_data.py `scrapers/import_setlist_data.py`	songnodes	ingest	i Non-final migration-repair	● Built	18,599	–
enrichment_queue_publisher.py `scrapers/pipelines/enrichment_queue_publisher.py`	songnodes	ingest	i Non-final superseded-prototype	● Built	10,538	–
004_create_target_track_searches.sql `scrapers/migrations/004_create_target_track_searches.sql`	songnodes	ingest	i Non-final superseded-prototype	● Built	1,785	–
load_283_tracks.sql `load_283_tracks.sql`	songnodes	ops	i Non-final migration-repair	● Built	69,728	–
assign_proper_artists.py `assign_proper_artists.py`	songnodes	ops	i Non-final migration-repair	● Built	8,341	–
analyze-adjacency-problem.js `analyze-adjacency-problem.js`	songnodes	ops	i Non-final dev-tool	● Built	4,419	–
backup_volumes.sh `backup_volumes.sh`	songnodes	ops	i Non-final superseded-prototype	● Built	7,525	–
monitor_pipeline.sh `monitor_pipeline.sh`	songnodes	ops	i Non-final superseded-prototype	● Built	5,355	–
setup-autostart.sh `setup-autostart.sh`	songnodes	ops	i Non-final release-installer	● Built	900	–
songnodes.service `songnodes.service`	songnodes	ops	i Non-final release-installer	● Built	602	–
start-docker.sh `start-docker.sh`	songnodes	ops	i Non-final superseded-prototype	● Built	7,437	–
skaffold.yaml `skaffold.yaml`	songnodes	ops	i Non-final superseded-prototype	● Built	8,011	–
QUICK_START_QUERIES.sql `QUICK_START_QUERIES.sql`	songnodes	ops	i Non-final doc-reference	● Built	10,392	–
backup_database.sh `scripts/backup_database.sh`	songnodes	ops	i Non-final superseded-prototype	● Built	2,644	–
restore_and_promote.sh `scripts/restore_and_promote.sh`	songnodes	ops	i Non-final migration-repair	● Built	8,297	–
upgrade-postgres-18.sh `scripts/upgrade-postgres-18.sh`	songnodes	ops	i Non-final migration-repair	● Built	3,548	–
remove_same_artist_edges.sql `scripts/remove_same_artist_edges.sql`	songnodes	ops	i Non-final migration-repair	● Built	3,618	–
clean_bracketed_artists.sql `scripts/clean_bracketed_artists.sql`	songnodes	ops	i Non-final migration-repair	● Built	4,987	–
load-library-targets.sh `scripts/load-library-targets.sh`	songnodes	library	i Non-final dev-tool	● Built	1,474	–
run_weekly.py `pipeline/run_weekly.py`	songnodes	ops	i Non-final trial-fork	● Built	7,022	–
helm `deploy/helm`	songnodes	ops	i Non-final superseded-prototype	● Built	36,560	–
k8s `k8s`	songnodes	ops	i Non-final superseded-prototype	● Built	9,886	–
start-dev `scripts/start-dev.sh`	songnodes	ops	● Production	● Built	6,324	–
import-music-library `scripts/import-music-library.sh`	songnodes	library	● Production	● Built	3,555	–

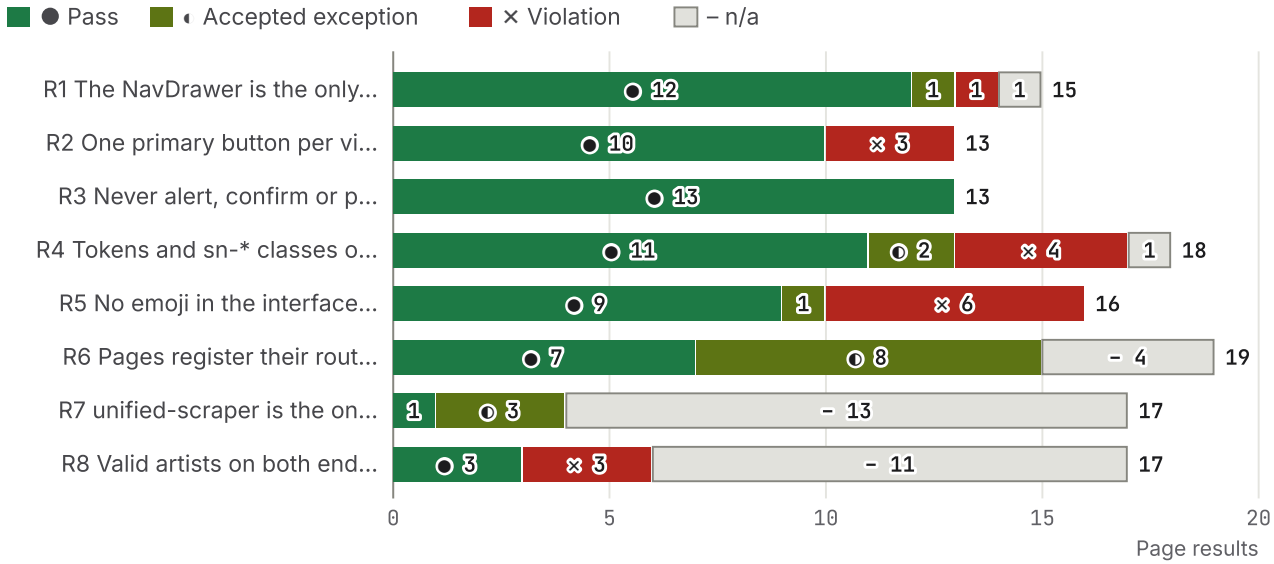
sync-design-system `scripts/sync-design-system.sh`	songnodes	design	● Production	● Built	3,079	–
scope_guard (G1) `scripts/scope_guard.py`	songnodes	ops	● Production	● Built	9,008	–
e2e-docker `scripts/e2e-docker.sh`	songnodes	ops	● Production	● Built	1,365	–
migrate.py `services/worker/migrate.py`	songnodes	ops	● Production	● Built	4,367	–
library reconcile CLI `services/library/reconcile/cli.py`	songnodes	library	● Production	● Built	2,014	–
Duo auth router (DUO1) `services/rest_api/routers/user_auth.py`	songnodes	services	i Non-final trial-fork	▲ Queued	-	–
cloudflared tunnel (DUO3) `compose.dev.yaml#cloudflared`	songnodes	services	i Non-final trial-fork	▲ Queued	-	–
Duo and tunnel guide (DUOAUTH) `docs/security/cisco-duo-auth-guide.md`	songnodes	services	i Non-final doc-reference	▲ Queued	-	–
Home `songnodes/frontend/src/pages/HomePage.tsx`	songnodes	ops	● Production	● Built	3,492	●
Adjacency `songnodes/frontend/src/pages/GraphPage.tsx`	songnodes	graph	● Production	● Built	2,074	●
Melbourne crate `songnodes/frontend/src/features/musicLibrary/LibraryPage.tsx`	songnodes	library	● Production	● Built	27,441	●
Set builder `songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx`	songnodes	sets	● Production	● Built	39,010	●
Route finder `songnodes/frontend/src/features/pathfinder/PathfinderPage.tsx`	songnodes	paths	● Production	● Built	14,782	●
Route finder (planned) `songnodes/frontend/src/pages/ModulePages.tsx`	songnodes	paths	i Non-final superseded-prototype	● Built	3,338	–
Camelot wheel `songnodes/frontend/src/features/camelot/CamelotPage.tsx`	songnodes	harmony	● Production	● Built	9,119	●
Target tracks `songnodes/frontend/src/pages/ModulePages.tsx`	songnodes	ingest	● Production	● Built	3,338	●
Import tracklist `songnodes/frontend/src/features/aiSetlists/AISetlistsPage.tsx`	songnodes	ingest	● Production	● Built	18,853	●
Telemetry `songnodes/frontend/src/pages/ScraperProgressPage.tsx`	songnodes	ingest	● Production	● Built	26,117	●
Settings `songnodes/frontend/src/pages/SettingsPage.tsx`	songnodes	services	● Production	● Built	44,111	●
Service sign-in return `songnodes/frontend/src/components/OAuthCallback.tsx`	songnodes	services	● Production	● Built	13,171	◐
Page not found `songnodes/frontend/src/pages/ModulePages.tsx`	songnodes	ops	● Production	● Built	3,338	●
SongNodes `songnodes/frontend/index.html`	songnodes	ops	● Production	● Built	1,304	–
Design system reference · SongNodes `songnodes/docs/design-system/reference.html`	songnodes	design	i Non-final doc-reference	● Built	93,789	–
Design system reference · SongNodes (standalone) `songnodes/docs/design-system/reference.standalone.html`	songnodes	design	i Non-final doc-reference	● Built	633,370	–
Component previews (98 files) `songnodes/design-system/components/*/preview.html`	songnodes	design	i Non-final doc-reference	● Built	1,068,167	–
Design-system source previews (39 files) `songnodes/design-system/source/previews/{navsrc,charts-src,pagesA/src,pagesb/src}/*.html`	songnodes	design	i Non-final doc-reference	● Built	403,625	–
Violations fixture `songnodes/design-system/fixtures/violations.html`	songnodes	design	i Non-final test-harness	● Built	568	–
SongNodes Work Queue `songnodes/plan/work-queue-2026-10-30/tracker.html`	songnodes	ops	i Non-final dev-tool	● Built	11,468	◔

Test DJ - Awesome Mix | MixesDB `songnodes/tests/unit/scrapers/fixtures/mixesdb_two_lists.html`	songnodes	ingest	i Non-final test-harness	● Built	849	–
Sign in `songnodes/frontend/src/features/auth/LoginPage.tsx`	songnodes	services	i Non-final trial-fork	▲ Queued	–	–

Category uses at least two independent signals per page (naming, live-data wiring, route registration or inbound links, docs status); hover a pill for the first two.

C8: **Standing-rule conformance**

17 of 128 page results are violations; R5 has the most (6).

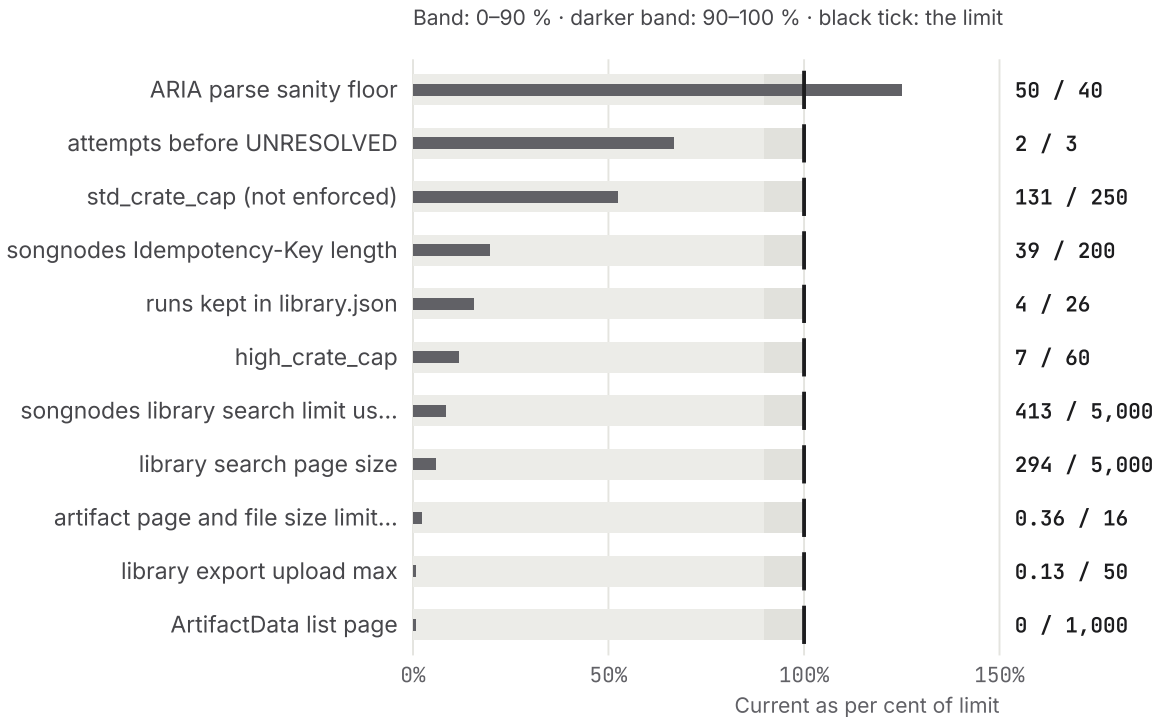


C8 · n = 128 · Unit: page results · ● Built

· Source: rule_results[] (rule, result), standing_rules[] · Method: Count of page results per rule R1–R8 (ordinal, not sorted) split by result.

C7: **Sizes against limits**

None of the 11 measured limits is within 10 % of its limit; the one with least headroom is ARIA parse sanity floor at 125 % of its floor.

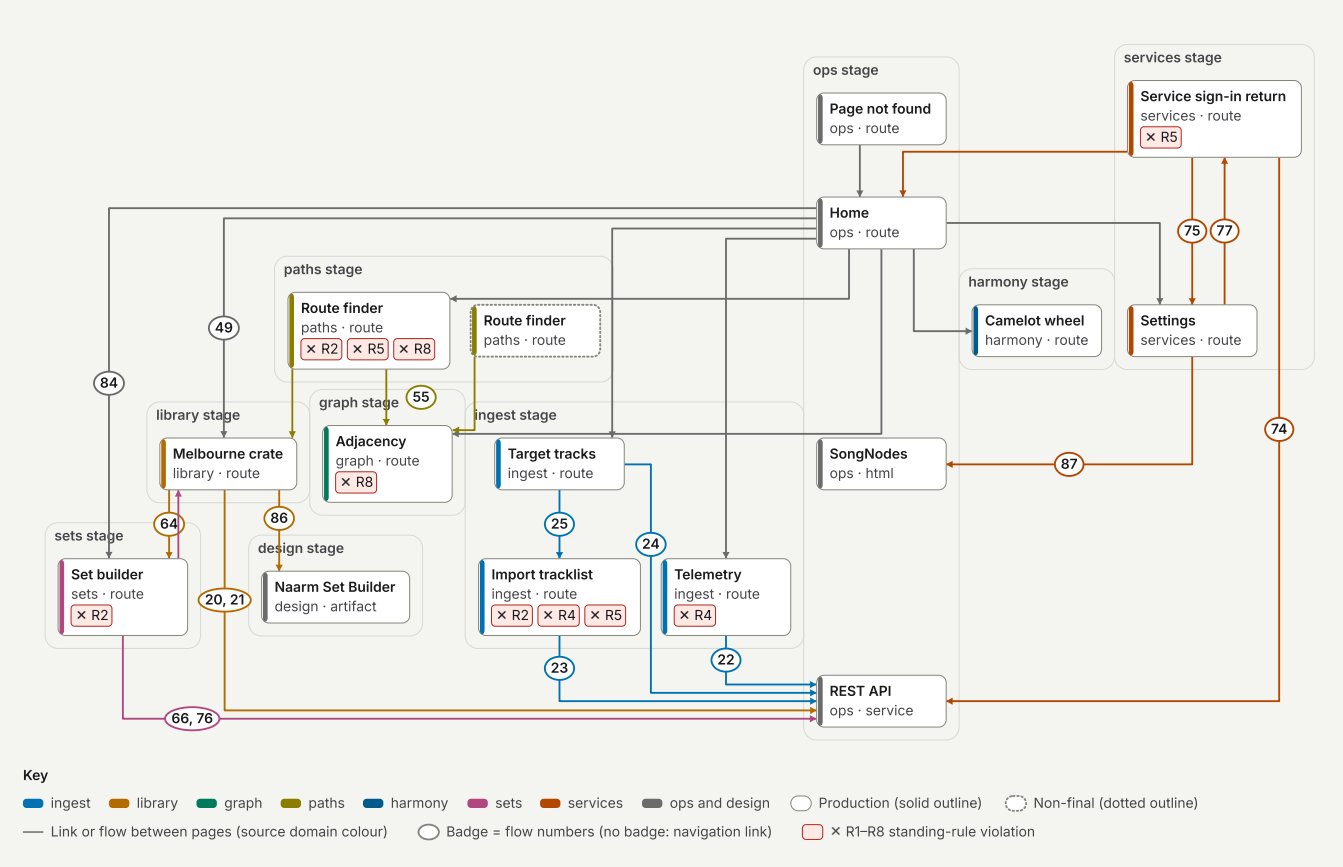


C7 · n = 11 · Unit: per cent of limit · ● Built

· Source: limits[] with current (value, unit, current) · Method: Bullet chart: current / limit. Text currents take their first number; a file map takes its largest file (the limit is per file), bytes to MB at 1,048,576. Floors (name says floor or minimum) are near when under 110 %, others when at or over 90 %.

D4a: **Page link graph – songnodes frontend**

7 of 17 songnodes frontend pages carry at least one standing-rule violation.

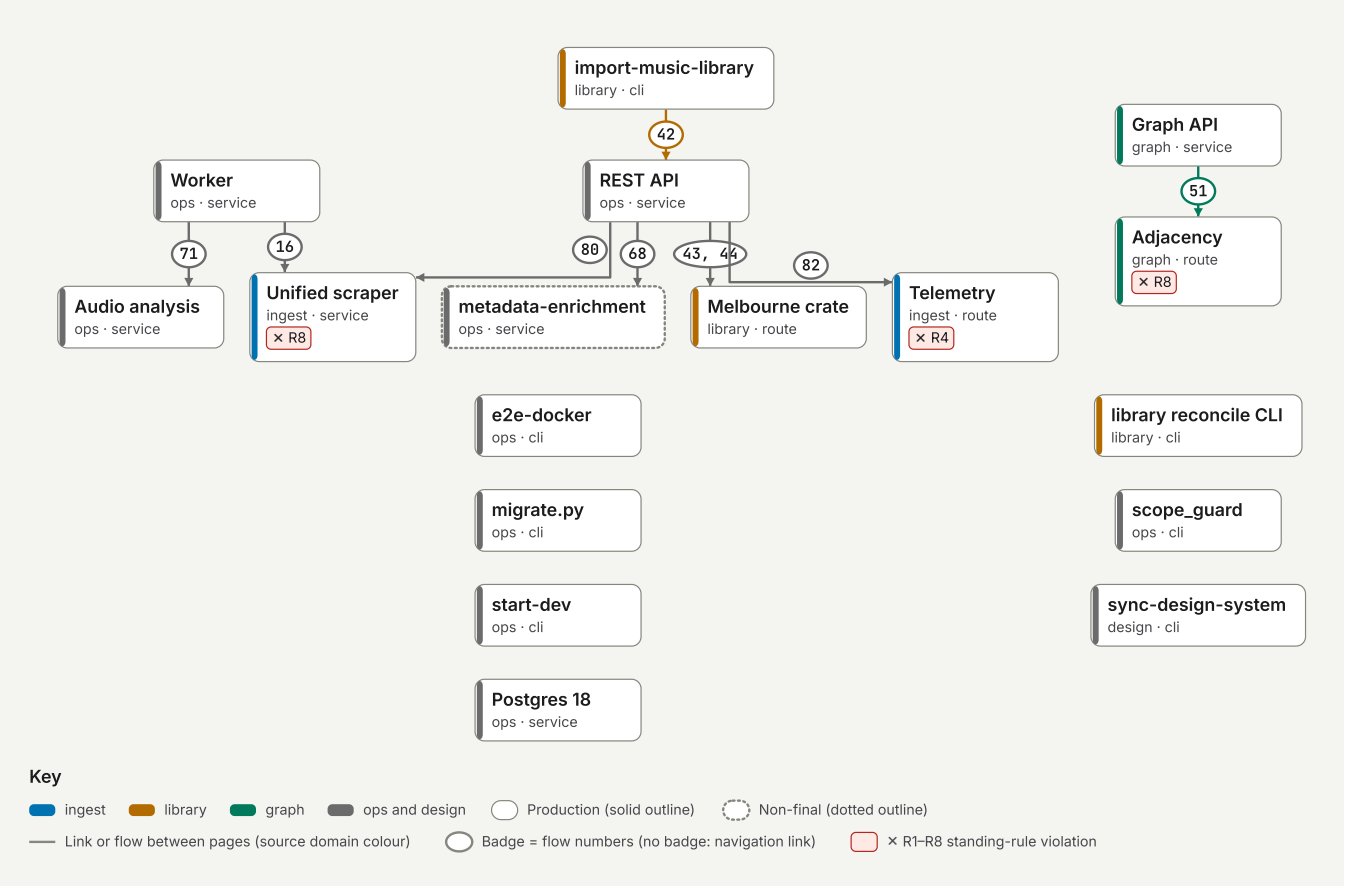


D4a · n = 17 · Unit: pages · **i built+queued**

· Source: pages[] (links_out, classification, state, domain); flows[] between pages; rule_results[] (result = violation) · Method: graphviz dot layered layout (TB, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 4 crossings, 0 edges through nodes

D4b: **Page link graph – songnodes backend**

3 of 20 songnodes backend pages carry at least one standing-rule violation.

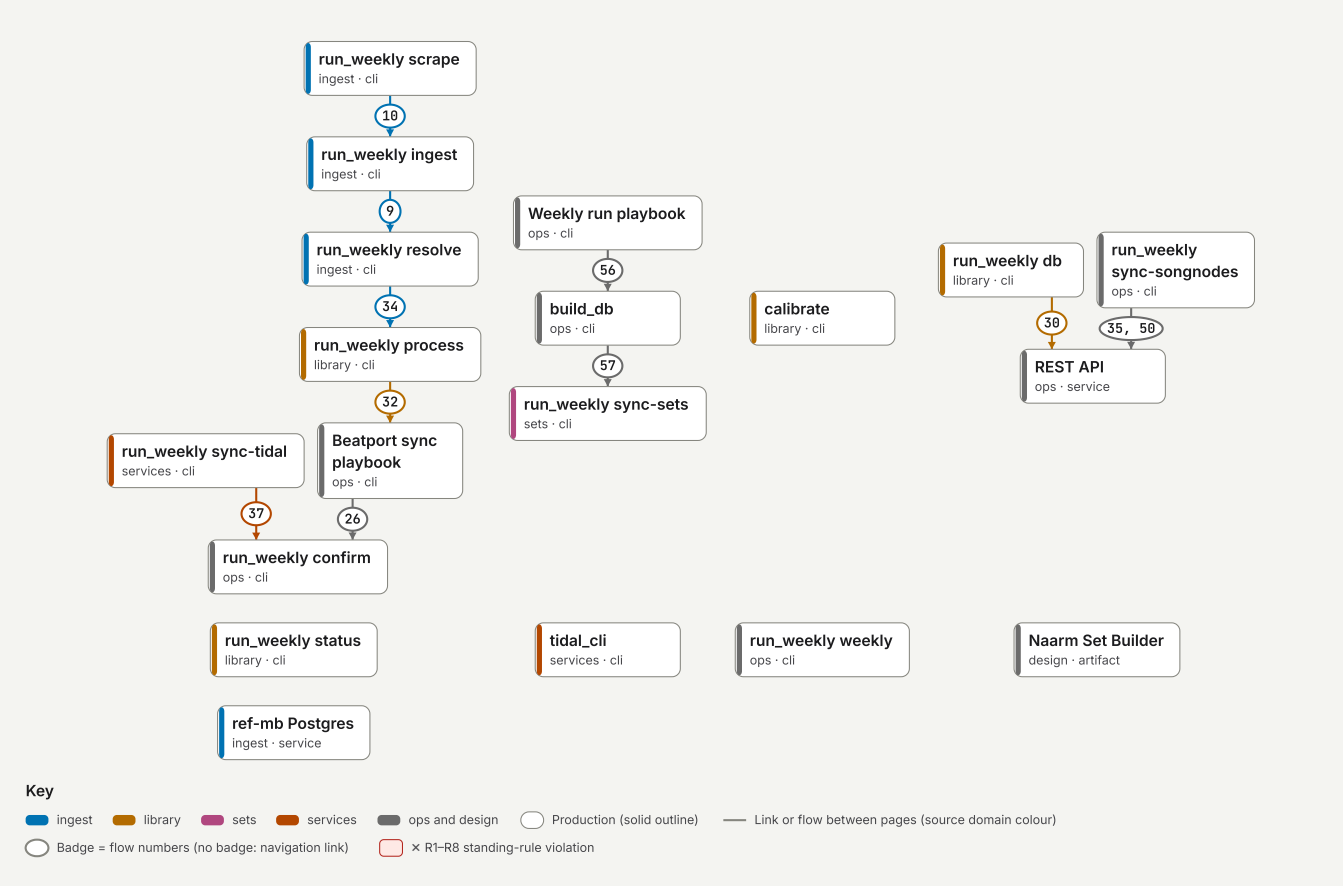


D4b · n = 20 · Unit: pages · **i built+queued**

· Source: pages[] (links_out, classification, state, domain); flows[] between pages; rule_results[] (result = violation) · Method: graphviz dot layered layout (TB, order degree, components packed 2 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D4c: **Page link graph – musicLibrary**

0 of 19 musicLibrary pages carry at least one standing-rule violation.



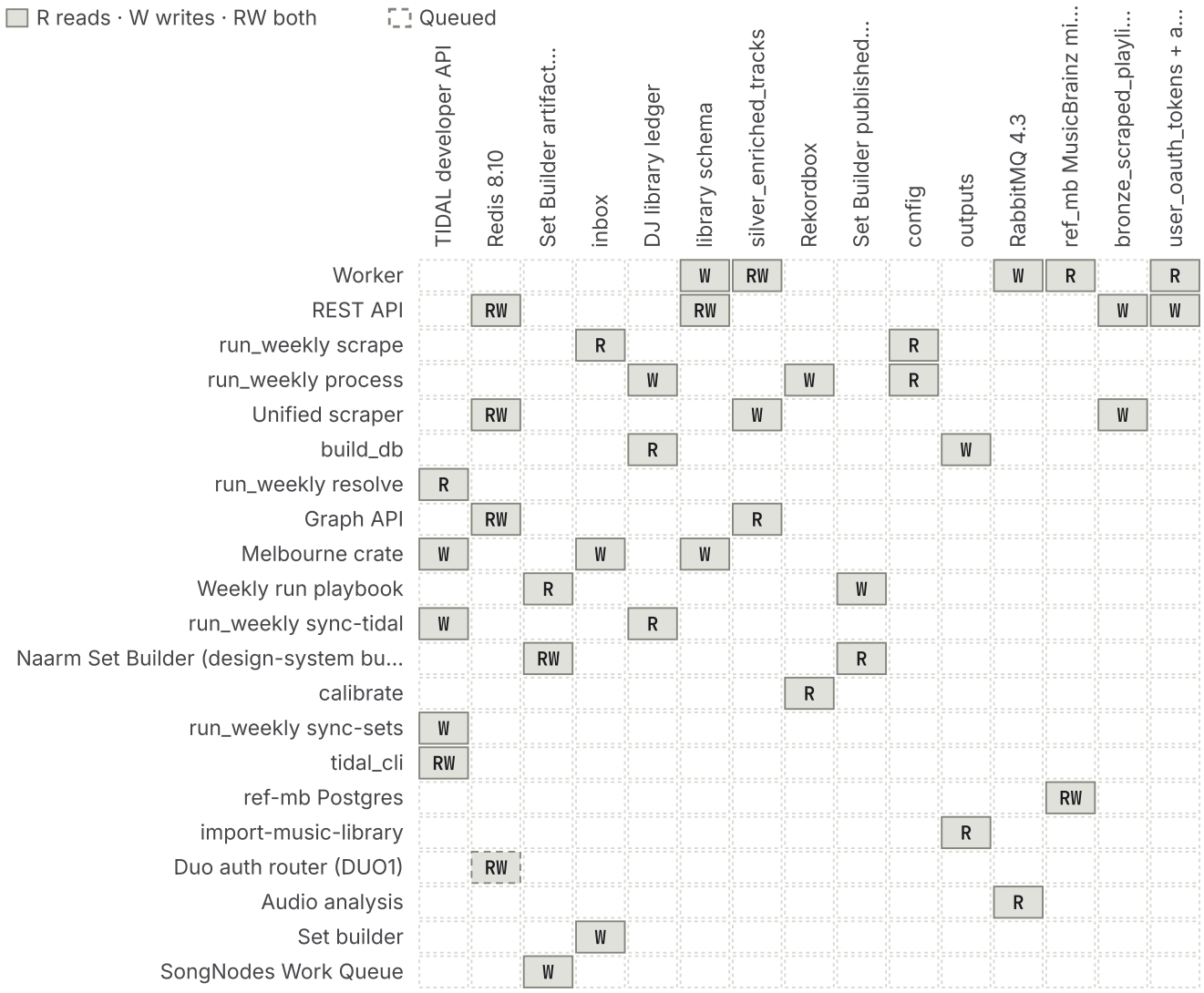
D4c · n = 19 · Unit: pages · ● Built

· Source: pages[] (links_out, classification, state, domain); flows[] between pages; rule_results[] (result = violation) · Method: graphviz dot layered layout (TB, order given, components packed 4 across, ungrouped; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

Section 4: **Stores, signals and contracts** 48 stores · 404 signals

C2: **Page and store access matrix**

The 21 most connected pages against the 15 most used stores (of 23 pages and 35 stores with 61 access links); letters give the access, R read, W write, RW both.

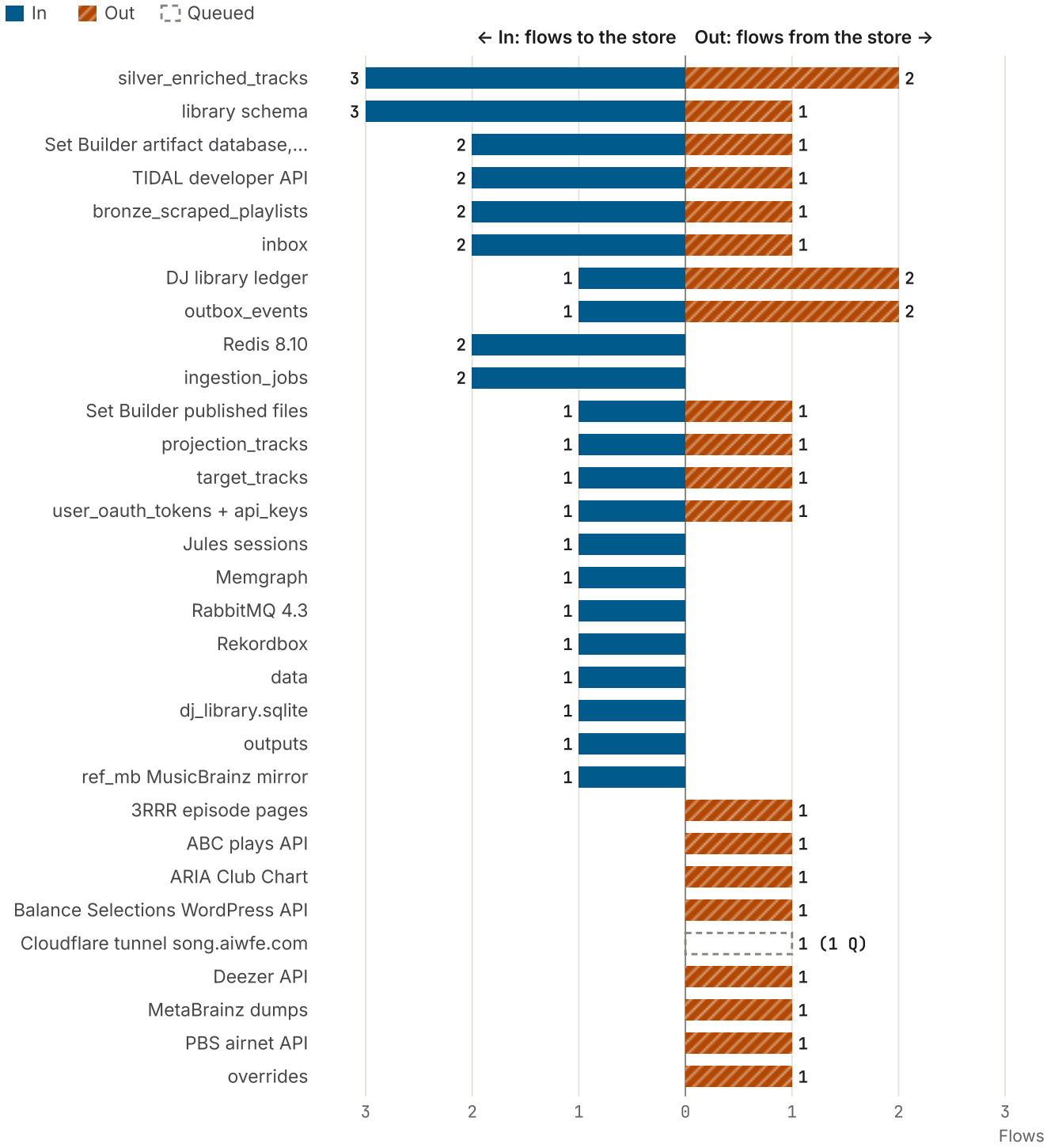


C2 · n = 61 · Unit: page–store links · **i Projected**

· Source: signals[] (from, to, direction, state), flows[] (from, to, state) · Method: A link is any signal or flow between a page and a store; signal direction gives R/W/RW, a flow into a store is W and out of a store is R. Rows and columns ranked by distinct connections (ties by id) and capped at 25 by 15.

C4: **Store fan-in and fan-out**

silver_enriched_tracks is the busiest store (3 flows in, 2 out); 31 of 48 stores sit on a numbered flow.

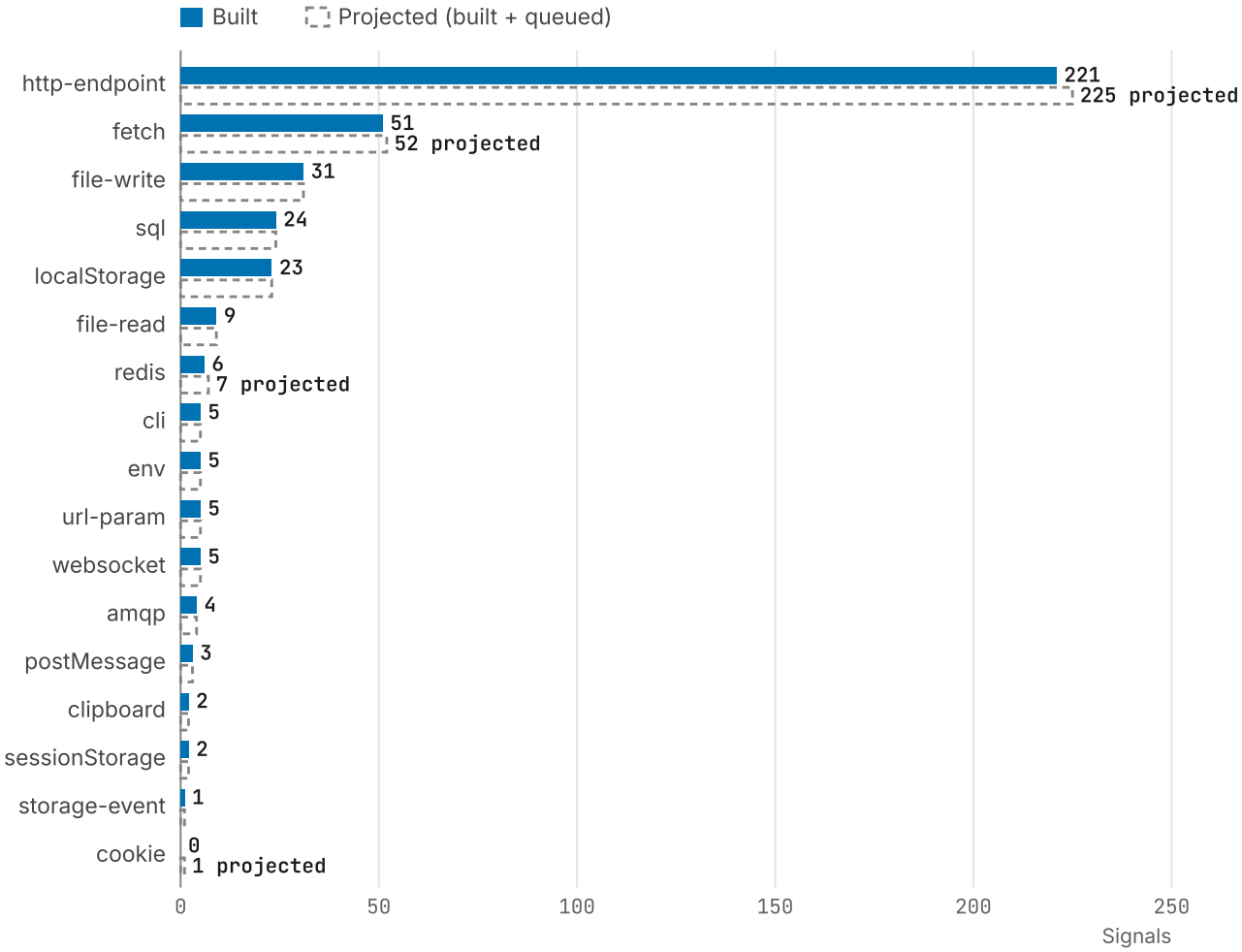


C4 · n = 49 · Unit: flows · i Projected

· Source: flows[] (from, to, channel, state), stores[], signals[] for the contract · Method: In = flows whose target is the store, out = flows whose source is the store; main contract = the signal named by the store's most frequent flow channel.

C3: **Signal primitives, built and projected**

http-endpoint carries 225 of 404 signals; queued work adds 7 signals (cookie, fetch, http-endpoint, redis).



C3 · n = 404 · Unit: signals · i Projected

· Source: signals[].type, signals[].state · Method: Count of signals per type; Built = state built, Projected = built + queued; sorted by projected count.

Table: **Stores and contracts** 48 stores

48 rows

STORE	KIND	LOCATION	WRITER	READERS
config `ml-store-config`	file	`config/rules.json`, `config/sources.json`	Markus (hand-edited; weekly agent must not edit thresholds)	ml-cli-scrape, ml cli-ingest, ml-cli-process, ml-cli-build-db, ml-cli-t pipeline
inbox `ml-store-inbox`	folder	`inbox/`	scrape, ingest, resolve, sync-tidal, weekly agent (artifact_sets.json), beatport agent (resolved/sync_result), Markus (manual_adds.csv)	ml-cli-ingest, ml-resolve, ml-cli-process, ml-cli-confirm, ml-cli-build-db, ml-cli-sync-sets, ml-cli sync-songnodes
DJ library ledger `ml-store-ledger`	file	`ledger/dj_library_ledger.json (+ ledger/history/)`	run_weekly ingest/process/confirm	ml-cli-build-db, r cli-calibrate, ml-(sync-tidal, ml-cli sync-sets, ml-cli status, ml-agent-beatport-sync

outputs `ml-store-outputs`	folder	`outputs/<date>/, outputs/crates/, outputs/artifact/library.json, outputs/rekordbox_musicLibrary.xml`	process, build_db, calibrate	ml-cli-build-db, r agent-weekly-ru ml-agent-beatpo sync, sn-scripts, persona-curator
overrides `ml-store-overrides`	folder	`overrides/energy_overrides.csv, overrides/rekordbox_collection.xml (absent)`	Markus	ml-cli-process, n cli-calibrate
dj_library.sqlite `ml-store-sqlite`	sqlite	`db/dj_library.sqlite`	build_db	ml-persona-cura (DB Browser)
secrets `ml-store-secrets`	folder	`secrets/ (tidal_app.json, tidal_user_token.json, tidal_pkce_pending.json, tidal_playlists.json); ref_mb/secrets/*.txt`	tidal_cli, sync-tidal, sync-sets; Markus	ml-cli-resolve, m cli-sync-tidal, ml sync-sets, ml-sv ref-mb
Set Builder artifact database, collection sets `ml-store-artifact-db`	artifact-db	`claude.ai artifact GcqZTGjk6obdpjWaU7WFRq`	Markus via the page (only-you-write rule)	ml-page-set-buil ml-agent-weekly run
Set Builder published files `ml-store-artifact-files`	artifact-files	`claude.ai artifact GcqZTGjk6obdpjWaU7WFRq: index.html, app.js, assets/*, library.json`	weekly agent (library.json), supervised rebuild (app.js)	ml-page-set-buil
ref_mb Postgres `ml-store-ref-mb`	postgres	`127.0.0.1:55433, data E:\musicLibrary\data\ref_mb_pgdata`	ref_mb load scripts	sn-be-refmb, ml-ref-mb-eval
data (58 GB, not copied) `ml-store-data`	folder	`data/: metabrainz/ (canonical, json, postgres dumps, logs), metabrainz_scratch/ (inventory, slice, sql benches, tools, a MusicBrainz server git checkout mbs/, PROGRESS.md), ref_mb_pgdata/pgdata (Postgres cluster, postmaster.pid 2026-11-06)`	fetch_metabrainz.*, ref_mb scripts and compose, MB investigation agents	ml-svc-ref-mb
plan `ml-store-plan`	folder	`plan/`	prompt builders and launchers	ext-jules
songnodes Postgres 18 `sn-be-store-postgres`	postgres	`compose service postgres, volume songnodes-dev_dev-postgres-18 (not published)`	rest-api, worker, unified-scraper, graph-api	sn-be-svc-rest-a sn-be-svc-graph api, sn-be-svc-worker, sn-be-sv unified-scraper
bronze_scraped_playlists / bronze_scraped_tracks / bronze_scraped_artists `sn-be-tbl-bronze`	postgres	`public schema (medallion 001)`	unified-scraper persistence pipeline; rest-api import-tracklist	sn-be-svc-worke (ingest), sn-be-s rest-api (catalog_reads, j
bronze_api_enrichments `sn-be-tbl-bronze-api-enrichments`	postgres	`medallion 001`	none	–
silver_enriched_tracks / _artists / _playlists / silver_playlist_tracks / silver_track_transitions / silver_track_metadata_evidence / silver_camelot_corrections `sn-be-tbl-silver`	postgres	`medallion 002, 006, migration 014, 270`	worker datalayer (ingest, evidence, identity); also scraper persistence pipeline and rest-api catalog writes	sn-be-svc-graph api, sn-be-svc-re api, sn-be-svc-worker
gold_track_graph (materialized view) `sn-be-tbl-gold-track-graph`	postgres	`medallion 006`	created at init; refreshed only by restore_and_promote.sh and remove_same_artist_edges.sql	–

gold_track_analytics / gold_playlist_analytics / gold_artist_analytics `sn-be-tbl-gold-analytics`	postgres	`medallion 003 (not applied by compose)`	data-transformer and gold_layer_processor (not deployed)	legacy gold-to-operational-etl, metadata-enrichment medallion_metric
projection_tracks `sn-be-tbl-projection-tracks`	postgres	`migration 014`	worker projection_consume / rebuild	sn-be-svc-rest-a (library_search, preview)
ingestion_jobs `sn-be-tbl-ingestion-jobs`	postgres	`migration 013`	rest-api (enqueue, retry), worker (schedule, claim, settle), start-dev.sh (transition_scoring seed)	sn-be-svc-rest-a sn-be-svc-worke import-music-library.sh via API
outbox_events / outbox_consumers / outbox_dead_letters `sn-be-tbl-outbox`	postgres	`migration 014, 220`	every catalog write in the same transaction	projection_tracks consumer, graph_projection consumer, GET /api/v1/outbox/st
library schema (tracks, track_xref, appearances, chart_positions, rotation_history, pending, rejected, beatport_check, sources, runs, sets, set_tracks, v_crates) `sn-be-tbl-library`	postgres	`migrations 101, 180-185`	worker library_ingest; rest-api sets routes	sn-be-svc-graph api, sn-be-svc-re api, sn-be-svc-worker
target_tracks `sn-be-tbl-target-tracks`	postgres	`migration 012, 280`	rest-api target_tracks router; library ingest auto-promotion; scrape_targets	sn-be-svc-worke sn-be-svc-rest-a
web_call_log `sn-be-tbl-web-call-log`	postgres	`migration 170`	common/providers/web_call_log.py	–
user_oauth_tokens + api_keys `sn-be-tbl-oauth`	postgres	`sql/init/07-api-keys.sql`	rest-api music_auth and api_keys routers	sn-be-svc-rest-a sn-be-svc-worke (saved keys)
song_adjacency `sn-be-tbl-song-adjacency`	postgres	`only in sql/init/05 and scrapers/music_schema.sql (not applied)`	scraper persistence pipeline	–
musicdb.* legacy schema (tracks, user_setlists, tracks_audio_analysis, target_track_searches) `sn-be-tbl-musicdb-legacy`	postgres	`not created by compose`	graph-api user-setlists, audio-analysis queue path, root SQL scripts	graph-api suggestions/test routes
Redis 8.10 `sn-be-store-redis`	redis	`compose service redis (anonymous volume /data)`	rest-api (OAuth state), graph-api (batch status), unified-scraper spiders (quota, dedupe)	same writers
RabbitMQ 4.3 `sn-be-store-rabbitmq`	rabbitmq	`compose service rabbitmq, volume dev-rabbitmq-4, guest/guest`	Celery beat in worker	Celery worker
Memgraph `sn-be-store-memgraph`	external-api	`bolt://memgraph:7687 (no service in compose)`	graph_projection consumer when enabled	–
ref_mb MusicBrainz mirror `sn-be-store-refmb`	postgres	`host.docker.internal:55433 db ref_mb (optional, MB_LOCAL_ENABLED=false)`	external (musicLibrary fetch_metabrainz)	sn-be-svc-worke sn-be-svc-rest-a
unified scraper in-memory progress `sn-be-store-scraper-progress`	memory	`OrderedDict in unified_scraper_api process`	unified-scraper	rest-api job stage
Cloudflare tunnel song.aiwfe.com `sn-be-q-store-cloudflare-tunnel`	external-api	`cloudflared container (queued)`	operator	public internet

useStore (Zustand, persist songnodes-store, devtools "SongNodes Store") `sn-fe-store-zustand`	localStorage	`browser localStorage songnodes-store`	frontend store slices (graph, credentials, pathfinding, search)	sn-fe-graph, sn- settings, sn-fe-s
useUnifiedTracksStore (in-memory zustand) `sn-fe-store-unified-tracks`	memory	`browser memory`	hooks/useUnifiedTracks.ts refresh()	sn-fe-paths, sn- harmony, sn-fe- sources-import, fe-library
loadLibrary() promise cache `sn-fe-store-library-cache`	memory	`module scope features/musicLibrary/data.ts`	loadLibrary	sn-fe-home, sn- library, sn-fe-set sn-fe-store-unifi tracks
per-device localStorage keys (sn-*, songnodes_*) `sn-fe-store-browser-ls`	localStorage	`browser origin http://127.0.0.1:3006`	see signals sn-fe-ls-*	all routes
frontend nginx (inline default.conf) `sn-fe-store-nginx`	proxy	`frontend image :8080, published 127.0.0.1:3006`	frontend/Dockerfile	browser
PBS airnet API `ext-pbs-airnet`	external-api	`internet`	third party	–
3RRR episode pages `ext-3rrr`	external-api	`internet`	third party	–
ABC plays API (triple j, Double J) `ext-abc-plays`	external-api	`internet`	third party	–
ARIA Club Chart `ext-aria`	external-api	`internet`	third party	–
Balance Selections WordPress API `ext-balance-wp`	external-api	`internet`	third party	–
TIDAL developer API `ext-tidal`	external-api	`internet`	third party	–
Deezer API `ext-deezer`	external-api	`internet`	third party	–
MetaBrainz dumps `ext-metabrainz`	external-api	`internet`	third party	–
Rekordbox / Lexicon (XML) `ext-rekordbox`	external-api	`internet`	third party	–
Jules sessions (Google) `ext-jules`	external-api	`internet`	third party	–

Table: **Signals** `404 signals`

404 rows

PRIMITIVE ↕	KEY OR ENDPOINT ↕	DIRECTION ↕	FROM ↕	TO ↕
file-read	`config/sources.json`	read	`ml-store-config`	`ml-c`
file-read	`config/rules.json`	read	`ml-store-config`	`ml-c`
fetch	`https://airnet.org.au/rest/stations/3pbs/programs/<slug>/episodes and <episodeRestUrl>/playlists`	read	`ext-pbs-airnet`	`ml-c`
fetch	`https://www.rrr.org.au/explore/programs/<slug>[/episodes/<id>-<slug>]`	read	`ext-3rrr`	`ml-c`
fetch	`https://music.abcradio.net.au/api/v1/plays/search.json?station&from&to&limit=100`	read	`ext-abc-plays`	`ml-c`
fetch	`ARIA Club Chart page (sources.json url)`	read	`ext-aria`	`ml-c`
fetch	`Balance Selections WordPress REST posts (sources.json api)`	read	`ext-balance-wp`	`ml-c`

file-write	inbox/manual_adds.csv	both	ml-persona-curator	ml-c
file-write	inbox/raw_<date>.json	both	ml-cli-scrape	ml-c
file-write	outputs/<date>/scrape_log.txt	write	ml-cli-scrape	ml-a week
file-write	inbox/to_resolve_<date>.json	both	ml-cli-ingest	ml-c
file-write	inbox/resolved_<date>.json	both	ml-cli-resolve	ml-c
fetch	https://openapi.tidal.com/v2 /searchResults, /tracks, /tracks/{id}/relationships/genres (client credentials)	read	ext-tidal	ml-c
fetch	https://api.deezer.com /search, /track/{id}, /album/{id}	read	ext-deezer	ml-c
fetch	https://auth.tidal.com/v1/oauth2/token and https://login.tidal.com/authorize (PKCE, redirect http://127.0.0.1:3006/oauth/callback)	both	ml-cli-tidal	ext-
file-read	secrets/tidal_app.json	read	ml-store-secrets	ml-c
file-write	secrets/tidal_user_token.json and tidal_pkce_pending.json	both	ml-cli-tidal	ml-c tida
file-write	secrets/tidal_playlists.json	both	ml-cli-sync-tidal	ml-c sets
fetch	openapi.tidal.com/v2 POST /playlists, GET/POST/DELETE /playlists/{id}/relationships/items (user token, Idempotency-Key uuid5)	write	ml-cli-sync-tidal	ext-
file-write	ledger/dj_library_ledger.json	both	ml-cli-process	ml-c db
file-write	ledger/history/ledger_before_<date>.json	write	ml-cli-process	ml-p deve
file-read	overrides/energy_overrides.csv	read	ml-persona-curator	ml-c
file-write	outputs/<date>/sync_plan.json	write	ml-cli-process	ml-a beat
file-write	outputs/crates/<crate>.csv	write	ml-cli-process	ml-p cura
file-write	outputs/<date>/buy_list.csv	write	ml-cli-process	ml-a beat
file-write	outputs/<date>/review_queue.csv	write	ml-cli-process	ml-p cura
file-write	outputs/<date>/rejected.csv	write	ml-cli-process	ml-p cura
file-write	outputs/<date>/weekly_report.md	both	ml-cli-process	ml-c db
file-write	outputs/rekordbox_musicLibrary.xml	write	ml-cli-process	ext-
file-read	overrides/rekordbox_collection.xml	read	ext-rekordbox	ml-c cali
file-write	outputs/calibration_suggestions.json	write	ml-cli-calibrate	ml-p cura
file-write	inbox/sync_result_<date>.json	both	ml-cli-sync-tidal	ml-c
file-write	outputs/<date>/ALERT_tidal.txt, ALERT_tidal_sets.txt	write	ml-cli-sync-tidal	ml-a week
file-write	inbox/artifact_sets.json	both	ml-agent-weekly-run	ml-c db
sql	db/dj_library.sqlite	write	ml-cli-build-db	ml-p cura
file-write	outputs/artifact/library.json	write	ml-cli-build-db	ml-p buil

file-write	artifact/build/library.json	both	ml-agent-weekly-run	ml-p buil
fetch	library.json (relative, cache no-cache)	read	ml-store-artifact-files	ml-p buil
postMessage	window.claude.use('db') collection 'sets' (get, doc.set, doc.delete; legacy page also onSnapshot and use('user').can('data.write'))	both	ml-page-set-builder	ml-s arti
cli	ArtifactData list collection 'sets' limit 1000 (weekly agent)	read	ml-store-artifact-db	ml-a week
cli	Artifact publish url=GcqZTGjk6obdpjWaU7WFRq file_path=weekly_publish/index.html files{library.json}	write	ml-agent-weekly-run	ml-s arti
localStorage	sn-music-library-sets	both	ml-page-set-builder	ml-p buil
localStorage	sn-set-draft	both	ml-page-set-builder	ml-p buil
localStorage	sn-filters-library	both	ml-page-set-builder	ml-p buil
localStorage	sn-density	both	ml-page-set-builder	ml-p buil
localStorage	sn-banner-keyed—<page>	both	ml-page-set-builder	ml-p buil
localStorage	nsb:view, nsb:filters, nsb:draft	both	ml-page-set-builder-legacy	ml-p buil
clipboard	navigator.clipboard.writeText tracklist	write	ml-page-set-builder-legacy	ext-
http-endpoint	POST http://127.0.0.1:8082/api/v1/library/ingest (Idempotency-Key import —<md5>)	write	ml-cli-sync-songnodes	sn-b rest
http-endpoint	POST /api/v1/tracks/import-tracklist with songnodes.setlists.v1 documents	write	ml-cli-sync-songnodes	sn-b rest
file-read	JohnCourse.json, kiki.json (root), JohnCourse.json.bak	read	ml-persona-developer	ml-c song
cli	songnodes scripts/import-music-library.sh (MUSIC_LIBRARY_DIR default ../musicLibrary)	read	sn-be-cli-import-music-library	ml-s outp
http-endpoint	GET /api/v1/library/search?limit=5000 (songnodes /library, /sets)	read	sn-be-svc-rest-api	sn-f
http-endpoint	GET/PUT/DELETE /api/v1/library/sets[/id], GET /api/v1/library/sets/export	both	sn-fe-library	sn-s libr
file-write	artifact_sets.json download from songnodes /sets (Export sets for the weekly run)	write	sn-fe-library	ml-s arti expo
http-endpoint	POST /api/v1/tidal/sets/send (songnodes Send to TIDAL dialog)	write	sn-fe-library	ext-
sql	postgres://ref_mb_reader@host.docker.internal:55433/ref_mb (MB_LOCAL_*)	read	ml-svc-ref-mb	sn-b refm
fetch	https://data.metabrainz.org/pub/musicbrainz (dumps 20261024-102842, 20261024-001001, 20261017-080003)	read	ext-metabrainz	ml-c meta
fetch	https://www.beatport.com/search/tracks?q=… via Claude in Chrome (supervised)	both	ml-agent-beatport-sync	ext-
fetch	POST https://jules.googleapis.com/v1alpha/sessions (X-Goog-Api-Key from JULES_API_KEY env)	write	ml-plan-jules	ext-
env	SONGNODES_API_URL (run_weekly) vs SONGNODES_API (import script)	read	ml-store-env	ml-c song

http-endpoint	GET /health	both	client	sn-b dlq-r
http-endpoint	GET /metrics	both	client	sn-b dlq-r
http-endpoint	GET /dlq/messages	both	client	sn-b dlq-r
http-endpoint	GET /dlq/stats	both	client	sn-b dlq-r
http-endpoint	POST /dlq/replay/{message_id}	both	client	sn-b dlq-r
http-endpoint	POST /dlq/replay/batch	both	client	sn-b dlq-r
http-endpoint	DELETE /dlq/message/{message_id}	both	client	sn-b dlq-r
http-endpoint	GET /dlq/errors/grouped	both	client	sn-b dlq-r
http-endpoint	GET /health	both	client	sn-b nlp-p
http-endpoint	GET /metrics	both	client	sn-b nlp-p
http-endpoint	POST /analyze	both	client	sn-b nlp-p
http-endpoint	POST /extract-artists	both	client	sn-b nlp-p
http-endpoint	POST /extract_tracklist	both	client	sn-b nlp-p
http-endpoint	POST /analyze_text	both	client	sn-b nlp-p
http-endpoint	POST /extract_timestamps	both	client	sn-b nlp-p
http-endpoint	POST /llm/extract_tracklist	both	client	sn-b nlp-p
http-endpoint	POST /llm/classify_genre	both	client	sn-b nlp-p
http-endpoint	POST /llm/generate	both	client	sn-b nlp-p
http-endpoint	GET /health	both	client	sn-b data tran
http-endpoint	POST /transform	both	client	sn-b data tran
http-endpoint	GET /tasks/{task_id}	both	client	sn-b data tran
http-endpoint	GET /tasks	both	client	sn-b data tran
http-endpoint	GET /stats	both	client	sn-b data tran
http-endpoint	GET /metrics	both	client	sn-b data tran
http-endpoint	POST /normalize	both	client	sn-b data tran
http-endpoint	GET /fingerprints/duplicates	both	client	sn-b data tran

http-endpoint	GET /api/graph/nodes	both	client	sn-b graph
http-endpoint	GET /api/graph/edges	both	client	sn-b graph
http-endpoint	GET /api/graph/data	both	client	sn-b graph
http-endpoint	GET /api/graph/stats	both	client	sn-b graph
http-endpoint	GET /api/graph/search	both	client	sn-b graph
http-endpoint	GET /api/v1/visualization/search	both	client	sn-b graph
http-endpoint	GET /test/adjacency	both	client	sn-b graph
http-endpoint	GET /test/nodes	both	client	sn-b graph
http-endpoint	GET /health	both	client	sn-b graph
http-endpoint	GET /metrics	both	client	sn-b graph
http-endpoint	POST /api/user-setlists	both	client	sn-b graph
http-endpoint	POST /api/v1/visualization/graph	both	client	sn-b graph
http-endpoint	POST /api/v1/visualization/nodes/batch	both	client	sn-b graph
http-endpoint	GET /api/v1/visualization/batch/{batch_id}/status	both	client	sn-b graph
websocket	WEBSOCKET /api/graph/ws/{room_id}	both	client	sn-b graph
http-endpoint	GET /api/graph/neighborhood/{node_id}	both	client	sn-b graph
http-endpoint	GET /api/graph	both	client	sn-b graph
http-endpoint	GET /api/v1/graph	both	client	sn-b graph
http-endpoint	GET /health	both	client	sn-b rest
http-endpoint	PATCH /api/tracks/{track_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/tracks/{track_id}/preview	both	client	sn-b rest
http-endpoint	GET /api/graph/nodes	both	client	sn-b rest
http-endpoint	GET /api/graph/edges	both	client	sn-b rest
http-endpoint	GET /api/v1/music-search/{service}	both	client	sn-b rest
http-endpoint	GET /api/v1/library/search	both	client	sn-b rest
http-endpoint	GET /api/v1/library/tracks/{track_id}/preview	both	client	sn-b rest
http-endpoint	GET /api/v1/tracks	both	client	sn-b rest
http-endpoint	GET /api/search/tracks	both	client	sn-b rest
http-endpoint	GET /api/tracks/{track_id}	both	client	sn-b rest

http-endpoint	GET /api/v1/tracks/{track_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/artists	both	client	sn-b rest
http-endpoint	GET /api/v1/artists/search	both	client	sn-b rest
http-endpoint	GET /api/v1/artists/{artist_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/setlists	both	client	sn-b rest
http-endpoint	GET /api/v1/playlists	both	client	sn-b rest
http-endpoint	GET /api/v1/setlists/{playlist_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/playlists/{playlist_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/setlists/{playlist_id}/tracks	both	client	sn-b rest
http-endpoint	POST /api/v1/scrape/trigger	both	client	sn-b rest
http-endpoint	GET /api/v1/jobs	both	client	sn-b rest
http-endpoint	GET /api/v1/jobs/{job_id}	both	client	sn-b rest
http-endpoint	POST /api/v1/jobs/{job_id}/retry	both	client	sn-b rest
http-endpoint	GET /api/v1/jobs/{job_id}/stages	both	client	sn-b rest
http-endpoint	POST /api/v1/spotify/playlists/create	both	client	sn-b rest
http-endpoint	GET /api/v1/spotify/playlists/list	both	client	sn-b rest
http-endpoint	POST /api/v1/spotify/playlists/{playlist_id}/tracks	both	client	sn-b rest
http-endpoint	POST /api/v1/spotify/tracks/search	both	client	sn-b rest
http-endpoint	POST /api/v1/spotify/tracks/check-availability	both	client	sn-b rest
http-endpoint	GET /api/v1/artists/dirty	both	client	sn-b rest
http-endpoint	POST /api/v1/artists/{artist_id}/rename	both	client	sn-b rest
http-endpoint	POST /api/v1/artists/{duplicate_id}/merge	both	client	sn-b rest
http-endpoint	DELETE /api/v1/artists/{artist_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/tracks/missing-artist	both	client	sn-b rest
http-endpoint	POST /api/v1/tracks/{track_id}/assign-artist	both	client	sn-b rest
http-endpoint	POST /api/v1/artists	both	client	sn-b rest
http-endpoint	POST /api/v1/tracks	both	client	sn-b rest
http-endpoint	POST /api/v1/tracks/import-tracklist	both	client	sn-b rest
http-endpoint	GET /api/v1/api-keys/requirements	both	client	sn-b rest

http-endpoint	GET /api/v1/api-keys	both	client	sn-b rest
http-endpoint	POST /api/v1/api-keys	both	client	sn-b rest
http-endpoint	DELETE /api/v1/api-keys/{service_name}/{key_name}	both	client	sn-b rest
http-endpoint	POST /api/v1/api-keys/test	both	client	sn-b rest
http-endpoint	POST /api/v1/library/ingest	both	client	sn-b rest
http-endpoint	GET /api/v1/library/sets	both	client	sn-b rest
http-endpoint	PUT /api/v1/library/sets/{set_id}	both	client	sn-b rest
http-endpoint	DELETE /api/v1/library/sets/{set_id}	both	client	sn-b rest
http-endpoint	GET /api/v1/library/sets/export	both	client	sn-b rest
http-endpoint	POST /api/v1/pathfinder/find-path	both	client	sn-b rest
http-endpoint	POST /api/v1/jobs	both	client	sn-b rest
http-endpoint	GET /api/v1/metadata/review	both	client	sn-b rest
http-endpoint	GET /api/v1/tracks/{track_id}/canonical	both	client	sn-b rest
http-endpoint	POST /api/v1/tracks/{track_id}/merge	both	client	sn-b rest
http-endpoint	GET /api/v1/merges	both	client	sn-b rest
http-endpoint	POST /api/v1/merges/{history_id}/revert	both	client	sn-b rest
http-endpoint	GET /api/v1/outbox/status	both	client	sn-b rest
http-endpoint	POST /api/v1/projections/tracks/reconcile	both	client	sn-b rest
http-endpoint	GET /api/v1/attributions	both	client	sn-b rest
http-endpoint	GET /api/v1/resolution/review	both	client	sn-b rest
http-endpoint	POST /api/v1/resolution/{decision_id}/review	both	client	sn-b rest
http-endpoint	GET /api/v1/scrape/window	both	client	sn-b rest
http-endpoint	PUT /api/v1/scrape/window	both	client	sn-b rest
http-endpoint	GET /api/v1/target-tracks/suggestions	both	client	sn-b rest
http-endpoint	GET /api/v1/target-tracks	both	client	sn-b rest
http-endpoint	POST /api/v1/target-tracks	both	client	sn-b rest
http-endpoint	PUT /api/v1/target-tracks/{track_id}	both	client	sn-b rest
http-endpoint	DELETE /api/v1/target-tracks/{track_id}	both	client	sn-b rest
http-endpoint	POST /api/v1/target-tracks/{track_id}/search	both	client	sn-b rest

http-endpoint	POST /api/v1/music-auth/test/tidal	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/test/spotify	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/spotify/authorize	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/spotify/callback	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/store-token	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/spotify/token	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/spotify/refresh	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/spotify/search	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/tidal/device/init	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/tidal/device/poll	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/tidal/oauth/init	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/tidal/oauth/callback	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/reddit/callback	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/lastfm/callback	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/tidal/oauth/refresh	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/tidal/refresh	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/health	both	client	sn-b rest
http-endpoint	GET /api/v1/music-auth/spotify/token/status	both	client	sn-b rest
http-endpoint	POST /api/v1/music-auth/spotify/token/refresh-manual	both	client	sn-b rest
http-endpoint	GET /api/v1/youtube/quota	both	client	sn-b rest
http-endpoint	GET /api/v1/youtube/search	both	client	sn-b rest
http-endpoint	POST /api/v1/youtube/quota/reset	both	client	sn-b rest
http-endpoint	POST /api/v1/ai/complete	both	client	sn-b rest
http-endpoint	POST /api/v1/tidal/playlists/create	both	client	sn-b rest
http-endpoint	GET /api/v1/tidal/playlists/list	both	client	sn-b rest
http-endpoint	POST /api/v1/tidal/playlists/{playlist_id}/tracks	both	client	sn-b rest
http-endpoint	POST /api/v1/tidal/tracks/search	both	client	sn-b rest
http-endpoint	POST /api/v1/tidal/tracks/check-availability	both	client	sn-b rest
http-endpoint	POST /api/v1/tidal/sets/send	both	client	sn-b rest

http-endpoint	GET /health	both	client	sn-b meta enri
http-endpoint	GET /metrics	both	client	sn-b meta enri
http-endpoint	POST /enrich	both	client	sn-b meta enri
http-endpoint	POST /enrich/batch	both	client	sn-b meta enri
http-endpoint	POST /enrich/trigger	both	client	sn-b meta enri
http-endpoint	POST /enrich/reset-circuit-breaker/{service}	both	client	sn-b meta enri
http-endpoint	GET /stats	both	client	sn-b meta enri
http-endpoint	GET /admin/config	both	client	sn-b meta enri
http-endpoint	POST /admin/config/reload	both	client	sn-b meta enri
http-endpoint	GET /admin/config/field/{field_name}	both	client	sn-b meta enri
http-endpoint	PUT /admin/config/field/{field_name}	both	client	sn-b meta enri
http-endpoint	GET /admin/config/providers	both	client	sn-b meta enri
http-endpoint	POST /enrich/multi-tier-resolve	both	client	sn-b meta enri
http-endpoint	POST /enrich/hunt-labels	both	client	sn-b meta enri
http-endpoint	POST /enrich/cooldown/migrate	both	client	sn-b meta enri
http-endpoint	POST /enrich/cooldown/process	both	client	sn-b meta enri
http-endpoint	GET /enrich/cooldown/stats	both	client	sn-b meta enri
http-endpoint	POST /enrich/probabilistic-match	both	client	sn-b meta enri
http-endpoint	GET /api/v1/search/{service}	both	client	sn-b meta enri
http-endpoint	GET /health	both	client	sn-b api- inte

http-endpoint	POST /api/{provider}/search	both	client	sn-b api- inte
http-endpoint	GET /api/{provider}/track/{track_id}	both	client	sn-b api- inte
http-endpoint	GET /admin/circuit-breakers	both	client	sn-b api- inte
http-endpoint	POST /admin/circuit-breakers/{provider}/reset	both	client	sn-b api- inte
http-endpoint	GET /admin/cache/stats	both	client	sn-b api- inte
http-endpoint	GET /health	both	client	sn-b audi
http-endpoint	GET /metrics	both	client	sn-b audi
http-endpoint	POST /analyze-url	both	client	sn-b audi
http-endpoint	POST /analyze	both	client	sn-b audi
http-endpoint	GET /health	both	client	sn-b webs
http-endpoint	GET /stats	both	client	sn-b webs
http-endpoint	GET /metrics	both	client	sn-b webs
websocket	WEBSOCKET /ws/public	both	client	sn-b webs
websocket	WEBSOCKET /ws/{user_id}	both	client	sn-b webs
http-endpoint	POST /api/v1/graph/broadcast	both	client	sn-b webs
http-endpoint	POST /api/v1/graph/nodes/added	both	client	sn-b webs
http-endpoint	POST /api/v1/graph/nodes/updated	both	client	sn-b webs
http-endpoint	POST /api/v1/graph/nodes/removed	both	client	sn-b webs
http-endpoint	POST /api/v1/broadcast/{room_id}	both	client	sn-b webs
http-endpoint	GET /api/v1/rooms/{room_id}/history	both	client	sn-b webs
http-endpoint	GET /health	both	client	sn-b data
http-endpoint	POST /validate	both	client	sn-b data
http-endpoint	POST /validate/sync	both	client	sn-b data
http-endpoint	GET /tasks/{task_id}	both	client	sn-b data
http-endpoint	GET /tasks	both	client	sn-b data
http-endpoint	GET /rules	both	client	sn-b data
http-endpoint	GET /stats	both	client	sn-b data

http-endpoint	GET /metrics	both	client	sn-b data
http-endpoint	GET /health	both	client	sn-b brow coll
http-endpoint	POST /collect	both	client	sn-b brow coll
http-endpoint	POST /extract/{raw_data_id}	both	client	sn-b brow coll
http-endpoint	GET /metrics	both	client	sn-b brow coll
http-endpoint	GET /health	both	client	sn-b db-c pool
http-endpoint	GET /stats	both	client	sn-b db-c pool
http-endpoint	GET /metrics	both	client	sn-b db-c pool
http-endpoint	POST /reload	both	client	sn-b db-c pool
http-endpoint	POST /query	both	client	sn-b db-c pool
http-endpoint	GET /pool-info	both	client	sn-b db-c pool
http-endpoint	GET /health	both	client	sn-b scra real
http-endpoint	GET /metrics	both	client	sn-b scra real
http-endpoint	POST /scrape	both	client	sn-b scra real
http-endpoint	GET /	both	client	sn-b scra serv
http-endpoint	GET /health	both	client	sn-b scra serv
http-endpoint	GET /spiders	both	client	sn-b scra serv
http-endpoint	POST /run/{spider_name}	both	client	sn-b scra serv
http-endpoint	GET /status/{spider_name}	both	client	sn-b scra serv
http-endpoint	POST /migrate	both	client	sn-b scra serv
http-endpoint	GET /config	both	client	sn-b scra serv

http-endpoint	GET /health	both	client	sn-b unif scra
http-endpoint	GET /metrics	both	client	sn-b unif scra
http-endpoint	GET /scrape/progress/{job_id}	both	client	sn-b unif scra
http-endpoint	GET /sources	both	client	sn-b unif scra
http-endpoint	POST /scrape	both	client	sn-b unif scra
amqp	celery task songnodes.drain (beat every JOB_POLL_SECONDS=5s, expires 15s)	write	sn-be-svc-worker	sn-b rabb
amqp	celery task songnodes.schedule (beat every SCHEDULE_SECONDS=30s)	write	sn-be-svc-worker	sn-b rabb
amqp	audio_analysis_queue (vhost musicdb)	read	sn-be-store-rabbitmq	sn-b audi
amqp	metadata-enrichment / dlq-manager / websocket-api queues	both	sn-be-legacy-metadata-enrichment	sn-b dlq-r
redis	oauth:spotify:{state} / oauth:tidal:{state} / oauth:tidal:device:{device_code} (setex, deleted on callback)	both	sn-be-svc-rest-api	sn-b redi
redis	batch:{batch_id} (setex 3600)	both	sn-be-svc-graph-api	sn-b redi
redis	cache_result keys (CACHE_TTL 300s)	both	sn-be-svc-graph-api	sn-b redi
redis	scraped:setlists:{mixesdb|1001tracklists}:last_run (TTL SCRAPER_RUN_TTL_HOURS=24h)	both	sn-be-svc-unified-scraper	sn-b redi
redis	scraped:setlists:{source}:{sha digest} (TTL SCRAPER_SOURCE_TTL_DAYS=30d)	both	sn-be-svc-unified-scraper	sn-b redi
redis	spotify spider token/expiry keys	both	sn-be-svc-unified-scraper	sn-b redi
websocket	/api/graph/ws/{room_id} (graph_update, cursor_position, ping)	both	client	sn-b grap
http-endpoint	POST {UNIFIED_SCRAPER_URL}/scrape	write	sn-be-svc-worker	sn-b unif scra
http-endpoint	GET {UNIFIED_SCRAPER_URL}/scrape/progress/{job_id}	read	sn-be-svc-rest-api	sn-b unif scra
http-endpoint	POST {AUDIO_ANALYSIS_URL}/analyze-url	write	sn-be-svc-worker	sn-b audi
http-endpoint	GET {METADATA_ENRICHMENT_URL=http://metadata-enrichment:8086}/search...	read	sn-be-svc-rest-api	sn-b meta enri
cli	scripts/import-music-library.sh → POST /api/v1/library/ingest (Idempotency-Key import—<md5>)	write	ml-cli-db	sn-b rest
file-read	$MUSIC_LIBRARY_DIR/outputs/artifact/library.json	read	ml-cli-db	sn-b impo libr
cli	python -m library.importer --sqlite <library.db> --api-url (default http://localhost:8080/...)	write	ml-library-db	sn-b rest

file-write	library.reconcile.cli --out <csv>	write	sn-be-store-postgres	oper
file-write	pipeline/build_db.py → db/library.db and outputs/artifact/library.json	write	sn-be-cli-pipeline	sn-b read json
file-write	backup-YYYYMMDD.dump via compose exec pg_dump -U songnodes	write	sn-be-store-postgres	oper
file-write	scripts/backup_database.sh → /backups/postgres/musicdb_backup_*.sql.gz; backup_volumes.sh → ./backups/volumes/*.tar.gz	write	sn-be-store-postgres	oper
file-write	/tmp/worker-heartbeat	both	sn-be-svc-worker	dock heal
env	TIDAL_CLIENT_ID/SECRET, SPOTIFY_CLIENT_ID/SECRET, LASTFM_API_KEY, DISCOGS_TOKEN, GETSONGBPM_API_KEY, MB_LOCAL_*	read	host-env/.env	sn-b work
env	ENRICHMENT_PROVIDERS (default none)	read	host-env/.env	sn-b work
sql	api_keys table via get_api_key(service,key,secret) → overrides env	read	sn-be-store-postgres	sn-b work
env	SONGNODES_API_TOKEN	read	host-env/.env	sn-b rest
sql	INSERT bronze_scraped_playlists / bronze_scraped_tracks / bronze_scraped_artists	write	sn-be-svc-unified-scraper	sn-b bron
sql	INSERT silver_enriched_tracks / silver_enriched_playlists / silver_playlist_tracks directly from the scraper	write	sn-be-svc-unified-scraper	sn-b silv
sql	INSERT song_adjacency (scraper)	write	sn-be-svc-unified-scraper	sn-b song
sql	INSERT bronze_* from POST /api/v1/tracks/import-tracklist, then ingest_playlist synchronously	write	sn-be-svc-rest-api	sn-b bron
sql	ingest_pending: bronze playlists without silver marker datalayer-ingest-v1 → silver_enriched_* + silver_track_transitions (N→N+1) + outbox	both	sn-be-svc-worker	sn-b silv
sql	projection_consume: outbox_events (track/artist) → projection_tracks	both	sn-be-svc-worker	sn-b proj trac
sql	graph-api reads silver_enriched_tracks + silver_track_transitions + library.* (REPEATABLE READ snapshot)	read	sn-be-tbl-silver	sn-b grap
sql	library_search / preview read projection_tracks + library.* + evidence	read	sn-be-tbl-projection-tracks	sn-b rest
sql	library_ingest job → library.tracks/track_xref/appearances/rotation_history/runs/sets/set_tracks + target_tracks auto-promotion + outbox library.track.*	write	sn-be-svc-worker	sn-b libr
sql	library.sets / library.set_tracks via PUT/DELETE /api/v1/library/sets/{id}; save appends library.set.saved.v1	both	sn-be-svc-rest-api	sn-b libr
sql	target_tracks CRUD (REST), scheduler stamps last_searched (SKIP LOCKED), scrape result writes playlists_found	both	sn-be-svc-rest-api	sn-b targ
sql	scrape_window single row (PUT /api/v1/scrape/window) read by scrape_targets	both	sn-be-svc-rest-api	sn-b work
sql	ingestion_jobs enqueue/claim/complete/fail/retry/prune	both	sn-be-svc-rest-api	sn-b work
sql	outbox_events append / outbox_consumers offsets / outbox_dead_letters	both	sn-be-svc-rest-api	sn-b work
sql	enrich_track / enrichment_backfill / audio_analysis → silver_track_metadata_evidence + silver_enriched_tracks consensus + backfill_checkpoints	write	sn-be-svc-worker	sn-b silv

sql	er_shadow → er_decisions; merges → entity_merge_history / entity_redirects	write	sn-be-svc-worker	sn-b silv
sql	INSERT web_call_log (provider calls)	write	sn-be-svc-worker	sn-b call
sql	user_oauth_tokens upsert/update (plain text tokens, user_id default_user)	write	sn-be-svc-rest-api	sn-b oaut
sql	POST /api/user-setlists → CREATE TABLE IF NOT EXISTS musicdb.user_setlists (+tracks)	write	sn-be-svc-graph-api	sn-b musi
sql	graph_projection_consume → Memgraph bolt://memgraph:7687 (only if MEMGRAPH_ENABLED/URI)	write	sn-be-svc-worker	sn-b memg
sql	MusicBrainz local mirror ref_mb (MB_LOCAL_* on host.docker.internal:55433, statement timeout 2000ms)	read	sn-be-store-refmb	sn-b work
fetch	https://openapi.tidal.com, auth.tidal.com	both	sn-be-svc-worker	sn-b stor
fetch	api.deezer.com + cdnt-preview.dzcdn.net previews	read	sn-be-svc-worker	sn-b stor
fetch	api.discogs.com	read	sn-be-svc-worker	sn-b stor
fetch	ws.audioscrobbler.com	read	sn-be-svc-worker	sn-b stor
fetch	musicbrainz.org (MUSICBRAINZ_CONTACT UA)	read	sn-be-svc-worker	sn-b stor musi
fetch	getsongbpm.com (GETSONGBPM_ENABLED false)	read	sn-be-svc-worker	sn-b stor gets
fetch	api.spotify.com / accounts.spotify.com (rest-api OAuth and playlists)	both	sn-be-svc-rest-api	sn-b stor
fetch	TIDAL user playlists (Bearer from client)	write	sn-be-svc-rest-api	sn-b stor user
fetch	www.googleapis.com YouTube Data API (quota counter)	read	sn-be-svc-rest-api	sn-b stor
fetch	api.anthropic.com / api.openai.com / generativelanguage.googleapis.com	both	sn-be-svc-rest-api	sn-b stor
fetch	www.mixesdb.com (Scrapy, robots crawl-delay)	read	sn-be-svc-unified-scraper	sn-b stor
fetch	www.1001tracklists.com (Scrapy + Playwright)	read	sn-be-svc-unified-scraper	sn-b stor 1001
fetch	Spotify/MusicBrainz/Last.fm/Discogs/Reddit from scraper item pipelines	read	sn-be-svc-unified-scraper	sn-b stor enri
fetch	Ollama CAPTCHA/HTML analysis (CLAUDE.md 5.1)	read	sn-be-svc-unified-scraper	sn-b stor
http-endpoint	POST /api/v1/auth/login	both	client	sn-b auth
http-endpoint	GET /api/v1/auth/duo/callback	both	client	sn-b auth
http-endpoint	GET /api/v1/auth/me	both	client	sn-b auth
http-endpoint	POST /api/v1/auth/logout	both	client	sn-b auth
redis	duo:auth_state:{state} (setex 600s; in-memory fallback dict)	both	sn-be-q-user-auth-router	sn-b redi

cookie	songnodes_session (HS256 JWT, 7 days)	write	sn-be-q-user-auth-router	brow
localStorage	sn-theme, sn-text-size, sn-density, sn-motion	both	sn-fe-settings	sn-f html
localStorage	songnodes-store	both	sn-fe-store-zustand	sn-f zust
localStorage	sn-music-library-sets	both	sn-fe-sets	sn-f
localStorage	sn-set-draft	both	sn-fe-library	sn-f
localStorage	sn-filters-library	both	sn-fe-library	sn-f
localStorage	sn-banner-keyed-library, sn-banner-keyed-sets	both	sn-fe-library	sn-f
localStorage	sn-favs-library-crate, sn-favs-pathfinder-waypoints	both	sn-fe-library	sn-f
localStorage	sn-recent-quick-search, sn-recent-music-service (and per-namespace AutocompleteSearchField)	both	sn-fe-graph	sn-f
localStorage	sn-graph-view	both	sn-fe-graph	sn-f
localStorage	sn-target-tracks-v1	both	sn-fe-library	sn-f
localStorage	sn-dismissed-suggestions	both	sn-fe-sources-targets	sn-f targ
localStorage	songnodes_ai_setlists_ledger	both	sn-fe-sources-import	sn-f impo
localStorage	songnodes_ai_key_config, songnodes_ai_keys_server_v1	both	sn-fe-settings	sn-f impo
localStorage	songnodes_ai_prompt_history	both	sn-fe-graph	sn-f
localStorage	tidal_oauth_tokens, spotify_oauth_tokens	read	sn-fe-oauth-callback	sn-f
localStorage	mutex_lock_token_refresh_<service>	both	sn-fe-graph	sn-f
localStorage	sn-disclosure:<name>, sn-skip-moved-targets	both	sn-fe-ds-component-previews	sn-f comp prev
sessionStorage	spotify_auth_state (also checks <service>_auth_state, tidal_auth_state)	both	sn-fe-settings	sn-f call
sessionStorage	<service>_oauth_tokens	write	sn-fe-oauth-callback	–
postMessage	oauth-success | oauth-error	write	sn-fe-oauth-callback	sn-f
storage-event	sn-favs-library-crate	read	sn-fe-library	sn-f
url-param	/library?q,rot,pillar,tier,tags,src,checks,rev,sort,focus=search	both	sn-fe-sets	sn-f
url-param	/sets?open=<setId>&pool=crate&<crate filters>	read	sn-fe-library	sn-f
url-param	/graph?mode=play|plan&tab=analysis|keymood|pathfinder|tidal|spotify|targets	read	sn-fe-paths	sn-f
url-param	/settings/:settingId and /settings#display	read	sn-fe-settings	sn-f
url-param	?code&state (callback query)	read	–	sn-f call
file-read	library export .json (musicLibrary outputs/artifact/library.json)	read	–	sn-f

file-read	tracklist .txt/.csv	read	–	sn-f
file-write	<set>.m3u8 | .csv | .json | .xml (Rekordbox)	write	sn-fe-sets	–
file-write	artifact_sets.json	write	sn-fe-sets	ml-s
file-write	pathfinder-result.json | pathfinder-playlist.m3u | pathfinder-result.csv	write	sn-fe-graph	–
clipboard	navigator.clipboard.writeText	write	sn-fe-library	–
env	VITE_API_BASE_URL, VITE_ALLOWED_ORIGINS, VITE_OAUTH_REDIRECT_ORIGIN	read	–	sn-f
fetch	GET /api/v1/library/search?limit=5000	read	sn-fe-library	sn-b rest
fetch	GET/PUT/DELETE /api/v1/library/sets[/{id}] (PUT /0 creates)	both	sn-fe-sets	sn-b rest
fetch	POST /api/v1/library/ingest (Idempotency-Key) + GET /api/v1/jobs/{id} + POST /api/v1/jobs/{id}/retry	both	sn-fe-library	sn-b rest
fetch	GET /api/v1/library/tracks/{id}/preview	read	sn-fe-library	sn-b rest
fetch	GET/POST/PUT/DELETE /api/v1/target-tracks[/{id}], POST /{id}/search, GET /suggestions	both	sn-fe-sources-targets	sn-b rest
fetch	GET /api/v1/music-search/{service}	read	sn-fe-sources-targets	sn-b rest
fetch	GET /api/graph/data?limit=10000 (fallback GET /api/graph)	read	sn-fe-graph	sn-b grap
fetch	GET /api/graph/edges?node_ids=…&limit=2	read	sn-fe-graph	sn-b grap
fetch	POST /api/v1/pathfinder/find-path	both	sn-fe-paths	sn-b rest
fetch	POST /api/v1/scrape/trigger?source&search_query (Idempotency-Key)	both	sn-fe-pipeline	sn-b rest
fetch	GET/PUT /api/v1/scrape/window	both	sn-fe-pipeline	sn-b rest
fetch	GET /api/v1/jobs?limit=200, ?kind=scrape&limit=50, GET /api/v1/outbox/status, GET /api/v1/jobs/{id}/stages, POST /jobs/{id}/retry	both	sn-fe-pipeline	sn-b rest
fetch	GET/POST /api/v1/api-keys, GET /requirements, POST /test, DELETE /{service}/{key}	both	sn-fe-settings	sn-b rest
fetch	POST /api/v1/music-auth/tidal/oauth/init, GET spotify/authorize (navigation), GET spotify/callback | tidal/oauth/callback, POST store-token, POST {service}/refresh, POST test/{tidal|spotify}	both	sn-fe-settings	sn-b rest
fetch	/api/v1/{tidal|spotify}/playlists/list|create|{id}/tracks, /tracks/search, /tracks/check-availability	both	sn-fe-graph	sn-b rest
fetch	POST /api/v1/tidal/sets/send (Bearer browser-held TIDAL token)	both	sn-fe-sets	sn-b rest
fetch	POST /api/v1/ai/complete	both	sn-fe-sources-import	sn-b rest
fetch	GET /api/v1/setlists	read	sn-fe-sources-import	sn-b rest
fetch	POST /api/v1/tracks/import-tracklist	both	sn-fe-sources-import	sn-b rest
fetch	GET /api/v1/tracks/missing-artist, /artists/search, /artists/dirty; POST /tracks/{id}/assign-artist, /artists, /artists/{id}/merge|rename; DELETE /artists/{id}	both	sn-fe-graph	sn-b rest
fetch	GET /api/v1/attributions	read	sn-fe-home	sn-b rest
fetch	POST /api/v1/scrapers/target-tracks/search, POST /api/v1/scrapers/queue/clear	both	sn-fe-graph	sn-b rest

fetch	/api/v1/search/*, /api/v1/pathfinding/*, /api/v1/tracks/batch, /api/v1/setlists POST/DELETE/export, /api/graph/tracks, /api/graph/positions, /api/v1/tracks/{id}/preview	both	sn-fe-graph	sn-b rest
websocket	/api/ws	read	sn-fe-graph	sn-b rest
fetch	PATCH /api/tracks/{id}	write	sn-fe-graph	sn-b rest
fetch	GET /api/v1/auth/me, POST /api/v1/auth/login, POST /api/v1/auth/logout (cookie, credentials: include)	both	sn-fe-login	sn-b rest
postMessage	window CustomEvent 'sn-toast'	both	any page (showToast)	sn-f

Section 5: **Synchronisation protocols, timings and limits** 21 protocols

Protocol: **Resumable resolve**

Participants	ml-cli-resolve, ml-agent-weekly-run
How it syncs	Load inbox/resolved_<date>.json if present; todo = to_resolve items whose norm_key is not yet a key; 4 worker threads; results merged in order and the whole file re-dumped every 20 songs and at the end. The agent loops 'timeout 160 ... resolve' until the final line appears.
Conflicts	first result per norm_key wins; an empty result (both APIs failed or no match) is stored and never retried for that date
Stale data	no TIDAL app secret: Deezer only
State	● Built

Evidence 2

```
musicLibrary/pipeline/run_weekly.py:414-463
musicLibrary/agent/weekly_run.md:3
```

Protocol: **Ledger save and snapshot**

Participants	ml-cli-ingest, ml-cli-process, ml-cli-confirm
How it syncs	json.dump to <ledger>.tmp; process first copy2's the existing ledger to ledger/history/ledger_before_<date>.json; os.replace tmp over the ledger.
Conflicts	last writer wins; no lock or version check
Stale data	history snapshots for manual restore
State	● Built

Evidence 1

```
musicLibrary/pipeline/djlib/ledger.py:62-70
```

Protocol: **TIDAL crate reconcile**

Participants	ml-cli-sync-tidal, ext-tidal, ml-cli-confirm
How it syncs	Desired = every active crate's tidal ids sorted by energy then BPM; playlists in tidal_playlists.json without 'set:' prefix and no longer a crate are emptied. For each: create if missing (Idempotency-Key uuid5('create:'+name)), list items, remove extras, add missing in chunks of 50 (onDuplicates SKIP, key uuid5(add:pid:date:name:n:i)). Then writes sync_result and runs confirm.
Conflicts	ledger is the source of truth; TIDAL order is not enforced for crates
Stale data	no user token: step skipped; TidalError: ALERT_tidal.txt and partial results
State	● Built

Evidence 2

```
musicLibrary/pipeline/run_weekly.py:467-521
musicLibrary/pipeline/djlib/tidal.py:237-272
```

Protocol: **Saved set to TIDAL**

Participants	ml-store-artifact-db, ml-agent-weekly-run, ml-cli-sync-sets, ext-tidal
How it syncs	Weekly agent copies ArtifactData rows into inbox/artifact_sets.json; load_sets flattens {id,data}; sets with tidal truthy map track ids through the ledger (fallback tr.tidal_id); playlist key 'set:<id>'; if current item list != wanted ids, remove all then add all.
Conflicts	page copy wins (full replace)
Stale data	no token: skipped with message
State	● Built

Evidence 3

```
musicLibrary/pipeline/run_weekly.py:524-564
musicLibrary/pipeline/build_db.py:93-106
musicLibrary/agent/weekly_run.md:5
```

Protocol: **Artifact saved-set mirror**

Participants	ml-page-set-builder, ml-store-artifact-db
How it syncs	initSavedSets: mem = localStorage; race use('db') and collection.get against 4 s; mem = remote + local-only; upload local-only. writeSavedSets: localStorage at once; changed docs set() and missing ids delete() on a serial promise queue with errors swallowed.
Conflicts	shared copy wins by id at load; local deletions made while the store was unreachable are never replayed
Stale data	no db: localStorage only
State	● Built

Evidence 1

```
musicLibrary/artifact/app/artifact-data.ts:38-72
```

Protocol: **library.json to songnodes**

Participants	ml-cli-sync-songnodes, sn-scripts, sn-be-library-ingest
How it syncs	POST body = file bytes, Idempotency-Key 'import-'+md5(file); enqueue library_ingest job; worker maps short keys, upserts tracks, writes library.track_xref(source_id → canonical uuid), appearances, rotation_history, then queues projections and enrichment backfill. The songnodes script polls /api/v1/jobs/<id> and retries a dead job; run_weekly does not poll.
Conflicts	same file never imports twice; a changed file is a new job
Stale data	run_weekly prints a note on failure and carries on
State	● Built

Evidence 3

```
musicLibrary/pipeline/run_weekly.py:567-597
songnodes/scripts/import-music-library.sh:29-70
songnodes/services/library/ingest/handler.py:6-17
```

Protocol: **TIDAL PKCE login through the songnodes app**

Participants	ml-cli-tidal, ext-tidal, songnodes frontend :3006
How it syncs	login-start stores verifier/state in tidal_pkce_pending.json and prints the authorize URL with redirect 127.0.0.1:3006/oauth/callback; the user copies the landing address; login-finish checks state and age (15 min) and exchanges the code with client_secret in the body.
Conflicts	if songnodes is running, its /oauth/callback route (App.tsx) consumes the single-use code with its own verifier, so musicLibrary's exchange fails
Stale data	refresh_token on expiry; else re-login
State	● Built

Evidence 3

```
musicLibrary/pipeline/djlib/tidal.py:87-114
songnodes/frontend/src/App.tsx:33
songnodes/frontend/src/components/OAuthCallback.tsx:168
```

Protocol: **Cloud to laptop file transfer**

Participants	ml-agent-weekly-run
How it syncs	device_stage_files to bring files into the cloud workspace; device_commit_files with a unique staged name per run and md5 check both sides.
Conflicts	unique filenames avoid stale cached copies
Stale data	stop and report
State	● Built

Evidence 1

```
musicLibrary/agent/weekly_run.md:File transfer
```

Protocol: **Manual adds processed stamp**

Participants	ml-persona-curator, ml-cli-scrape
How it syncs	Rows without 'processed' are appended to the scrape rows, then the CSV is rewritten with processed=<run date> before raw_<date>.json is written.
Conflicts	stamped rows are never re-read
Stale data	none
State	● Built

Evidence 1

```
musicLibrary/pipeline/run_weekly.py:378-395
```

Protocol: **Durable job queue (ingestion_jobs)**

Participants	sn-be-svc-rest-api, sn-be-svc-worker, sn-be-tbl-ingestion-jobs
How it syncs	enqueue INSERT ON CONFLICT (kind,idempotency_key) DO NOTHING returning the original row; claim_next first dead-letters expired leases with no attempts left, then selects queued (run_after<=now) or lease-expired running rows FOR UPDATE SKIP LOCKED and increments attempts; handler and follow-up enqueues and complete run in one transaction; failure → fail() requeues with backoff 30s*2^(attempts-1) capped 3600s (x4 for transient), dead when attempts>=max_attempts or permanent 4xx (except 408/429).
Conflicts	idempotency key returns existing job; SKIP LOCKED prevents double claim; lease reclaim for crashed workers
Stale data	dead jobs revived by POST /api/v1/jobs/{id}/retry; succeeded pruned after 14 days
State	● Built

Evidence 3

```
songnodes/services/common/datalayer/jobs.py:26
songnodes/services/common/datalayer/jobs.py:39
songnodes/services/common/datalayer/runner.py:25
```

Protocol: **Recurring jobs via time-bucketed keys**

Participants	sn-be-svc-worker
How it syncs	songnodes.schedule every 30s enqueues projection_consume, graph_projection_consume (15s buckets), ingest_pending (60s), scrape_targets (3600s), audio_analysis (600s) with key tick:{kind}:{now//interval}.
Conflicts	overlapping beats collide on the same key
Stale data	missed buckets are simply skipped
State	● Built

Evidence 1

```
songnodes/services/worker/celery_app.py:83
```

Protocol: **Transactional outbox to projections**

Participants	all catalog writers, projection_tracks consumer, graph_projection consumer
How it syncs	append in the writer transaction; consumers read (txid,id) > offset and txid < pg_snapshot_xmin; handler + advance in one transaction; 3 attempts then outbox_dead_letters and advance; rebase_after_restore resets txids after dump/restore.
Conflicts	keyed upserts, at-least-once
Stale data	projection_reconcile / rebuild jobs; POST /api/v1/projections/tracks/reconcile
State	● Built

Evidence 2

```
songnodes/services/common/datalayer/outbox.py:1
songnodes/services/common/datalayer/projection.py:84
```

Protocol: **Library ingest idempotency**

Participants	import-music-library.sh, sn-be-svc-rest-api, sn-be-svc-worker
How it syncs	script computes md5 of library.json → Idempotency-Key import-<md5>; POST /api/v1/library/ingest enqueues library_ingest; script polls GET /api/v1/jobs/{id} and retries dead once; worker processes payload in one transaction and enqueues projection/backfill follow-ups.
Conflicts	same key returns the original job regardless of body (contract promises 409 for a different body: not implemented)
Stale data	none
State	● Built

Evidence 3

```
songnodes/scripts/import-music-library.sh:45
songnodes/services/rest_api/routers/library_ingest.py:10
songnodes/contracts/library_ingest.md:16
```

Protocol: **Scrape -> ingest -> projection ordering**

Participants	sn-be-svc-worker, sn-be-svc-unified-scraper
How it syncs	scrape handler (sync HTTP) → follow-up ingest_pending keyed by job → if changed, projection_consume, graph_projection_consume, enrichment_backfill; ingest_pending chains itself while examined>=limit; independent 60s tick also runs ingest_pending.
Conflicts	ingest_playlist locks bronze and silver playlist rows FOR UPDATE; INGEST_MARKER prevents re-ingest
Stale data	60s tick catches anything a follow-up missed
State	● Built

Evidence 2

```
songnodes/services/common/datalayer/handlers.py:96
songnodes/services/common/datalayer/handlers.py:111
```

Protocol: **OAuth state in Redis**

Participants	sn-be-svc-rest-api, sn-be-store-redis, Spotify, TIDAL
How it syncs	authorize stores PKCE verifier under oauth:{svc}:{state} with TTL; callback reads and deletes; tokens returned to client and optionally persisted to user_oauth_tokens (default_user); access log redacts code/state.
Conflicts	single-use state key
Stale data	TTL expiry → user restarts flow
State	● Built

Evidence 2

```
songnodes/services/rest_api/routers/music_auth.py:413
songnodes/services/rest_api/routers/access_log.py:1
```

Protocol: **Scheduled-scrape window**

Participants	sn-be-svc-rest-api, sn-be-svc-worker
How it syncs	scrape_targets reads one scrape_window row; outside Melbourne-day window it returns outside_window; manual scrapes ignore it.
Conflicts	single row
Stale data	missing table/row = no window
State	● Built

Evidence 1

```
songnodes/services/common/datalayer/target_scrape.py:12
```

Protocol: **Saved sets server/device merge**

Participants	sn-fe-sets, sn-fe-home, sn-be-rest-api, localStorage sn-music-library-sets, sn-set-draft
How it syncs	readSavedSetsAsync: GET /api/v1/library/sets; pair each server set with a local copy by id, else by content (name + ordered track ids) for local ids (non-numeric); server fields win; unpaired local-id sets kept; numeric ids missing on server dropped (deleted elsewhere); write merged list to localStorage and remap draft id. writeSavedSetsAsync diffs against serverSetsCache: DELETE removed, PUT changed (updated or name differs), PUT /0 creates and expects {id}; pendingCreates map dedupes concurrent creates in-tab.
Conflicts	server wins on name, description, tracks; local-only fields (shape, minutes, mix, pool, seed) kept from local copy
Stale data	GET failure → device copy, source "device"
State	● Built

Evidence 1

```
songnodes/frontend/src/features/musicLibrary/data.ts:166-337
```

Protocol: **Library target flag sync**

Participants	sn-fe-library, sn-fe-store-unified-tracks, sn-be-rest-api
How it syncs	syncTargetTrack POSTs /api/v1/target-tracks (or DELETEs by UUID) and records {isTarget,priority,pendingSync} in sn-target-tracks-v1; useUnifiedTracks hydrates library rows from local state first, then backend targets matched by lower-case artist|title.
Conflicts	local entry always beats backend state
Stale data	failure leaves pendingSync=true; no retry loop
State	● Built

Evidence 2

```
songnodes/frontend/src/services/trackSyncAdapter.ts:31-100
songnodes/frontend/src/hooks/useUnifiedTracks.ts:88-110
```

Protocol: **OAuth connect and token handoff**

Participants	sn-fe-settings, sn-fe-oauth-callback, sn-be-rest-api, sn-fe-store-zustand
How it syncs	Spotify: random 32-byte state in sessionStorage, navigate to /api/v1/music-auth/spotify/authorize; TIDAL: POST tidal/oauth/init then navigate. Callback page validates state, GETs backend callback, stores tokens in sessionStorage <service>_oauth_tokens, POSTs store-token, then postMessage to opener (same origin) or redirects to /.
Conflicts	n/a
Stale data	useTokenRefresh refreshes under navigator.locks (localStorage mutex fallback)
State	● Built

Evidence 3

```
songnodes/frontend/src/pages/SettingsPage.tsx:329-394
songnodes/frontend/src/components/OAuthCallback.tsx:60-290
songnodes/frontend/src/hooks/useTokenRefresh.ts:43-210
```

Protocol: **Pipeline telemetry polling**

Participants	sn-fe-pipeline, sn-be-rest-api
How it syncs	fetch jobs (200), scrape jobs (50), outbox status in parallel; setTimeout chain 10 s (30 s intended on error); refetch on visibilitychange.
Conflicts	n/a
Stale data	source 'unavailable' with explicit error
State	● Built

Evidence 1

```
songnodes/frontend/src/features/pipeline/usePipelineTelemetry.ts:270-
335
```

Protocol: **Idempotent library ingest and scrape trigger**

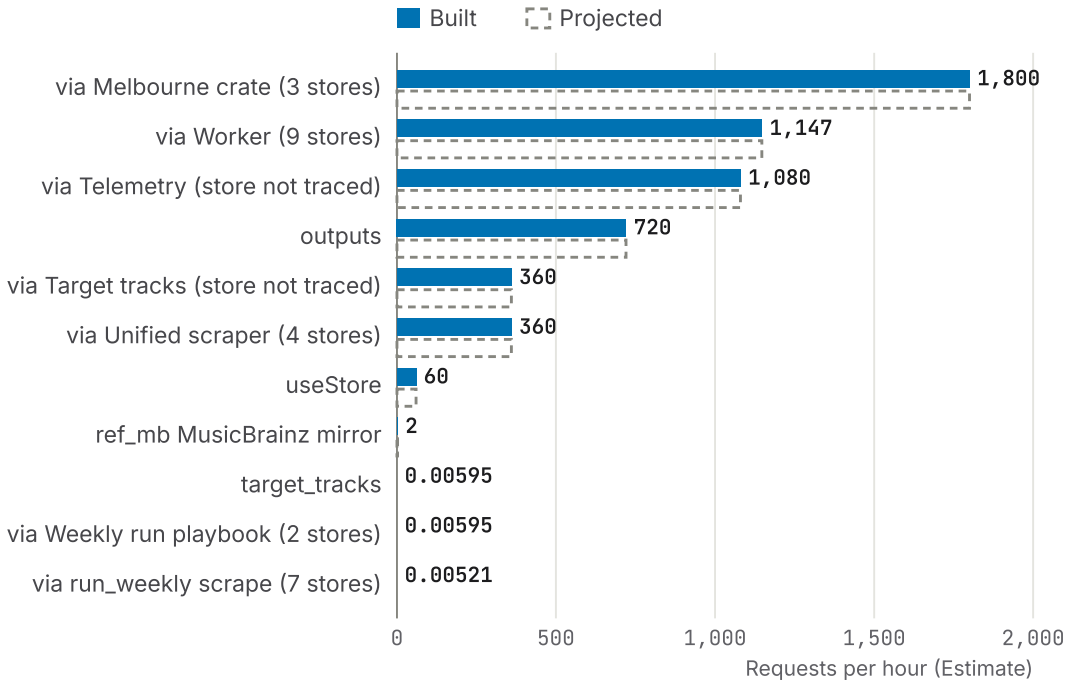
Participants	sn-fe-library, sn-fe-pipeline, sn-be-rest-api
How it syncs	library: Idempotency-Key import-sha256-<file hash> so the same file is never imported twice (matches scripts/import-music-library.sh scheme intent); scrape: one crypto.randomUUID per form submission reused on retry; 202 created=false shows "Already queued".
Conflicts	server-side idempotency (backend scope)
Stale data	job retry endpoint
State	● Built

Evidence 3

```
songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:15-21,114
songnodes/frontend/src/pages/ScraperProgressPage.tsx:94-128
songnodes/plan/agents/L8_library_import_dialog.md:9
```

C6: **Polling load by target store (Estimate)**

Estimate: the heaviest polling target is via Melbourne crate (3 stores), 1,800 requests per hour; queued work adds no polling.



C6 · n = 17 · Unit: requests per hour (Estimate) · ● Built

· Source: timings[] (kind poll|schedule, value_ms, where), signals[] and flows[] to resolve the store · Method: Estimate: requests per hour = 3,600,000 / value_ms per poll or schedule timing, summed by target store; the target is the timing's own store, else the single store linked to its component by signals or flows, else the component is named with its store count. Assumes every tick makes one request.

Table: **Timing constants** `105 constants`

105 rows

CONSTANT	KIND	VALUE	WHERE	STATE	EVIDENCE
weekly run schedule	schedule	Sunday 7:37am (weekly)	ml-agent-weekly-run	● Built	musicLibrary/agent/weekly_run.m
scrape look-back window	schedule	--days 8	ml-cli-scrape	● Built	musicLibrary/pipeline/run_weekl
scraper polite delay	debounce	delay=1.0	ml-cli-scrape	● Built	musicLibrary/pipeline/djlib/scr
3RRR episode delay	debounce	delay=2	ml-cli-scrape	● Built	musicLibrary/pipeline/djlib/scr

scraper HTTP timeout	timeout	timeout=30	ml-cli-scrape	Built	musicLibrary/pipeline/djlib/scr
Deezer per-request sleep	debounce	time.sleep(0.15)	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/dee
Deezer quota backoff	retry	2 ** attempt s, 4 attempts (1,2,4,8 s)	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/dee
Deezer HTTP timeout	timeout	timeout=20	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/dee
TIDAL API and token HTTP timeout	timeout	timeout=30	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/tic
TIDAL 429/5xx retry	retry	Retry-After or 2 ** attempt s, 5 attempts	ml-cli-sync-tidal	Built	musicLibrary/pipeline/djlib/tic
TIDAL user token refresh margin	expiry	expires_in - 120	ml-store-secrets	Built	musicLibrary/pipeline/djlib/tic
TIDAL client token refresh margin	expiry	expires_in - 60	ml-cli-resolve	Built	musicLibrary/pipeline/djlib/tic
TIDAL token default lifetime	expiry	tok.get('expires_in', 3600)	ml-store-secrets	Built	musicLibrary/pipeline/djlib/tic
PKCE login link max age	expiry	> 900	ml-cli-tidal	Built	musicLibrary/pipeline/djlib/tic
one-step login callback wait	timeout	600 x 0.5 s	ml-cli-tidal	Built	musicLibrary/pipeline/djlib/tic
resolve foreground chunk	timeout	timeout 160	ml-agent-weekly-run	Built	musicLibrary/agent/weekly_run.m
songnodes ingest POST timeout	timeout	timeout=30	ml-cli-sync-songnodes	Built	musicLibrary/pipeline/run_weekl
import-music-library job poll	poll	sleep 5, 120 polls (10 min)	sn-be-cli-import-music-library	Built	songnodes/scripts/import-music-
artifact shared-store init timeout	timeout	setTimeout(..., 4000)	ml-page-set-builder	Built	musicLibrary/artifact/app/artif
legacy toast	timeout	2200	ml-page-set-builder-legacy	Built	musicLibrary/artifact/index.htm
legacy delete confirm arm	timeout	3000	ml-page-set-builder-legacy	Built	musicLibrary/artifact/index.htm
promotion window	ttl	promote_window_days 30	ml-cli-process	Built	musicLibrary/config/rules.json
early chart entry window	ttl	chart_entry_window_days 21	ml-cli-process	Built	musicLibrary/config/rules.json
chart last-seen freshness	ttl	timedelta(days=14)	ml-cli-process	Built	musicLibrary/pipeline/djlib/lec
High Rotation tenure cap	ttl	high_tenure_max_days 42	ml-cli-process	Built	musicLibrary/config/rules.json
High Rotation silence demotion	ttl	high_silence_days 45	ml-cli-process	Built	musicLibrary/config/rules.json
Standard archive minimum age	ttl	std_archive_min_age_days 180	ml-cli-process	Built	musicLibrary/config/rules.json
Standard archive silence	ttl	std_archive_silence_days 60	ml-cli-process	Built	musicLibrary/config/rules.json
ref-mb stop grace period	timeout	stop_grace_period: 10m	ml-svc-ref-mb	Built	musicLibrary/ref_mb/compose.yam
ref-mb healthcheck	heartbeat	interval 30s, timeout 5s, retries 5	ml-svc-ref-mb	Built	musicLibrary/ref_mb/compose.yam

ref-mb checkpoint_timeout	schedule	checkpoint_timeout=30min	ml-svc-ref-mb	Built	musicLibrary/ref_mb/compose.yam
songnodes ref_mb statement timeout	timeout	MB_LOCAL_STATEMENT_TIMEOUT_MS 2000	sn-be-store-refmb	Built	songnodes/services/common/refmb
MetaBrainz HEAD retry	retry	5 tries, sleep 5, timeout 60	ml-cli-fetch-metabrainz	Built	musicLibrary/pipeline/fetch_met
MetaBrainz curl retry	retry	--retry 8 --retry-delay 5	ml-cli-fetch-metabrainz	Built	musicLibrary/pipeline/fetch_met
Celery beat drain-jobs	poll	JOB_POLL_SECONDS default '5'	sn-be-svc-worker	Built	songnodes/services/worker/celer
drain task expires	expiry	max(POLL_SECONDS * 3, 15)	sn-be-svc-worker	Built	songnodes/services/worker/celer
Celery beat schedule-recurring	schedule	SCHEDULE_SECONDS default '30'	sn-be-svc-worker	Built	songnodes/services/worker/celer
schedule task expires	expiry	SCHEDULE_SECONDS * 3	sn-be-svc-worker	Built	songnodes/services/worker/celer
ingest_pending bucket	schedule	INGEST_INTERVAL_SECONDS '60'	sn-be-svc-worker	Built	songnodes/services/worker/celer
projection_consume and graph_projection_consume bucket	schedule	PROJECTION_INTERVAL_SECONDS '15'	sn-be-svc-worker	Built	songnodes/services/worker/celer
scrape_targets bucket	schedule	SCRAPE_TARGETS_INTERVAL_SECONDS 3600	sn-be-svc-worker	Built	songnodes/compose.dev.yaml:134
audio_analysis bucket	schedule	AUDIO_ANALYSIS_INTERVAL_SECONDS 600	sn-be-svc-worker	Built	songnodes/compose.dev.yaml:128
job lease	lease	DEFAULT_LEASE_SECONDS = 300	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datal
scrape job lease	lease	KIND_LEASES = {'scrape': 1800.0}	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datal
job retry backoff base (x4 for transient), doubled per attempt	retry	BACKOFF_SECONDS = 30; 30*2^(attempts-1)	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datal
job retry backoff cap	retry	MAX_BACKOFF_SECONDS = 3600	sn-be-tbl-ingestion-jobs	Built	songnodes/services/common/datal
succeeded job retention	ttl	JOB_RETENTION_DAYS '14'	sn-be-tbl-ingestion-jobs	Built	songnodes/services/worker/celer
target re-search interval	schedule	TARGET_RESEARCH_DAYS 7	sn-be-tbl-target-tracks	Built	songnodes/compose.dev.yaml:136
worker → /scrape HTTP timeout	timeout	SCRAPE_TIMEOUT_SECONDS = 960	sn-be-svc-worker	Built	songnodes/services/common/datal
unified scraper spider timeout	timeout	asyncio.wait_for(..., timeout=900.0)	sn-be-svc-unified-scraper	Built	songnodes/scrapers/unified_scra
scrapy LOGSTATS interval feeding progress	poll	LOGSTATS_SECONDS = 10	sn-be-svc-unified-scraper	Built	songnodes/scrapers/unified_scra
rest-api scrape progress fetch timeout	timeout	httpx.AsyncClient(timeout=2.0)	sn-be-svc-rest-api	Built	songnodes/services/rest_api/rou
worker heartbeat staleness	heartbeat	time.time()-mtime < 90	sn-be-svc-worker	Built	songnodes/services/worker/Docke
worker Docker HEALTHCHECK interval	heartbeat	--interval=30s --timeout=10s --start-period=40s	sn-be-svc-worker	Built	songnodes/services/worker/Docke

postgres healthcheck	heartbeat	interval 3s, retries 30	sn-be-store-postgres	● Built	songnodes/compose.dev.yaml:21
redis healthcheck	heartbeat	interval 3s, retries 20	sn-be-store-redis	● Built	songnodes/compose.dev.yaml:31
rest-api healthcheck	heartbeat	interval 5s, timeout 6s, retries 20	sn-be-svc-rest-api	● Built	songnodes/compose.dev.yaml:80
rabbitmq healthcheck	heartbeat	interval 10s, retries 12	sn-be-store-rabbitmq	● Built	songnodes/compose.dev.yaml:91
unified-scraper healthcheck	heartbeat	interval 10s, start_period 20s	sn-be-svc-unified-scraper	● Built	songnodes/compose.dev.yaml:179
audio-analysis healthcheck	heartbeat	interval 10s, start_period 30s	sn-be-svc-audio-analysis	● Built	songnodes/compose.dev.yaml:192
start-dev compose up wait timeout	timeout	--wait-timeout 120	sn-be-cli-start-dev	● Built	songnodes/scripts/start-dev.sh:
ref_mb statement timeout	timeout	MB_LOCAL_STATEMENT_TIMEOUT_MS 2000	sn-be-store-refmb	● Built	songnodes/compose.dev.yaml:69
graph-api CACHE_TTL (unused decorator)	ttl	CACHE_TTL default 300	sn-be-svc-graph-api	● Built	songnodes/services/graph-visual
graph-api batch status key TTL	ttl	setex(batch:{id}, 3600)	sn-be-store-redis	● Built	songnodes/services/graph-visual
graph-api pool and query timeout	timeout	pool_timeout=30, command_timeout 30	sn-be-svc-graph-api	● Built	songnodes/services/graph-visual
graph-api pool recycle	ttl	pool_recycle=3600	sn-be-svc-graph-api	● Built	songnodes/services/graph-visual
Spotify/TIDAL OAuth state TTL	ttl	setex(..., 600)	sn-be-store-redis	● Built	songnodes/services/rest_api/rou
music search proxy timeout	timeout	ClientTimeout(total=5)	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/mai
AI proxy upstream timeout	timeout	ClientTimeout(total=60)	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/rou
401 rate-limit window	timeout	BLOCK_DURATION_SEC = 60	sn-be-svc-rest-api	● Built	songnodes/services/rest_api/sec
spider daily quota key TTL	ttl	SCRAPER_RUN_TTL_HOURS '24'	sn-be-store-redis	● Built	songnodes/scrapers/spiders/mixe
setlist dedupe key TTL	ttl	SCRAPER_SOURCE_TTL_DAYS '30'	sn-be-store-redis	● Built	songnodes/scrapers/spiders/mixe
Scrapy default DOWNLOAD_DELAY (randomised 0.5-1.5x)	debounce	DOWNLOAD_DELAY = 2	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/settings.py:
1001tracklists DOWNLOAD_DELAY	debounce	'DOWNLOAD_DELAY': 90.0	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/spiders/1001
AUTOTHROTTLE_MAX_DELAY	retry	AUTOTHROTTLE_MAX_DELAY = 60	sn-be-svc-unified-scraper	● Built	songnodes/scrapers/settings.py:
Memgraph connection timeout	timeout	connection_timeout=5.0	sn-be-store-memgraph	● Built	songnodes/services/common/datal
Duo session cookie expiry (queued)	expiry	SESSION_EXPIRY_SECONDS = 7 * 24 * 60 * 60	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/route

Duo auth state TTL (queued)	ttl	STATE_EXPIRY_SECONDS = 600	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/route
Duo router Redis connect timeout (queued)	timeout	socket_connect_timeout=0.2	sn-be-q-user-auth-router	▲ Queued	PR #578 services/rest_api/route
pipeline telemetry poll	poll	startPolling(interval = 10000)	sn-fe-pipeline	● Built	songnodes/frontend/src/features
pipeline poll backoff on error (never engages, stale closure)	retry	error ? 30000 : 10000	sn-fe-pipeline	● Built	songnodes/frontend/src/features
job stall threshold	timeout	STALL_MS = 10 * 60 * 1000	sn-fe-pipeline	● Built	songnodes/frontend/src/features
scrape job stages poll (tab visible)	poll	STAGES_POLL_MS = 5000	sn-fe-pipeline	● Built	songnodes/frontend/src/features
target tracks search-state poll	poll	SEARCH_POLL_MS = 10_000	sn-fe-sources-targets	● Built	songnodes/frontend/src/componer
library ingest job poll	poll	pollMs = 2000	sn-fe-library	● Built	songnodes/frontend/src/features
tracklist import stall	timeout	STALL_MS = 60_000	sn-fe-graph	● Built	songnodes/frontend/src/componer
ApiClient default request timeout	timeout	timeout: 30000 (DEFAULT_CONFIG.api.timeout)	sn-fe-home	● Built	songnodes/frontend/src/types/ir
token refresh buffer before expiry	expiry	REFRESH_BUFFER_MS = 5 * 60 * 1000	sn-fe-store-zustand	● Built	songnodes/frontend/src/hooks/us
token expiry check interval	poll	CHECK_INTERVAL_MS = 60 * 1000	sn-fe-store-zustand	● Built	songnodes/frontend/src/hooks/us
localStorage mutex lease	lease	lockTimeoutMs = 10000	sn-fe-store-browser-ls	● Built	songnodes/frontend/src/hooks/us
mutex acquire give-up	timeout	Date.now() - start > 12000	sn-fe-store-browser-ls	● Built	songnodes/frontend/src/hooks/us
mutex retry spin	retry	setTimeout(res, 100)	sn-fe-store-browser-ls	● Built	songnodes/frontend/src/hooks/us
default token lifetime when expires_in missing	expiry	(tokenData.expires_in || 3600) * 1000	sn-fe-oauth-callback	● Built	songnodes/frontend/src/componer
callback pop-up close / redirect delay	timeout	setTimeout(..., 1500)	sn-fe-oauth-callback	● Built	songnodes/frontend/src/componer
callback error redirect delay	timeout	setTimeout(..., 3000)	sn-fe-oauth-callback	● Built	songnodes/frontend/src/componer
toast visible time (8 s with action)	timeout	left = current.action ? 8000 : 4000	sn-fe-home	● Built	songnodes/frontend/src/componer
"slow" loading hint	timeout	setTimeout(() ⇒ setSlow(true), 1000)	sn-fe-library	● Built	songnodes/frontend/src/features
clear-draft double-press window	timeout	setTimeout(() ⇒ setArmedNew(false), 3000)	sn-fe-sets	● Built	songnodes/frontend/src/features
minimap viewport debounce	debounce	VIEWPORT_UPDATE_DEBOUNCE_MS = 100	sn-fe-graph	● Built	songnodes/frontend/src/componer
InfoTip hover delay	debounce	HOVER_DELAY = 500	sn-fe-home	● Built	songnodes/design-system/compone
PIXI init retry delay	retry	setTimeout(() ⇒ initializePixi(), 100)	sn-fe-graph	● Built	songnodes/frontend/src/componer
copied/saved status reset	timeout	setTimeout(..., 2000)	sn-fe-sources-import	● Built	songnodes/frontend/src/features

nginx DNS re-resolve TTL	ttl	resolver 127.0.0.11 valid=10s	sn-fe-store-nginx	Built	songnodes/frontend/Dockerfile:3
frontend container healthcheck	heartbeat	interval: 10s, timeout: 5s, retries: 6	sn-fe-store-nginx	Built	songnodes/compose.dev.yaml:204-
hashed asset cache	ttl	expires 1y	sn-fe-store-nginx	Built	songnodes/frontend/Dockerfile:4

Table: **Size and count limits** 76 limits

76 rows

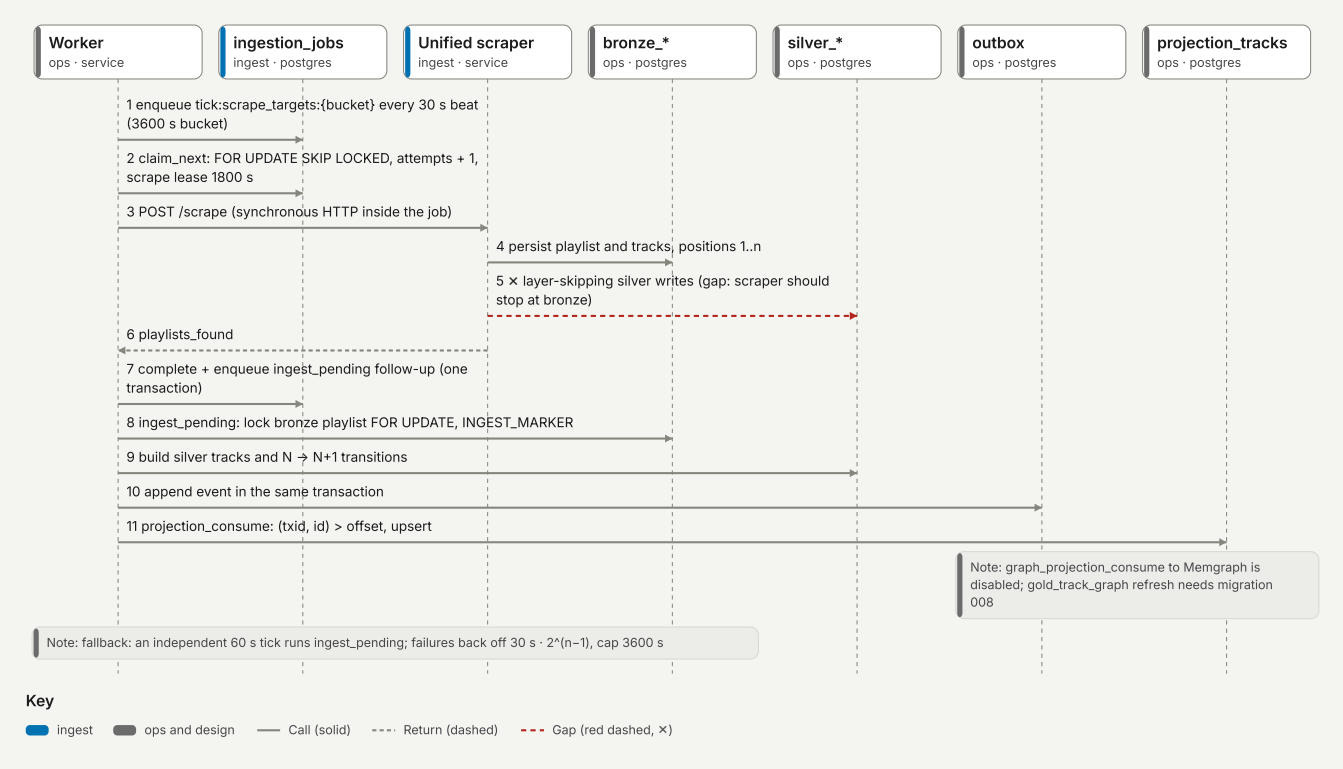
LIMIT	VALUE	CURRENT	WHERE	STATE	EVIDE
ABC plays page size	100 items	–	ml-cli-scrape	Built	music
episodes per program default	2 episodes	–	ml-cli-scrape	Built	music
ARIA parse sanity floor	40 rows	50	ml-cli-scrape	Built	music
TIDAL search results kept	20 tracks	–	ml-cli-resolve	Built	music
TIDAL /tracks filter[id] and ISRC batch	20 ids	–	ml-cli-resolve	Built	music
TIDAL playlist add/remove chunk	50 tracks	–	ml-cli-sync-tidal	Built	music
Deezer search limit per query	25 tracks	–	ml-cli-resolve	Built	music
resolve thread pool	4 workers	–	ml-cli-resolve	Built	music
resolve flush interval	20 songs	–	ml-cli-resolve	Built	music
attempts before UNRESOLVED	3 attempts	max 2 in pending	ml-cli-process	Built	music
rejected re-check threshold	2 appearances	–	ml-cli-ingest	Built	music
high_crate_cap	60 tracks per crate	7	ml-cli-process	Built	music
std_crate_cap (not enforced)	250 tracks per crate	131	ml-cli-process	Built	music
promote_min_sources	2 sources	–	ml-cli-process	Built	music
promote_min_saves	50 saves	–	ml-cli-process	Built	music
high_chart_saturation_weeks	6 weeks	–	ml-cli-process	Built	music
duration floors	{'MEL_House': 285, 'MEL_Techno': 285, 'UK_Garage': 225} seconds	–	ml-cli-process	Built	music
SET playlist name length	250 characters	–	ml-cli-sync-sets	Built	music
runs kept in library.json	26 runs	4	ml-cli-build-db	Built	music
ArtifactData list page	1000 documents	0	ml-agent-weekly-run	Built	music
Beatport actions before check-in	25 actions	–	ml-agent-beatport-sync	Built	music
artifact page and file size limit (platform)	16 MB	{'app.js': 372921, 'library.json': 185443, 'style.css': 262326}	ml-store-artifact-files	Built	Artif
songnodes library search limit used by the page	5000 tracks	413	sn-fe-library	Built	songn
songnodes Idempotency-Key length	200 characters	39	sn-be-svc-rest-api	Built	songn

ref-mb memory and buffers	{'mem_limit': '4g', 'shm_size': '1g', 'shared_buffers': '1GB', 'max_wal_size': '8GB', 'work_mem': '32MB'} bytes	–	ml-svc-ref-mb	Built	music
MetaBrainz parallel downloads	1 connections	–	ml-cli-fetch-metabrainz	Built	music
data folder size	58 GB	–	ml-store-data	Built	music
MusicBrainz web API rate (brief rule)	1 request per second	–	ml-queued-wp7-pipeline-client	Queued	music
MAX_JOBS_PER_DRAIN	25 jobs	–	sn-be-svc-worker	Built	songn
Celery worker processes	2 processes	–	sn-be-svc-worker	Built	songn
worker_prefetch_multiplier	1 tasks	–	sn-be-svc-worker	Built	songn
job max_attempts default	3 attempts	–	sn-be-tbl-ingestion-jobs	Built	songn
outbox handler attempts before dead letter	3 attempts	–	sn-be-tbl-outbox	Built	songn
stored last_error length	2000 characters	–	sn-be-tbl-ingestion-jobs	Built	songn
ingest_pending playlists per job (standalone tracks x4)	25 playlists	–	sn-be-svc-worker	Built	songn
playlist entries accepted	5000 tracks	–	sn-be-svc-worker	Built	songn
enrichment_backfill batch_size	25 tracks	–	sn-be-svc-worker	Built	songn
camelot_repair / transition_scoring batch	500 rows	–	sn-be-svc-worker	Built	songn
projection_consume batch x max_batches	200 events (x50)	–	sn-be-svc-worker	Built	songn
graph_projection_consume batch x max_batches	100 events (x50)	–	sn-be-svc-worker	Built	songn
SCRAPE_TARGETS_PER_RUN	2 targets (x sources)	–	sn-be-svc-worker	Built	songn
AUDIO_ANALYSIS_PER_RUN	10 tracks	–	sn-be-svc-worker	Built	songn
scraper persistence batch_size	50 items	–	sn-be-svc-unified-scraper	Built	songn
CONCURRENT_REQUESTS	8 requests	–	sn-be-svc-unified-scraper	Built	songn
GET /api/graph/nodes and /data limit max	15000 nodes	–	sn-be-svc-graph-api	Built	songn
GET /api/graph/edges limit max	50000 edges	–	sn-be-svc-graph-api	Built	songn
neighbourhood view max tracks / steps	2000 tracks (steps ⩽ 6)	–	sn-be-svc-graph-api	Built	songn
WebSocket connections per room / total	100 connections (total 1000)	–	sn-be-svc-graph-api	Built	songn
rest-api asyncpg pool	15 connections (min 5)	–	sn-be-svc-rest-api	Built	songn
graph-api SQLAlchemy pool + overflow	30 connections (15+15)	–	sn-be-svc-graph-api	Built	songn
Idempotency-Key length	200 characters	–	sn-be-svc-rest-api	Built	songn
scrape search_query length	255 characters	–	sn-be-svc-rest-api	Built	songn
AI proxy body size	102400 bytes	–	sn-be-svc-rest-api	Built	songn

AI proxy max_tokens cap	8000 tokens	–	sn-be-svc-rest-api	● Built	songn
401 attempts before block	5 attempts/60s/IP	–	sn-be-svc-rest-api	● Built	songn
audio-analysis mem_limit	1 GiB	–	sn-be-svc-audio-analysis	● Built	songn
CLAUDE.md resource limits (DB 1-2GB, APIs 512MB, scrapers 1GB, frontend 256MB)	None not enforced	–	compose.dev.yaml	● Built	songn
Duo router PBKDF2 iterations (queued)	100000 iterations	–	sn-be-q-user-auth-router	▲ Queued	PR #5
library search page size	5000 tracks	294 tracks in the pinned fixture (frontend/tests/e2e/data/library.fixture.json)	sn-fe-library	● Built	songn
library rows rendered first	150 rows	–	sn-fe-library	● Built	songn
library export upload max	50 MB	fixture export 132,486 bytes	sn-fe-library	● Built	songn
frontend nginx request body (default, unset)	1 MB	–	sn-fe-store-nginx	● Built	songn
tracklist upload max	5 MB	–	sn-fe-graph	● Built	songn
graph nodes/edges per load	10000 rows	–	sn-fe-graph	● Built	songn
neighbourhood view cap (graph-hood-max token)	200 nodes	–	sn-fe-graph	● Built	songn
jobs listed on pipeline page	200 jobs	–	sn-fe-pipeline	● Built	songn
scrape jobs listed	50 jobs	–	sn-fe-pipeline	● Built	songn
scrape stage cards kept per visit	5 cards	–	sn-fe-pipeline	● Built	songn
artist attribution page size	50 rows	–	sn-fe-graph	● Built	songn
recent searches kept per namespace	10 items	–	sn-fe-store-browser-ls	● Built	songn
autocomplete recent / suggestions	8 items (5 recent, 8 suggestions)	–	sn-fe-paths	● Built	songn
quick search results	12 items	–	sn-fe-graph	● Built	songn
AI prompt history kept	30 items	–	sn-fe-store-browser-ls	● Built	songn
edges fetched for details	2 edges	–	sn-fe-graph	● Built	songn
toasts queued	3 toasts	–	sn-fe-home	● Built	songn
unified-scraper memory (compose; frontend has none set, CLAUDE.md says 256MB)	1 GB	–	sn-fe-store-nginx	● Built	songn

D6a: **Sequence – Scrape to bronze, silver and projections**

11 messages across 7 participants, numbered in order; red marks where the protocol leaks.

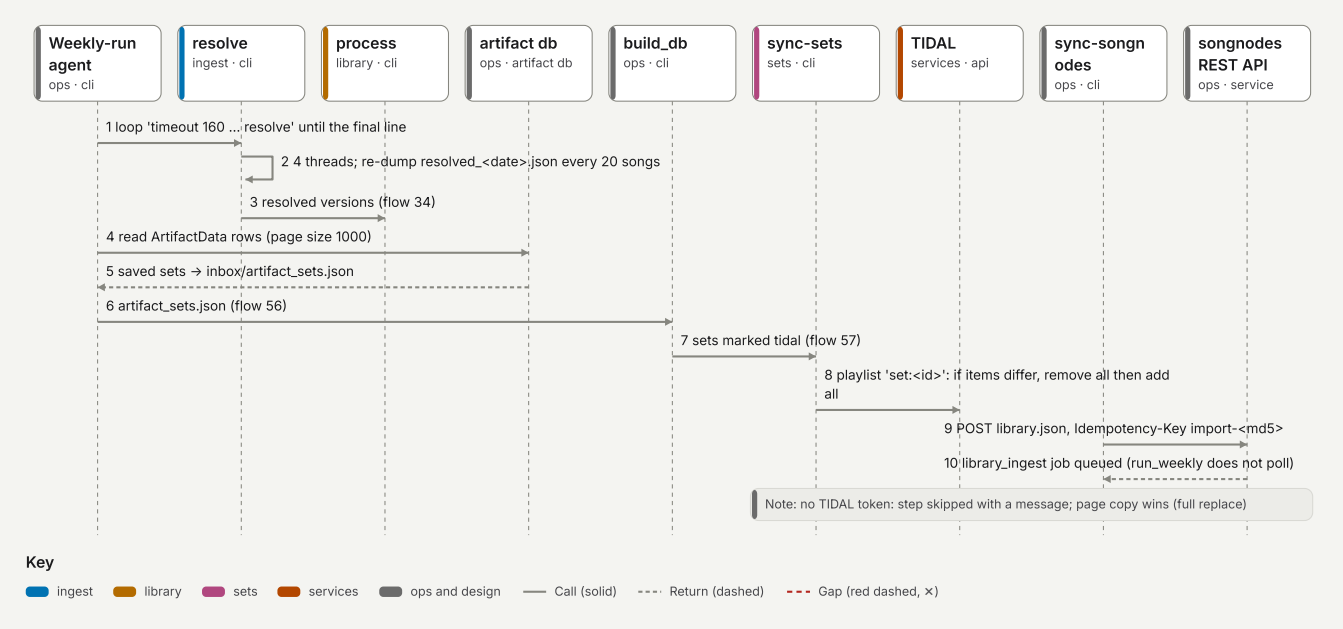


D6a · n = 11 · Unit: messages · ● Built

· Source: protocols[] sn-be-proto-recurring-schedule, sn-be-proto-job-queue, sn-be-proto-scrape-etl-order, sn-be-proto-outbox (algorithm, conflict_resolution, stale_fallback); timings[] · Method: hand-placed UML sequence (lifelines in first-use order, one row per message); protocols chosen per theme by flows touching their participants (sn-be-proto-recurring-schedule 6, sn-be-proto-job-queue 31, sn-be-proto-scrape-etl-order 10, sn-be-proto-outbox 0); 0 crossings

D6b: **Sequence – Weekly musicLibrary run to TIDAL and songnodes**

10 messages across 9 participants, numbered in order; red marks where the protocol leaks.

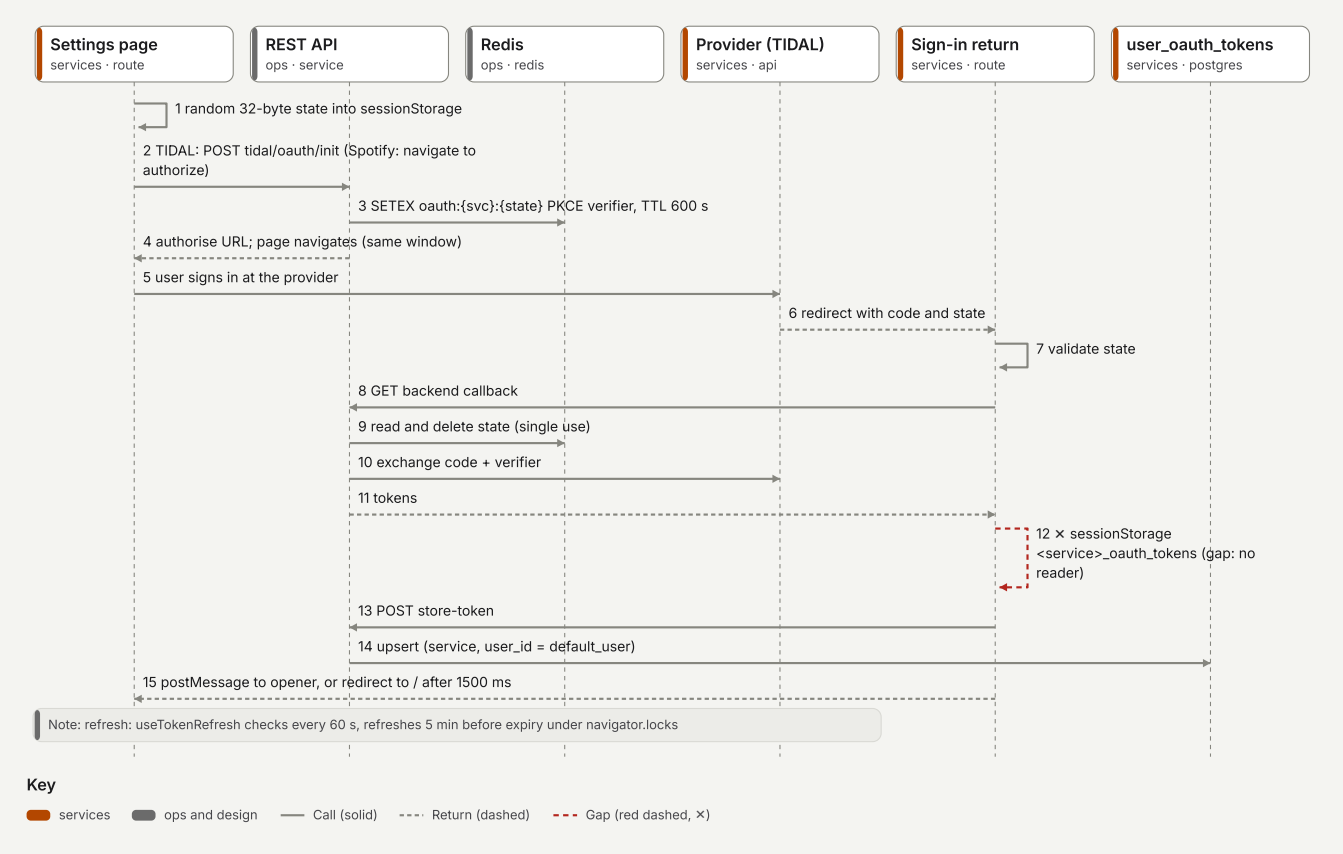


D6b · n = 10 · Unit: messages · ● Built

· Source: protocols[] ml-proto-resolve-resume, ml-proto-sync-sets, ml-proto-songnodes-ingest (algorithm, conflict_resolution, stale_fallback); timings[] · Method: hand-placed UML sequence (lifelines in first-use order, one row per message); protocols chosen per theme by flows touching their participants (ml-proto-resolve-resume 7, ml-proto-sync-sets 9, ml-proto-songnodes-ingest 2); 0 crossings

D6c: **Sequence – TIDAL and Spotify OAuth connect**

15 messages across 6 participants, numbered in order; red marks where the protocol leaks.

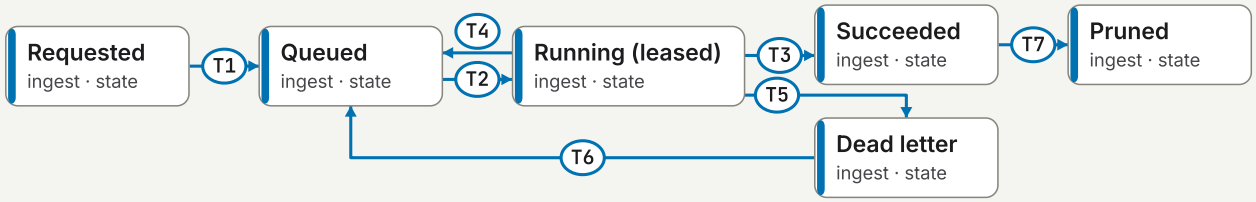


D6c · n = 15 · Unit: messages · ● Built

· Source: protocols[] sn-fe-proto-oauth, sn-be-proto-oauth-pkce (algorithm, conflict_resolution, stale_fallback); timings[] · Method: hand-placed UML sequence (lifelines in first-use order, one row per message); protocols chosen per theme by flows touching their participants (sn-fe-proto-oauth 4, sn-be-proto-oauth-pkce 25); 0 crossings

D7a: **State machine – Scrape and ingestion job**

6 states, 7 transitions; every timing on an arrow comes from the timings table.



Transitions

T1 Requested → Queued: enqueue: INSERT ON CONFLICT (kind, idempotency_key) DO NOTHING returns the existing job

T2 Queued → Running (leased): claim_next: run_after ≤ now, FOR UPDATE SKIP LOCKED, attempts + 1

T3 Running (leased) → Succeeded: handler, follow-up enqueues and complete in one transaction

T4 Running (leased) → Queued: fail: requeue with backoff 30 s · 2^(n−1), cap 3600 s (×4 transient); or lease expired (300 s, scrape 1800 s) and reclaimed

T5 Running (leased) → Dead letter: attempts ≥ max_attempts, or permanent 4xx (not 408/429)

T6 Dead letter → Queued: POST /api/v1/jobs/{id}/retry

T7 Succeeded → Pruned: pruned after 14 days

Key

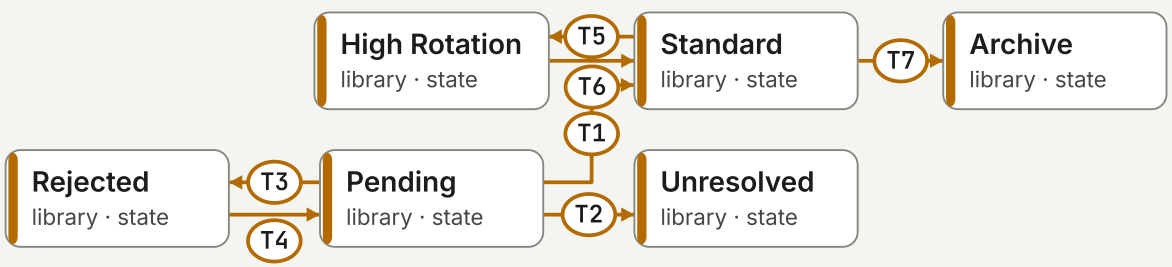
ingest — Transition (badge T = row in the transition list)

D7a · n = 7 · Unit: transitions · ● Built

· Source: protocols sn-be-proto-job-queue, sn-be-proto-scrape-etl-order; timings sn-be-t-lease-default, sn-be-t-lease-scrape, sn-be-t-backoff-base, sn-be-t-backoff-max · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D7b: **State machine – Track rotation**

6 states, 7 transitions; every timing on an arrow comes from the timings table.



Transitions

T1 Pending → Standard: resolved and passes the gates

T2 Pending → Unresolved: 3 resolve attempts fail (resolve_attempts)

T3 Pending → Rejected: fails a gate (for example a duration floor)

T4 Rejected → Pending: re-checked after 2 sightings (recheck threshold)

T5 Standard → High Rotation: promote: min 2 sources, min 50 saves, window 30 days; chart entry window 21 days; cap 60

T6 High Rotation → Standard: tenure > 42 days, silence 45 days or chart saturation 6 weeks

T7 Standard → Archive: age ≥ 180 days and silence 60 days

Key

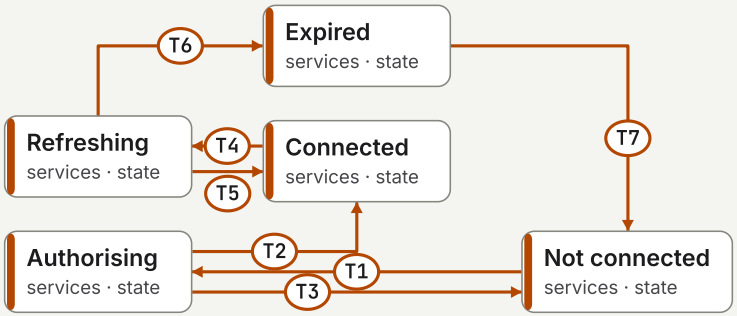
library — Transition (badge T = row in the transition list)

D7b · n = 7 · Unit: transitions · ● Built

· Source: timings ml-t-promote-window, ml-t-chart-entry, ml-t-high-tenure, ml-t-high-silence, ml-t-std-archive-age, ml-t-std-archive-silence; limits ml-l-promote-sources, ml-l-promote-saves, ml-l-high-cap, ml-l-chart-sat, ml-l-resolve-attempts, ml-l-recheck · Method: graphviz dot layered layout (LR, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

D7c: **State machine – OAuth token**

5 states, 7 transitions; every timing on an arrow comes from the timings table.



Transitions

T1 Not connected → Authorising: connect: state in sessionStorage, PKCE verifier in Redis (TTL 600 s)

T2 Authorising → Connected: callback validates state, exchanges code, stores token (default_user)

T3 Authorising → Not connected: state expired or error: redirect after 3000 ms; musicLibrary PKCE link > 900 s

T4 Connected → Refreshing: checked every 60 s; 5 min before expiry (default lifetime 3600 s)

T5 Refreshing → Connected: refresh under navigator.locks (localStorage mutex fallback)

T6 Refreshing → Expired: refresh fails

T7 Expired → Not connected: user signs in again

Key

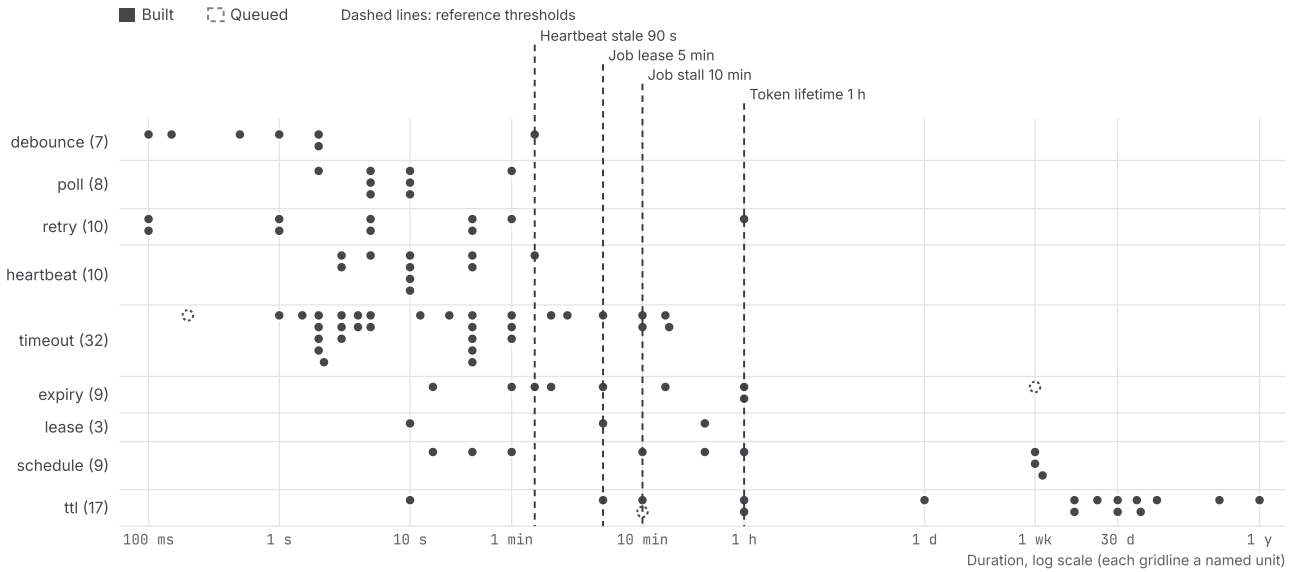
services — Transition (badge T = row in the transition list)

D7c · n = 7 · Unit: transitions · ● Built

· Source: protocols sn-fe-proto-oauth, sn-be-proto-oauth-pkce, ml-proto-tidal-login; timings sn-be-t-oauth-state, sn-fe-t-token-buffer, sn-fe-t-token-check, sn-fe-t-token-default-expiry, sn-fe-t-oauth-error-redirect, ml-t-pkce-age · Method: graphviz dot layered layout (LR, order reverse; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

C5: **Timing constants on a log axis**

105 timing constants span 100 ms to 365 d; reference lines mark Heartbeat stale (sn-be-t-heartbeat-stale), Job lease (sn-be-t-lease-default), Job stall (sn-fe-t-stall), Token lifetime (ml-t-tidal-user-default).



C5 · n = 105 · Unit: milliseconds (log scale) · **i Projected**

· Source: timings[].value_ms, timings[].kind, timings[].state · Method: Each constant is one dot on a log10 axis, one lane per kind (kinds ordered by median value); coincident dots stack. Reference lines are the timings matched by name: heartbeat staleness, default job lease, job stall threshold and default token lifetime.

DJ performing

Objective Pick the next track from proven transitions while a set is playing.

Device laptop in the DJ booth (desktop browser, booth theme available)

Entry point `sn-fe-graph`

Basis Inferred from pages and docs

Routine

- open /graph?mode=play
- search a track (QuickSearch / Ctrl K)
- read neighbours and edge details
- check key on /harmony

3 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	`sn-fe-graph`	load graph and select current track	songnodes-store viewState, sn-graph-view, sn-recent-quick-search	open /graph	–	/api/graph fallback, then error state
2	`sn-fe-graph`	open edge details	–	pointer on edge → GET /api/graph/edges	–	none
3	`sn-fe-harmony`	check Camelot moves	–	NavDrawer Harmony	track id in route / set draft via /paths or Library add-to-set	none

Failure modes and timeouts

- graph API down: useDataLoader falls back to /api/graph then shows error
- TIDAL token expired: useTokenRefresh refresh under lock

Set planner

Objective Shape a wave-shaped set from the crate, check every transition and send it to TIDAL.

Device browser on claude.ai (desktop or phone); desktop browser at home (127.0.0.1:3006)

Entry point `ml-page-set-builder`

Basis Inferred from pages and docs

Routine

- open the Set Builder artifact, filter Library, add tracks
- switch to Set builder, choose shape and minutes, Build set
- Save set, tick Send to TIDAL on the weekly run
- wait for the next weekly run to create 'SET · <name>'
- filter the crate on /library
- Build set from crate
- shape and reorder on /sets
- save, export, send to TIDAL

8 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF
1	`ml-page-set-builder`	filter and add to set	localStorage sn-filters-library, sn-set-draft	user	–
2	`ml-page-set-builder`	Save set with tidal flag	artifact db sets/<id>, localStorage sn-music-library-sets	user	SavedSet {name,shape,minutes,mix,pool,tidal,seed,tracks[{id,tgt,...,move}],updated,c via artifact db collection sets
3	`ml-agent-weekly-run`	export sets	inbox/artifact_sets.json	schedule	–
4	`ml-cli-sync-sets`	push to TIDAL	TIDAL playlist, secrets/tidal_playlists.json	schedule	–
5	`sn-fe-library`	filter and add tracks	sn-filters-library, sn-set-draft, sn-favs-library-crate	filter rail	–
6	`sn-fe-sets`	build from crate and shape energy	sn-set-draft	/sets?pool=crate&...	–
7	`sn-fe-sets`	save set	PUT /api/v1/library/sets/{id}; sn-music-library-sets	Save	–
8	`sn-fe-sets`	export or send	download .m3u8/.csv/.json/.xml or artifact_sets.json; POST /api/v1/tidal/sets/send	Export / Send to TIDAL	artifact_sets.json (SavedSet[]) via manual file copy to E:\musicLibrary\inbo›

Failure modes and timeouts

- up to a week of latency before a set reaches TIDAL
- Camelot moves and BPM gaps are inert: M3 render says none of the 413 tracks has a key and BPM
- a set edited while the db capability is unavailable is overwritten by the shared copy on next load (shared wins by id)
- songnodes /sets has its own Send to TIDAL that creates playlists directly, so the same set can end up as two TIDAL playlists
- server unreachable: sets kept on device and merged by content later
- set created twice if PUT /0 returns no id (content match fixes on next read)

Library curator

Objective — Keep the crates right each week: review, buy, override and import.

Device — laptop tuf-boi (E:\musicLibrary, Excel/DB Browser) and phone for the run message; desktop browser on the machine running the Compose stack

Entry point — `ml-agent-weekly-run`

Basis — Inferred from pages and docs

Routine

- read the weekly run message and outputs/<date>/weekly_report.md
- open review_queue.csv, fill energy_score_override, copy rows into overrides/energy_overrides.csv

- buy High Rotation tracks from buy_list.csv and put local_path into overrides
- add heard tracks to inbox/manual_adds.csv
- occasionally run calibrate.py and copy agreed values into config/rules.json
- browse db/dj_library.sqlite in DB Browser
- import the weekly library.json export
- review tracks needing an energy check
- mark missing tracks as targets
- manage targets on /sources/targets

8 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	`ml-agent-weekly-run`	read summary	–	scheduled task message	–	–
2	`ml-cli-process`	review_queue.csv to overrides CSV (manual copy)	overrides/energy_overrides.csv	weekly	energy_overrides.csv rows via file	none; rows ignored if not copied
3	`ml-cli-scrape`	manual adds	inbox/manual_adds.csv	heard a track	–	–
4	`ml-cli-calibrate`	fit rules	outputs/calibration_suggestions.json	ad hoc	–	–
5	`sn-fe-library`	Import library export	POST /api/v1/library/ingest; job polled	Import dialog FileDrop .json	–	Retry on dead job
6	`sn-fe-library`	review rev=true rows	sn-filters-library	Banner Review link	–	none
7	`sn-fe-library`	mark target	POST /api/v1/target-tracks; sn-target-tracks-v1	TargetActionButton	–	pendingSync=true kept locally
8	`sn-fe-sources-targets`	edit priorities and search	PUT/POST /api/v1/target-tracks	NavDrawer Sources	target_tracks rows via /api/v1/target-tracks -> scrape jobs	toast error

Failure modes and timeouts

- review queue lists all 413 tracks because no track has a BPM, so it cannot be worked through
- 0 overrides, 0 local_path, no rekordbox_collection.xml: none of the three levers has been used, so calibrate has nothing to fit
- review_queue.csv override column is not read by any code; rows must be copied by hand into a CSV with different columns
- failed target sync stays pendingSync=true forever
- ingest job dead: Retry button

Pipeline operator

Objective: Keep the stack, scrapers and imports healthy.

Device: laptop tuf-boi (Windows, Git Bash + Docker Desktop); 127.0.0.1 ports; desktop browser on the Docker host

Entry point: `sn-be-cli-start-dev`

Basis: Inferred from pages and docs

Routine

- bash scripts/start-dev.sh after git pull on main
- bash scripts/import-music-library.sh after the musicLibrary weekly run
- Set the scrape window (PUT /api/v1/scrape/window) and target tracks
- Watch /api/v1/jobs and /api/v1/jobs/{id}/stages; retry dead jobs
- Back up Postgres with the CLAUDE.md pg_dump one-liner
- queue a scrape for a target
- watch medallion stages and jobs
- retry dead jobs
- set scrape window
- connect TIDAL/Spotify and API keys in Settings
- import AI-found setlists

10 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	sn-be-cli-start-dev	git switch main && git pull; bash scripts/start-dev.sh	images, schema migrations, ingestion_jobs transition_scoring seed	code change on main	–	re-run; never reset the Postgres volume
2	sn-be-cli-import-music-library	bash scripts/import-music-library.sh after the musicLibrary weekly run	ingestion_jobs library_ingest	weekly musicLibrary run	library.json (export shape) via sn-be-sig-import-music-library	script retries a dead job once, then prints the error
3	sn-be-svc-worker	worker runs library_ingest, projection and backfill follow-ups	library.*, target_tracks, outbox, projection_tracks	beat drain	outbox events via sn-be-sql-outbox	projection_reconcile job
4	sn-be-svc-rest-api	PUT /api/v1/scrape/window and manage /api/v1/target-tracks	scrape_window, target_tracks	planning	–	no window = scrape every hour
5	sn-be-svc-rest-api	watch /api/v1/jobs and /api/v1/jobs/{id}/stages; POST /retry for dead jobs	ingestion_jobs	dead letter or warning	–	docker compose logs worker
6	sn-be-store-postgres	pg_dump -Fc via compose exec	backup-YYYYMMDD.dump	before risky change	–	none: scripts/backup_database.sh targets the wrong database
7	sn-fe-pipeline	queue scrape	POST /api/v1/scrape/trigger	scrape form	–	toast with HTTP detail
8	sn-fe-pipeline	retry job	POST /api/v1/jobs/{id}/retry	ConfirmDialog	–	none
9	sn-fe-settings	connect TIDAL	sessionStorage state; backend token store	/settings/tidal-connection	–	error banner on Settings
10	sn-fe-sources-import	find and import AI setlists	POST /api/v1/tracks/import-tracklist; songnodes_ai_setlists_ledger	/sources/import	job rows via REST → RabbitMQ/Celery (backend scope)	ledger kept locally

Failure modes and timeouts

- import-music-library.sh sends no API token, so it gets 401 once SONGNODES_API_TOKEN is set
- backup scripts in scripts/ and root target musicdb/songnodes_* names and back up nothing
- music search returns 503 because metadata-enrichment is not in compose
- hourly 1001tracklists target scrapes after the first of the day hit the Redis daily quota and report found_nothing
- polling backoff never engages (stale error closure)
- same-window OAuth leaves browser store without tokens

Developer

Objective — Ship one-job packages through PRs without breaking contracts or the design system.

Device — laptop or cloud agent session (Jules/Claude) with Docker for E2E; developer workstation and claude.ai artifact viewer; laptop tuf-boi (Git Bash / Linux VM shell)

Entry point — `sn-be-cli-scope-guard-g1`

Basis — Inferred from pages and docs

Routine

- branch <code>-<slug> off main
- add a router module with register(app) (G3) or a migration in the owned range (contracts/ownership.yaml)
- run scope_guard, pytest, e2e-docker
- update own STATUS.md slot
- read docs/design-system/reference.html
- check component previews

- run the violations fixture through the linter
- track work packages on the work-queue artifact
- edit pipeline/djlib, run test_pipeline.py
- fix a scraper after a 0-row source and log in outputs/<date>/fixes.md
- rebuild app.js with build.mjs from songnodes/frontend and publish
- load and evaluate ref_mb

8 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	`sn-be-cli-scope-guard-g1`	branch <code>-<slug>, edit owned paths only, run scope_guard.py	git branch	work-package brief	–	two attempts then draft PR
2	`sn-be-svc-rest-api`	add a router module with register(app) or a migration in the owned number range	new file	feature need	–	list follow-ups in the PR
3	`sn-be-cli-start-dev`	add the migration to start-dev.sh and rebuild the stack	compose stack	new migration	–	migration silently missing if not added
4	`sn-be-cli-e2e-docker`	run E2E and pytest; update own STATUS.md slot	STATUS.md slot	before PR	PR with checks run via git	report out-of-scope failures
5	`sn-fe-ds-reference`	look up tokens and components	sn-density	open file	–	none
6	`sn-fe-ds-violations-fixture`	lint regression	–	validate.py	–	none
7	`sn-fe-work-queue-tracker`	copy job prompt	artifact db tasks.note	artifact page	prompt text via clipboard	none
8	`ml-page-artifact-app-entry`	node build.mjs <fe> <out> <tools>	artifact/build/app.js, assets/*	songnodes Library or Set builder pages change	app.js IIFE via Artifact publish	–

Failure modes and timeouts

- a new migration is not applied unless someone adds it to start-dev.sh (no migration runner/ledger in compose)
- router modules without register(app) are silently skipped (data_quality)
- CLAUDE.md schema and API examples disagree with code
- previews fail standalone without injected window.SongNodes
- build.mjs against current songnodes main will not compile: SetBuilderPage imports readSavedSetsAsync and writeSavedSetsAsync, which artifact-data.ts does not export
- copy patch strings no longer match songnodes wording
- build pinned to design system 1.1.0 while songnodes is at 1.10.0

Scraper (system actor)

Objective — Find complete playlists containing target tracks and persist them to bronze.

Device — worker and unified-scraper containers on the laptop compose stack

Entry point — `sn-be-svc-unified-scraper`

Basis — Inferred from pages and docs

Routine

- songnodes.schedule enqueues scrape_targets hourly
- scrape_targets picks 2 targets not searched for 7 days, one scrape job per source
- worker drain claims scrape job (lease 1800s) and POSTs /scrape
- spider writes bronze (and silver) via item pipelines
- handler records playlists_found and enqueues ingest_pending

6 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	sn-be-svc-worker	songnodes.schedule enqueues tick:scrape_targets:{hour}	ingestion_jobs scrape_targets	30s beat	–	bucket collision skips duplicates
2	sn-be-svc-worker	scrape_targets picks 2 active targets older than 7 days (SKIP LOCKED), one scrape job per source	target_tracks.last_searched, ingestion_jobs scrape	job run	–	outside_window returns without work
3	sn-be-svc-worker	claim scrape (lease 1800s) and POST /scrape	ingestion_jobs running	drain	ScrapeRequest json via sn-be-http-worker-to-scraper	transport error or 5xx/429 → retry with backoff
4	sn-be-svc-unified-scraper	scrapy crawl subprocess; item pipelines persist	bronze_*, silver_*, Redis quota and dedupe keys	POST /scrape	–	daily_quota_reached closes the spider with nothing found; 900s timeout → 504
5	sn-be-svc-worker	record playlists_found, complete job, enqueue ingest_pending	target_tracks.playlists_found, ingestion_jobs	scraper response	follow-up job ingest-after-scrape:{job_id} via sn-be-sql-ingestion-jobs	60s ingest tick
6	sn-be-svc-worker	ingest_pending builds silver and N → N+1 transitions	silver_*, silver_track_transitions, outbox	follow-up	silver_track_transitions via sn-be-sql-graph-read-silver	rejected playlist outbox event

Failure modes and timeouts

- scraper restart loses live progress
- daily quota closes spider
- scrape timeout 900s → 504 → retry with backoff
- long scrape occupies one of two worker processes

Weekly-run agent (system actor)

Objective: Run scrape to republish every week without a person present.

Device: cloud session driving laptop tuf-boi through device_bash

Entry point: `ml-agent-weekly-run`

Basis: Inferred from pages and docs

Routine

- test_pipeline
- scrape and ingest
- resolve in 160 s foreground chunks until done
- process, sync-tidal
- ArtifactData list sets to inbox/artifact_sets.json via device_commit_files with md5 check
- db (also posts to songnodes)
- sync-sets
- copy library.json to artifact/build, stage, read artifact, publish
- check scrape_log and ALERT files, SendUserMessage

3 rows

STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF	FALLBACK
1	ml-cli-resolve	chunked resume	inbox/resolved_<date>.json	loop until 'resolved N songs →'	–	repeat chunks until 'resolved N songs →'
2	ml-cli-build-db	rebuild	db, library.json	step 6	short report: headline numbers, top 15 High Rotation, review count, alerts, Beatport actions, sets pushed, artifact version via SendUserMessage	–
3	ml-page-set-builder	republish	artifact library.json	step 8	–	live shell differs: stop and report

Failure modes and timeouts

- playbook says Sunday 7:37am but runs are dated 24 Oct (Sat), 27 Oct (Tue), 31 Oct (Fri), 3 Nov (Tue)
- resolution yield fell from 614/777 TIDAL hits (24 Oct) to 6/81 (3 Nov) with no alert
- songnodes ingest failures are only printed as a 'note', never raised as an ALERT file

Beatport sync (assisted)

Objective: Check Beatport for tracks to buy, with Markus at the laptop.

Device: laptop tuf-boi Chrome with Markus present

Entry point: `ml-agent-beatport-sync`

Basis: Inferred from pages and docs

Routine

- read latest sync_plan.json, buy_list.csv, ledger beatport_check
- in Markus's Chrome add and remove tracks per action, pause every 25
- write resolved_<date>.json beatport_versions and sync_result_<date>.json
- run process and confirm

3 rows

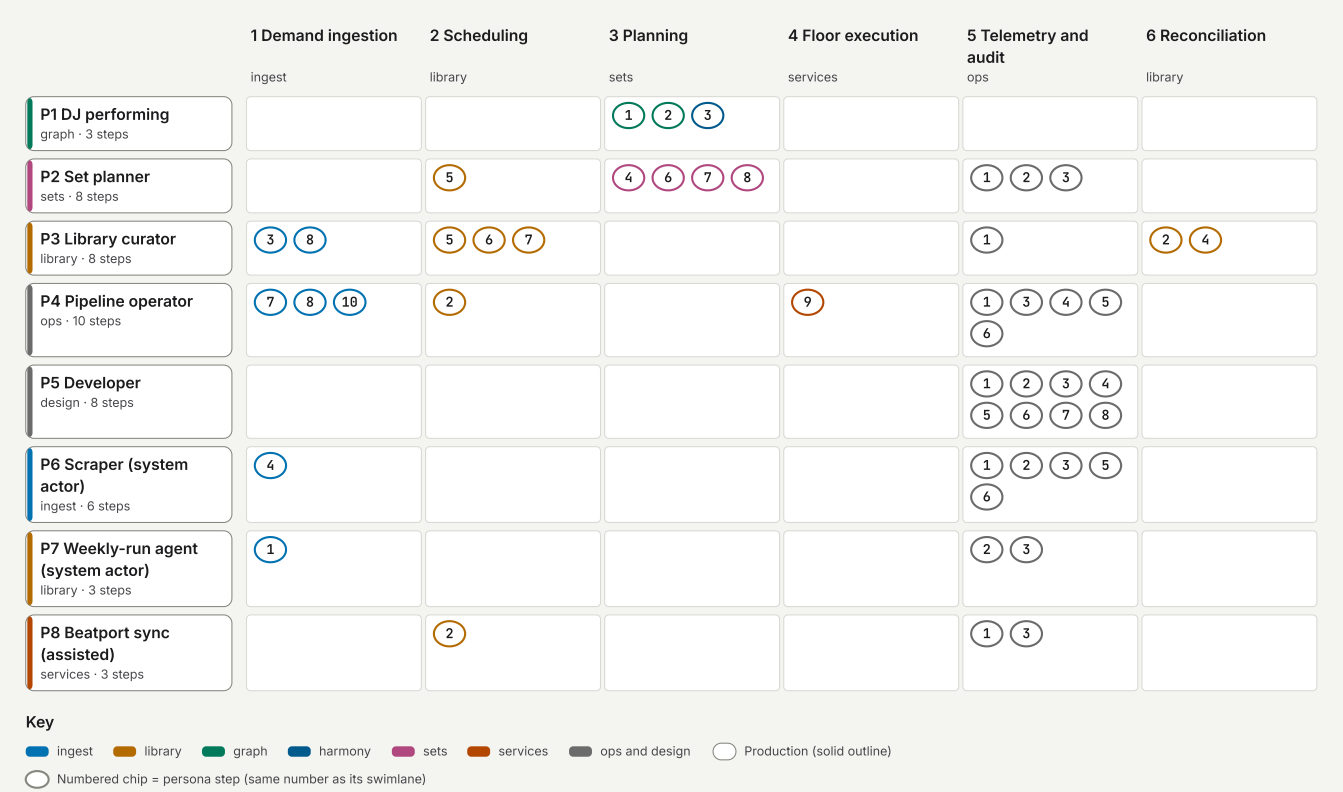
STEP	PAGE	ACTION	DATA WRITTEN	TRIGGER TO NEXT	HANDOFF
1	`ml-agent-beatport-sync`	playlist edits	Beatport playlists	Markus asks	{norm_key: {beatport_versions[{id,url,mix_name,duration_seconds,bpm,key,genre,label}]}} via inbox/resolved_<date>.json
2	`ml-cli-process`	re-gate short-only songs	ledger	after beatport_versions written	–
3	`ml-cli-confirm`	mark beatport synced	ledger platform_sync_state	after sync_result	–

Failure modes and timeouts

- writing resolved_<date>.json and sync_result_<date>.json for a date the weekly run already used overwrites TIDAL resolve data and the TIDAL sync result
- sync_plan only holds this run's diff, so skipped weeks are never caught up on Beatport (unlike TIDAL's full reconcile)
- 243 beatport_check songs waiting; no Beatport sync has run (0 tracks with beatport ids, no SYNCED_BOTH)

D5: **Persona overview**

8 personas take 49 steps; only 2 of them touch Reconciliation.



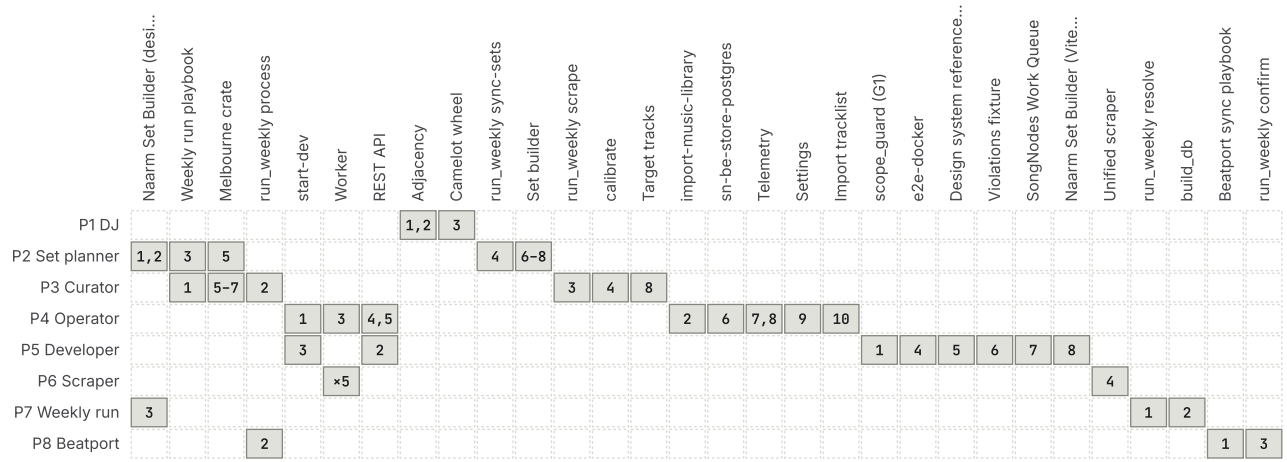
D5 · n = 49 · Unit: steps · ● Built

· Source: personas[] (steps[].page, action, writes); loop[]; pages[].domain · Method: hand-placed grid (lanes = personas, columns = loop stages); stage = page domain, override/calibrate/confirm steps = Reconciliation; 0 crossings

C9: **Personas by page, with step numbers**

8 personas take 49 steps over 30 pages; 7 pages serve more than one persona. No ease scores: findings.json holds no evidence for them.

Cell numbers are step numbers in that persona's journey; ×n = n steps (listed in the table)

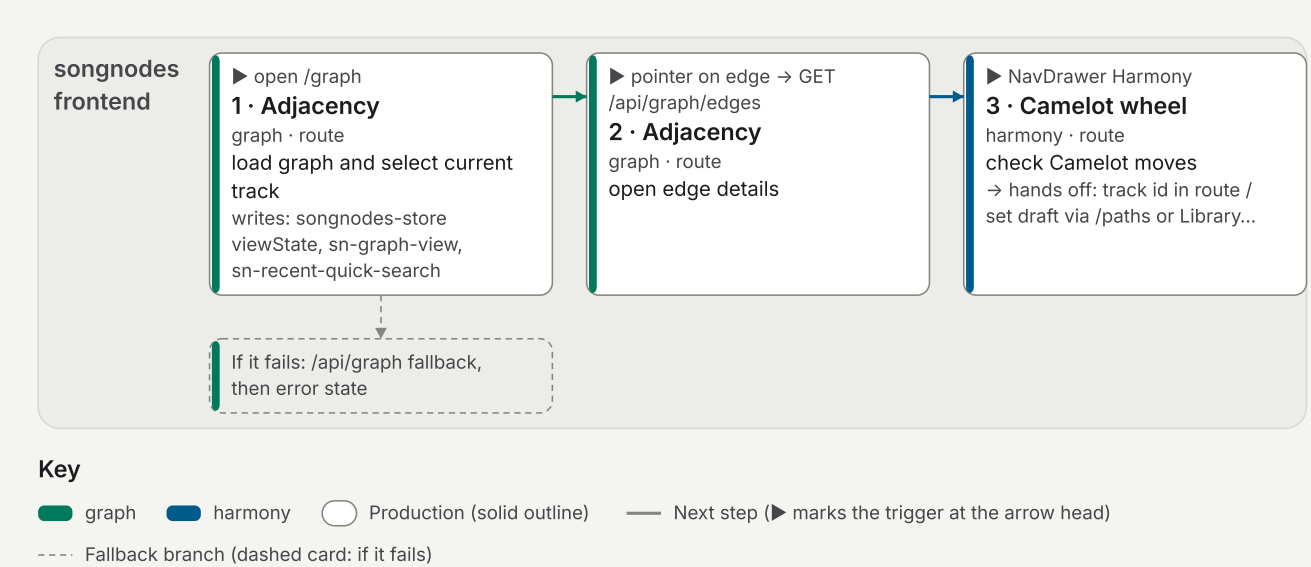


C9 · n = 49 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page) · Method: Each cell lists the step numbers a persona takes on a page; columns ordered by number of personas, then by first appearance. Ease scores omitted (no evidence).

D5-P1: **Swimlane – P1 DJ performing**

DJ performing: 3 steps across 1 lane; 1 step name a fallback.

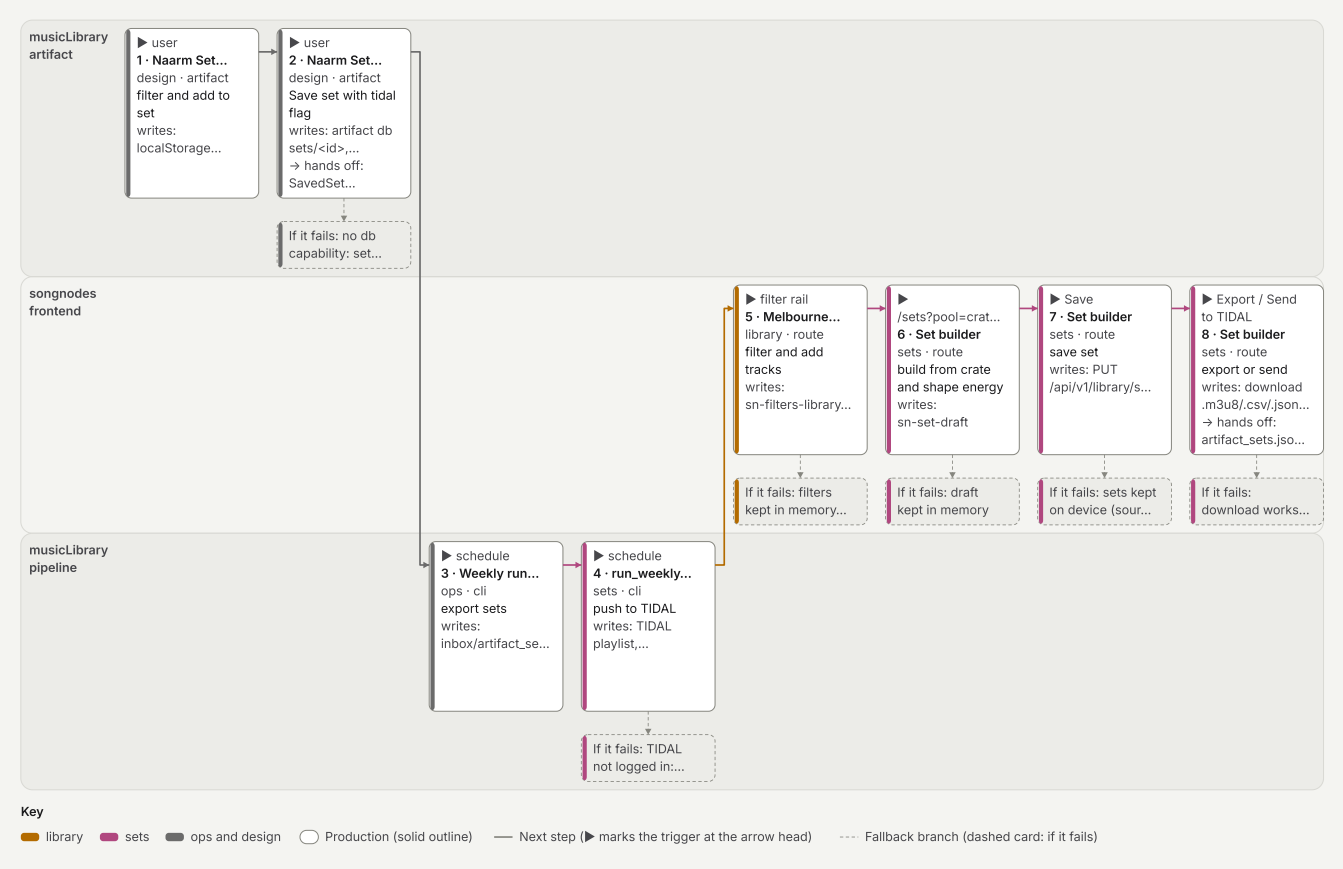


D5-P1 · n = 3 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 220); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P2: **Swimlane – P2 Set planner**

Set planner: 8 steps across 3 lanes; 6 steps name a fallback.

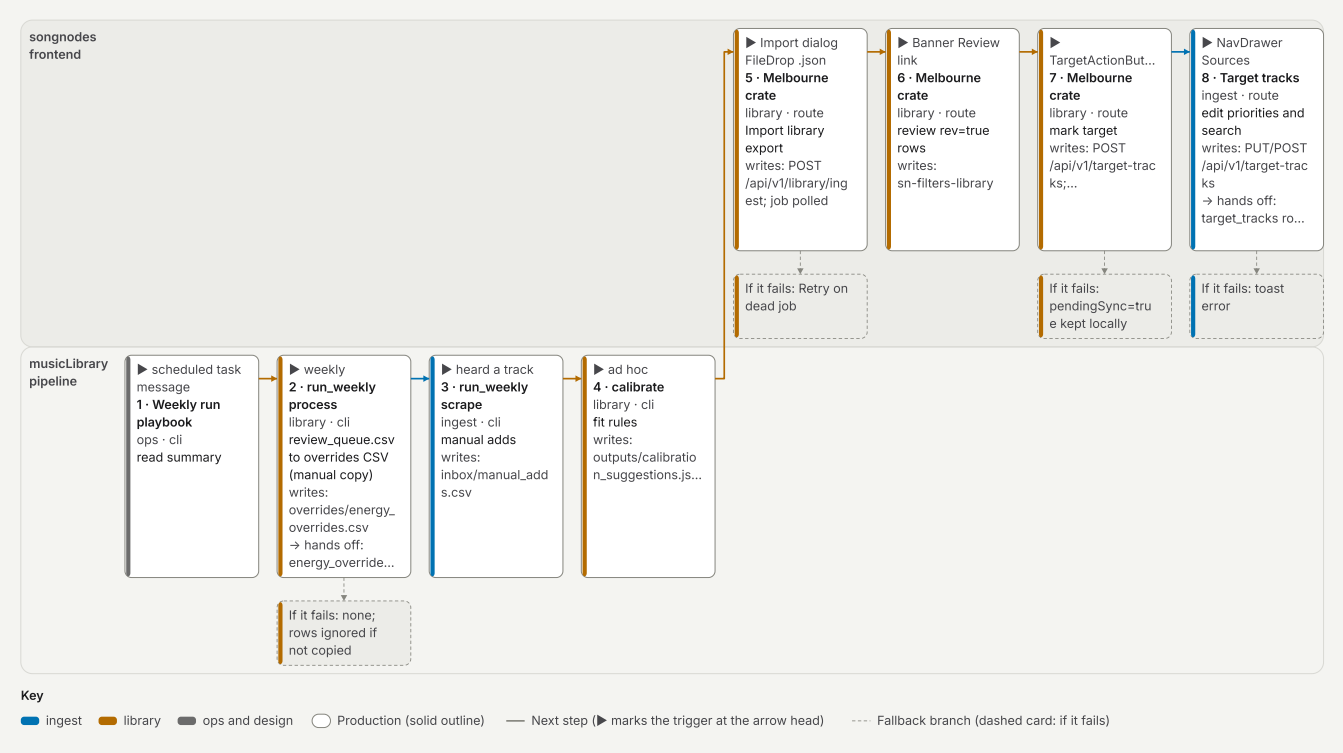


D5-P2 · n = 8 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P3: **Swimlane – P3 Library curator**

Library curator: 8 steps across 2 lanes; 4 steps name a fallback.

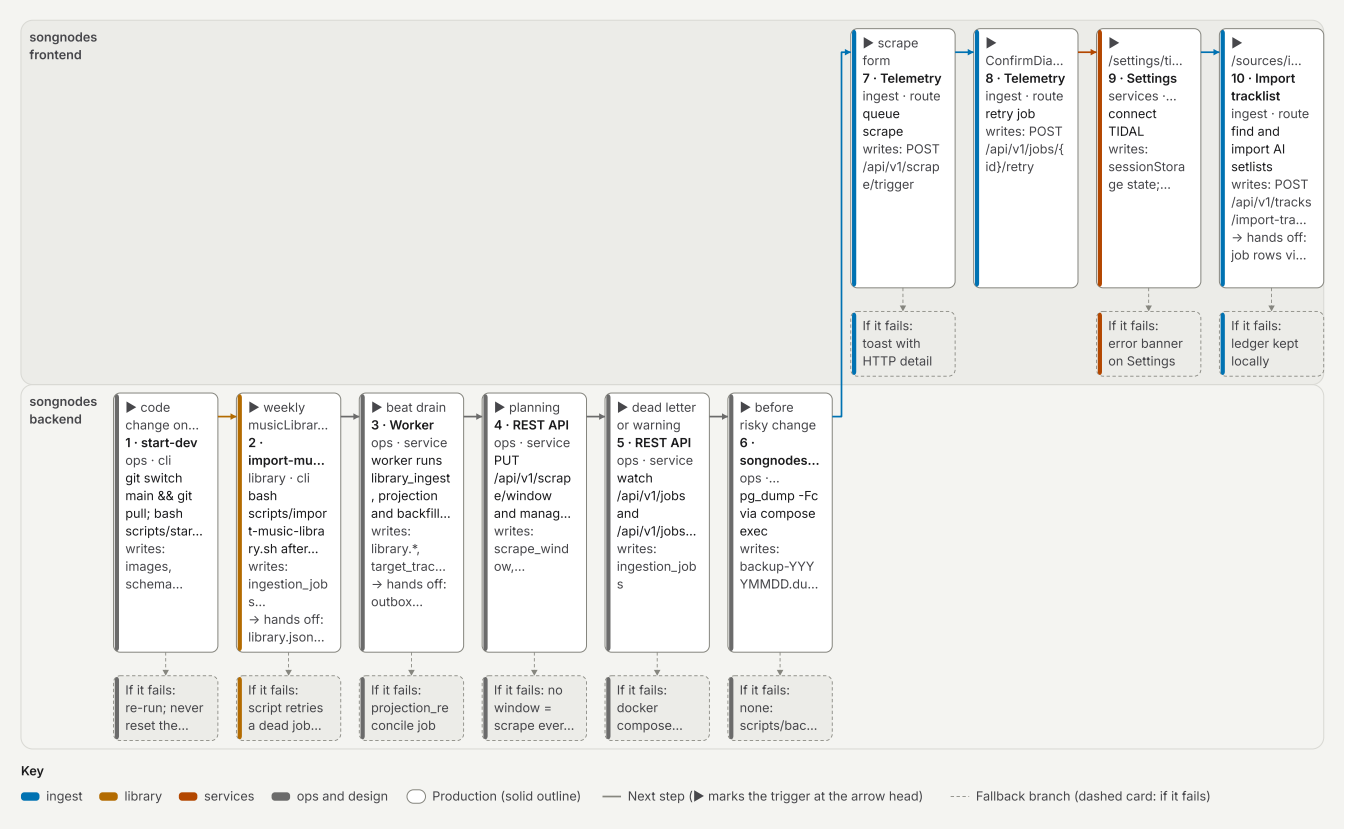


D5-P3 · n = 8 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P4: **Swimlane – P4 Pipeline operator**

Pipeline operator: 10 steps across 2 lanes; 9 steps name a fallback.

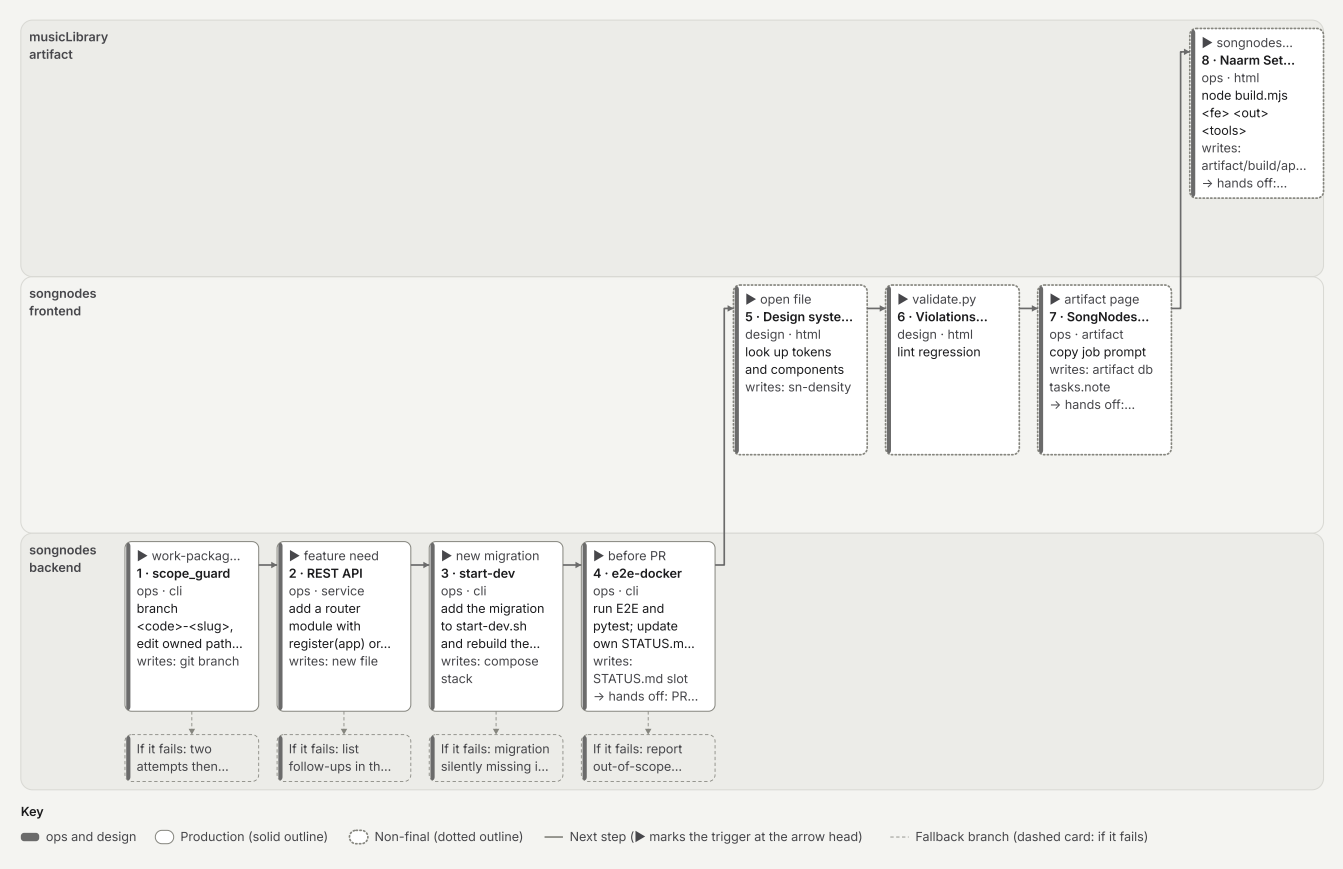


D5-P4 · n = 10 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 124); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P5: **Swimlane – P5 Developer**

Developer: 8 steps across 3 lanes; 4 steps name a fallback.

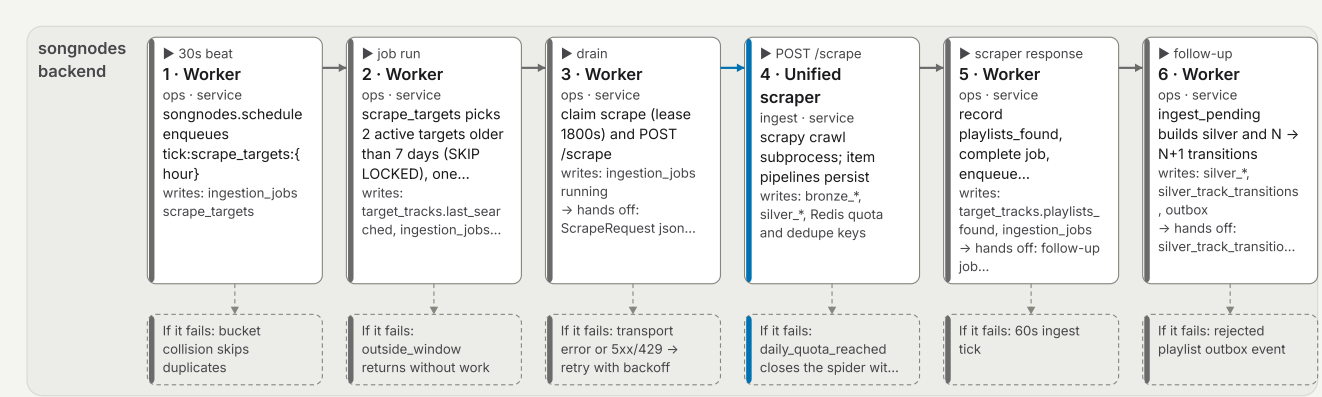


D5-P5 · n = 8 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P6: **Swimlane – P6 Scraper (system actor)**

Scraper (system actor): 6 steps across 1 lane; 6 steps name a fallback.

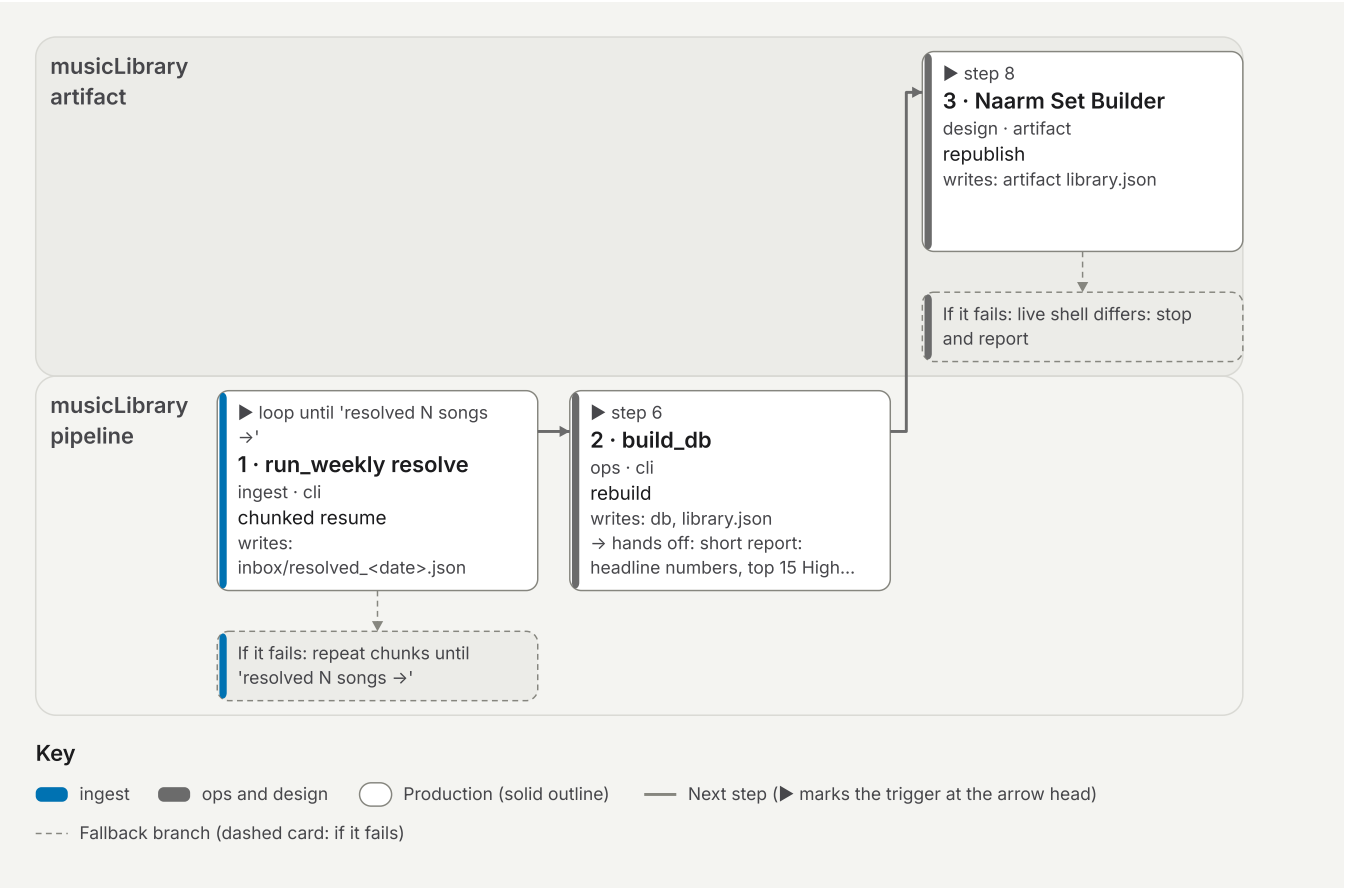


D5-P6 · n = 6 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 160); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P7: **Swimlane – P7 Weekly-run agent (system actor)**

Weekly-run agent (system actor): 3 steps across 2 lanes; 2 steps name a fallback.

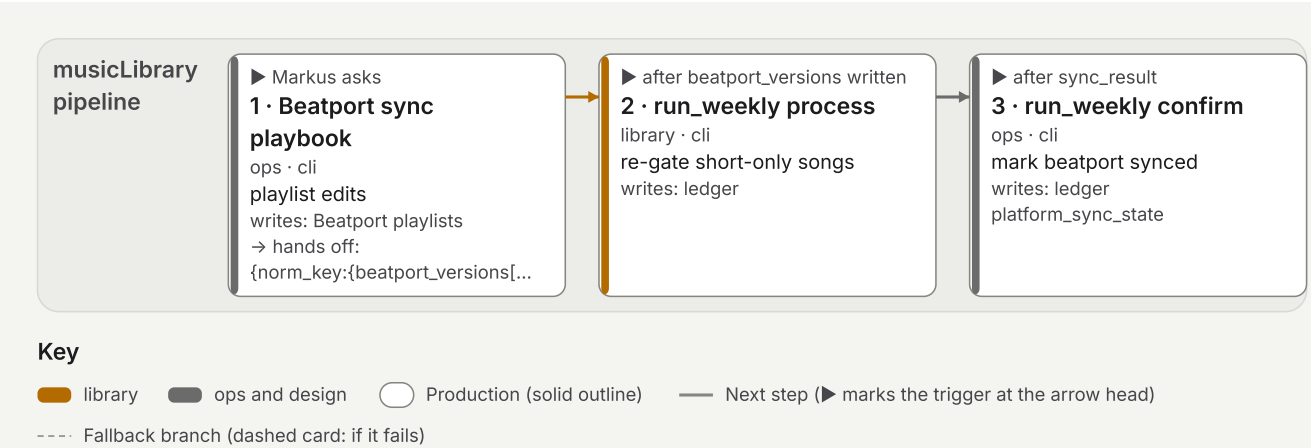


D5-P7 · n = 3 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 220); orthogonal arrows in the column gaps; 0 crossings by construction

D5-P8: **Swimlane – P8 Beatport sync (assisted)**

Beatport sync (assisted): 3 steps across 1 lane; 0 steps name a fallback.



D5-P8 · n = 3 · Unit: steps · ● Built

· Source: personas[].steps[] (n, page, action, writes, trigger, handoff, fallback); pages[].domain · Method: hand-placed swimlane (lanes = systems, columns = steps, card width 220); orthogonal arrows in the column gaps; 0 crossings by construction

Section 7: **Technical debt, risk and roadmap** 47 items

C10: **Risk matrix, severity by effort**

11 of 47 debt items are critical or high; 4 of those are small-effort fixes.

	Effort S (small)	Effort M (medium)	Effort L (large)
✕ Critical	0 items	1 item ml:no-bpm	1 item be:q-tunnel Q
▲ High	4 items ml:pending-rejected ml:resolve-yield be:open-admin fe:creds-localstorage	5 items ml:artifact-build be:backups-wrong be:daily-quota be:gold-operational be:scraper-silver	0 items
◖ Medium	7 items ml:idempotency-same ml:no-run ml:secrets-plain ml:songnodes-ingest be:auth-header be:migration-ledger be:worker-starvation	8 items be:default-encryption be:doc-drift be:library-events be:library-ingest be:long-transaction be:orphan-tables fe:api-ts fe:dead-graph	0 items
○ Low	17 items ml:build-charset ml:history-overwrite ml:legacy-page ml:schedule-drift ml:sqlite-copy ml:unused-std be:ingest-like be:memgraph-noop be:progress-polling be:route-shadowing be:set-delete be:two-setlist be:unmounted-router fe:poll-backoff fe:presets-only fe:test-key fe:tracker-hex	4 items be:q-sentinel Q fe:proxy-drift fe:source-label fe:unused-nginx	0 items

IDs: repo code (ml, be backend, fe frontend) and the first words of the debt id; Q = queued

C10 · n = 47 · Unit: debt items · i Projected

· Source: debt[] (severity, effort, state) · Method: Count of debt items per severity (ordinal rows) and effort S/M/L (ordinal columns), with each item's short id.

Table: **Technical debt, severity ranked** 47 items

47 rows

ID	SEVERITY	ITEM	KIND	EFFORT	DETAIL	STATE	EV
ml-debt-no-bpm-key	× Critical	**No track has a BPM or key, so two of the three energy methods and all Camelot logic are dead**	drift	M	TIDAL v2 returned 0 bpm/key across 1007 versions; Deezer BPM (38 versions) is only used as the cross-check 'other' and never copied onto the chosen TIDAL version. Result: ledger bpm=0/413, camelot_key=0/413, every track needs_review (v_review_queue 413), Set Builder warns that key and BPM moves are off (M3).	● Built	mu

sn-be-debt-q-tunnel-before-auth	× Critical	**Queued tunnel would publish open admin routes and a default admin/admin login**	security	L	PR #577 exposes the frontend (which proxies all /api/ to rest-api and graph-api) publicly; rest-api has no auth middleware on main (SONGNODES_API_TOKEN unset) and PR #578 adds a login router without any middleware gating other routes. PR #578 falls back to AUTH_USERNAME/AUTH_PASSWORD defaults of admin/admin and accepts plain-text password matches. Merging #577 before an enforced auth layer (Cloudflare Access or middleware) opens jobs, merges, API keys and AI proxy to the internet. cloudflared:latest is also an unpinned tag.	▲ Queued	PR
ml-debt-artifact-build-drift	▲ High	**Artifact bundle cannot be rebuilt from songnodes main**	drift	M	SetBuilderPage now imports readSavedSetsAsync and writeSavedSetsAsync and loads from /api/v1/library/search; artifact-data.ts exports neither, and the copy-patch text changed. Published app.js is the 24 Oct build on design system 1.1.0 (songnodes now 1.10.0).	● Built	mu
ml-debt-pending-rejected-dup	▲ High	**Rejected songs re-enter pending and stay there**	orphaned-data	S	ingest_raw setdefaults a pending entry for any replayed key even when it is in rejected, but queues it for resolve only at 2+ appearances. 297 of 442 pending keys are also in rejected with resolve_attempts 0; counts and the pending table are inflated.	● Built	mu
ml-debt-resolve-yield-silent	▲ High	**Resolve failures are silent**	other	S	Exceptions are printed and an empty result is saved and never retried; TIDAL hits fell 614/777 → 6/81 between 24 Oct and 3 Nov with no ALERT. resolve_log.txt exists only for 24 Oct and is empty.	● Built	mu
sn-be-debt-backups-wrong-target	▲ High	**Backup scripts target names that do not exist**	drift	M	scripts/backup_database.sh dumps -U musicdb_user musicdb; backup_volumes.sh tars songnodes_postgres_data etc. The compose project uses user/db songnodes and volume songnodes-dev_dev-postgres-18. Only the CLAUDE.md one-liner works.	● Built	so
sn-be-debt-daily-quota-vs-hourly-targets	▲ High	**1001tracklists daily Redis quota blocks hourly target scrapes**	race	M	scrape_targets enqueues a 1001tracklists job per target each hour; the 1001tracklists spider does not set force_run for search_query (MixesDB does), so after the first run in 24 h enforce_run_quota raises CloseSpider(daily_quota_reached), the job succeeds with found_nothing and the target is stamped last_searched for 7 days.	● Built	so
sn-be-debt-gold-operational-absent	▲ High	**Gold and operational layers do not exist in the running stack**	drift	M	CLAUDE.md describes Bronze → Silver → Gold → Operational; compose applies no gold migration (003) and runs no gold ETL; gold_track_graph is created but never read or refreshed; graph-api reads silver directly. Operational is effectively projection_tracks (library search) and silver reads.	● Built	so
sn-be-debt-open-admin	▲ High	**Write and admin REST routes open in compose**	security	S	SONGNODES_API_TOKEN is not passed in compose.dev.yaml, so register_security returns without a middleware. Port bound to 127.0.0.1 limits exposure to the laptop.	● Built	so

sn-be-debt-scraper-silver-layer-skip	▲ High	**Unified scraper writes silver tables directly, bypassing datalayer ingest**	race	M	PersistencePipeline maps tracks/playlists/playlist_tracks batches to silver_enriched_* and inserts with no ON CONFLICT on tracks, defaults artist to Unknown Artist and appends no outbox event. datalayer ingest_pending then creates its own silver rows for the same bronze playlist (lookup-then-insert, no unique key), so duplicates and Unknown Artist rows are possible and projection_tracks never hears of the scraper rows. Violates CLAUDE.md "no layer skipping" and artist attribution.	● Built	so
sn-fe-debt-creds-localstorage	▲ High	**OAuth tokens and client secrets persisted in localStorage and exposed on window.debugZustand**	security	S	songnodes-store partialize persists musicCredentials (access/refresh tokens, clientId/clientSecret); window.debugZustand.getMusicCredentials and injectTestCredentials ship in production builds. OAuthCallback deliberately moved tokens to sessionStorage, so the two disagree.	● Built	so
ml-debt-idempotency-same-length	◐ Medium	**sync-sets add uses an idempotency key that ignores content**	race	S	add_tracks key is uuid5(add:pid:<date>:<set id>:<len>:i). Re-running on the same date after editing a set to the same length reuses the key, so TIDAL may replay the earlier add. Remove-all-then-add also loses duplicates because adds use onDuplicates SKIP, so a set with a repeated track is rewritten every run.	● Built	mu
ml-debt-no-run-lock	◐ Medium	**No run lock between weekly run, Beatport agent and hand runs**	race	S	Ledger, resolved_<date>.json and sync_result_<date>.json are whole-file rewrites; the Beatport playbook writes the same date-keyed files the weekly run owns.	● Built	mu
ml-debt-secrets-plain	◐ Medium	**TIDAL tokens and ref_mb passwords are plain files on a synced drive**	security	S	secrets/*.json and ref_mb/secrets/*.txt; ref_mb/.gitignore covers secrets but musicLibrary has no git; deletes are blocked on the synced drive so pending PKCE files are blanked, not removed.	● Built	mu
ml-debt-songnodes-ingest-swallowed	◐ Medium	**songnodes ingest and setlist posts swallow errors**	other	S	cmd_sync_songnodes prints a 'note' on ingest failure and 'except Exception: pass' on setlists; it never polls the job, unlike import-music-library.sh.	● Built	mu
sn-be-debt-auth-header-collision	◐ Medium	**API token middleware reads Authorization Bearer before X-API-Token**	permission	S	TIDAL routes require Authorization: Bearer <tidal token>; when SONGNODES_API_TOKEN is set the middleware takes the TIDAL token as the API token and returns 401 even with a valid X-API-Token. SENSITIVE_GET_PREFIXES lists /api-keys but the route is /api/v1/api-keys, so GET of saved keys stays open.	● Built	so
sn-be-debt-default-encryption-key	◐ Medium	**Saved API keys encrypted with a hard-coded default key**	security	M	API_KEY_ENCRYPTION_SECRET is not set for rest-api or worker; store_api_key falls back to a hard-coded default key committed in the SQL file (value not reproduced here). OAuth tokens are plain text by design.	● Built	so

sn-be-debt-doc-drift	◐ Medium	**CLAUDE.md contracts disagree with code**	drift	M	Unified scraper body is {source, search_query, ...} not {spider_name, search_params}; transitions columns are from_track_id/to_track_id not source/target; bronze uses raw_json/scraped_at not raw_metadata; target_tracks not target_track_searches; enrichment does not delegate to metadata-enrichment:8020 (8020 is audio-analysis); Discogs env is DISCOGS_TOKEN in worker and DISCOGS_API_TOKEN for the scraper, which reads DISCOGS_TOKEN.	● Built	so
sn-be-debt-library-events-ignored	◐ Medium	**projection_tracks consumer ignores library_track events**	drift	M	import_library emits library.track.upserted.v1 / rotation_changed.v1 with aggregate_type library_track; projection.handle_event handles only track and artist, so library search reads stale projection rows until reconcile/rebuild. graph_projection does handle library_track.	● Built	so
sn-be-debt-library-ingest-contract	◐ Medium	**Library ingest does not meet its contract**	drift	M	Contract: same key + different body → 409, and per-row results in the response. Code: jobs.enqueue returns the original job for any body and the endpoint returns only {job_id,status,created}. GET /api/v1/library/tracks/{id} is documented but only /preview exists.	● Built	so
sn-be-debt-long-transaction-scrape	◐ Medium	**Scrape handler holds a DB transaction open for up to 16 minutes**	race	M	runner.run_one wraps the handler in conn.transaction(); the scrape handler awaits a synchronous POST /scrape (timeout 960s) inside it, leaving an idle-in-transaction session and blocking pg_snapshot_xmin, which also stalls outbox consumers (events from newer transactions stay invisible while this xmin is held).	● Built	so
sn-be-debt-migration-ledger	◐ Medium	**No migration ledger in compose; start-dev.sh is the list**	drift	S	Migrations run by hand-maintained psql lines with ad-hoc sentinels; 000 schema_migrations, 003-005/007-008 medallion, 001-011 and 250 are not applied; worker migrate.py only knows 012-014.	● Built	so
sn-be-debt-orphan-tables	◐ Medium	**Tables written but never read, or read but never written**	orphaned-data	M	web_call_log (written, no reader); gold_track_graph (created, no reader); musicdb.user_setlists (written by graph-api, no reader, schema absent); song_adjacency (written by scraper, table absent); bronze_api_enrichments (neither); target_track_searches (CLAUDE.md says scraper reads it; only legacy SQL writes it); library.set.saved.v1 and setlist ingested/rejected outbox events have no consumer.	● Built	so
sn-be-debt-worker-starvation	◐ Medium	**Two worker processes shared by 15-minute scrapes and 15-second projections**	other	S	Worker runs --concurrency=2 with beat in-process; two queued scrapes occupy both processes and projection/ingest ticks wait (their Celery tasks expire after 15s).	● Built	so
sn-fe-debt-api-ts-dead	◐ Medium	**services/api.ts declares many clients for endpoints that do not exist**	drift	M	searchApi (/api/v1/search/*; backend has /api/search/tracks), pathfindingApi (/api/v1/pathfinding/*; backend is /api/v1/pathfinder/find-path), setlistApi POST/DELETE/export, /api/v1/tracks/batch, /api/graph/tracks, /api/graph/positions have no backend route and no caller.	● Built	so

sn-fe-debt-dead-graph-modules	◐ Medium	**Unreachable modules still carrying endpoints**	orphaned-data	M	Import-graph walk from main.tsx: 50 of 184 non-test modules unreachable, including LiveDataLoader (/api/ws WebSocket), EditTrackModal (PATCH /api/tracks/{id}), NowPlayingDeck (trackApi preview), AIAssistantModal (songnodes_ai_prompt_history), StatsPanel (stats JSON/CSV download), PreferencesPanel, AITransitionHelper, useLiveDataLoader, useKeyboardShortcuts.	● Built	/h
ml-debt-build-charset	i Low	**Build shell has no charset**	other	S	artifact/build/index.html has no <meta charset>; rendered outside the claude.ai wrapper the middle dot and apostrophes become mojibake (M3: 'Â·'). Fine only while the platform wrapper adds a charset.	● Built	mu
ml-debt-history-overwrite	i Low	**Ledger snapshot is overwritten on a same-day re-run**	other	S	ledger_before_<date>.json is copied on every process save, so a second process on the same date replaces the real pre-run snapshot with the first run's output.	● Built	mu
ml-debt-legacy-page-ds	i Low	**Legacy artifact/index.html breaks the design system**	drift	S	48 hex colours, 181 px values, own Google fonts and tokens, 0 sn- classes; README still calls it the page source. The published build (app.js) has 0 hex and uses songnodes tokens.	● Built	mu
ml-debt-schedule-drift	i Low	**Run cadence does not match the documented schedule**	drift	S	Documented Sunday 7:37am; outputs exist for 24 Oct, 27 Oct, 31 Oct and 3 Nov (none on a Sunday), so the 8-day window overlaps and ARIA weeks are re-counted.	● Built	mu
ml-debt-sqlite-copy	i Low	**SQLite swap is a copy, not an atomic rename**	other	S	shutil.copyfile over db/dj_library.sqlite; an open DB Browser session can see a half-copied file. Docstring claims an atomic swap.	● Built	mu
ml-debt-unused-std-cap	i Low	**std_crate_cap is configured but never enforced**	drift	S	rules.rotation.std_crate_cap=250 has no reader; only high_crate_cap is applied.	● Built	mu
sn-be-debt-ingest-like-scan	i Low	**already_linked check scans transitions with LIKE on jsonb text**	other	S	Re-ingest of an existing playlist runs playlist_occurrences::text LIKE %id% across silver_track_transitions (no index), cost grows with the graph.	● Built	so
sn-be-debt-memgraph-noop	i Low	**graph_projection consumer runs every 15 s against a store that does not exist**	orphaned-data	S	No memgraph service or MEMGRAPH_URI in compose; handle_event returns immediately but offsets advance, so enabling Memgraph later needs a rebuild.	● Built	so
sn-be-debt-progress-polling	i Low	**Job stages poll in-memory scraper progress**	polling	S	Frontend polls GET /api/v1/jobs/{id}/stages, which polls GET /scrape/progress/{job_id} with a 2s timeout; progress lives in process memory and is lost on scraper restart.	● Built	so
sn-be-debt-q-sentinel-dead-route	i Low	**Queued PR #580 sanitises errors on routes no client can reach**	drift	M	PR #580 changes rest-api main.py /api/graph/nodes and /api/graph/edges error details; nginx sends /api/graph to graph-api, so those handlers are shadowed. The music-search change is effective.	▲ Queued	PR
sn-be-debt-route-shadowing	i Low	**Duplicate /api/graph/nodes and /api/graph/edges in rest-api are unreachable**	drift	S	nginx sends ^/api/graph to graph-api, so rest-api main.py versions never receive frontend traffic; two implementations can diverge.	● Built	so
sn-be-debt-set-delete-no-event	i Low	**Deleting a saved set emits no outbox event**	drift	S	save_set appends library.set.saved.v1; delete_set does not, so any projection of sets cannot remove it.	● Built	so
sn-be-debt-two-setlist-stores	i Low	**Two saved-setlist stores**	orphaned-data	S	graph-api POST /api/user-setlists (musicdb.user_setlists) and rest-api library.sets coexist; only library.sets has a reader.	● Built	so

sn-be-debt-unmounted-router	i Low	**data_quality router never mounted**	orphaned-data	S	No register(app) and not included by main.py; router_registry skips it by AST check. Its routes are also commented out.	● Built	so
sn-fe-debt-poll-backoff	i Low	**Pipeline poll backoff reads a stale error value**	polling	S	startPolling(error ? 30000 : 10000) captures error from the first render inside useEffect, so the 30 s backoff never engages.	● Built	so
sn-fe-debt-presets-only-cross-tab	i Low	**Only filter presets react to other tabs**	race	S	sn-set-draft and sn-music-library-sets are written by Library and Set builder without storage-event listeners; two tabs overwrite each other's draft (last write wins).	● Built	so 45
sn-fe-debt-proxy-drift	i Low	**Vite dev proxy and nginx route graph traffic differently**	drift	M	nginx sends /api/(graph|v1/visualization|user-setlists) to graph-api; Vite sends only /api/graph and /api/v1/graph (rewritten) to 8084, so /api/v1/visualization and /api/user-setlists hit rest-api in dev. rest-api also defines /api/graph/nodes and /api/graph/edges that nginx never reaches.	● Built	so
sn-fe-debt-source-label	i Low	**Library rows always labelled 'library-fixture'**	drift	M	fromLibraryRow sets source 'library-fixture' on every row although loadLibrary only reads the backend (_source 'backend'); CamelotPage shows 'Music library'. DB1 brief planned the fixture fallback; current code has neither the fallback nor public/data/music-library.json.	● Built	so
sn-fe-debt-test-key-drift	i Low	**registry.test seeds 'sn-saved-sets' but code reads 'sn-music-library-sets'**	drift	S	Test mock storage key does not match SAVED_SETS_KEY, so the test never exercises saved sets on Home.	● Built	so
sn-fe-debt-tracker-hex	i Low	**Work-queue tracker uses its own hex palette**	drift	S	plan/work-queue-2026-10-30/tracker.html defines --bg:#f6f5f9 etc. instead of design-system tokens, against CLAUDE.md §4.	● Built	so
sn-fe-debt-unused-nginx	i Low	**nginx/ and nginx-local-proxy.conf are not used by the Compose stack**	drift	M	compose.dev.yaml builds the frontend with an inline default.conf; nginx/conf.d/frontend.conf proxies to api-gateway:8080 and websocket-api:8083, nginx-local-proxy.conf to host.docker.internal:8088 — none exist in compose.dev.yaml. Design-system docs (13-page-patterns, 16-platform) still cite nginx/nginx.conf for error pages and frame headers.	● Built	so

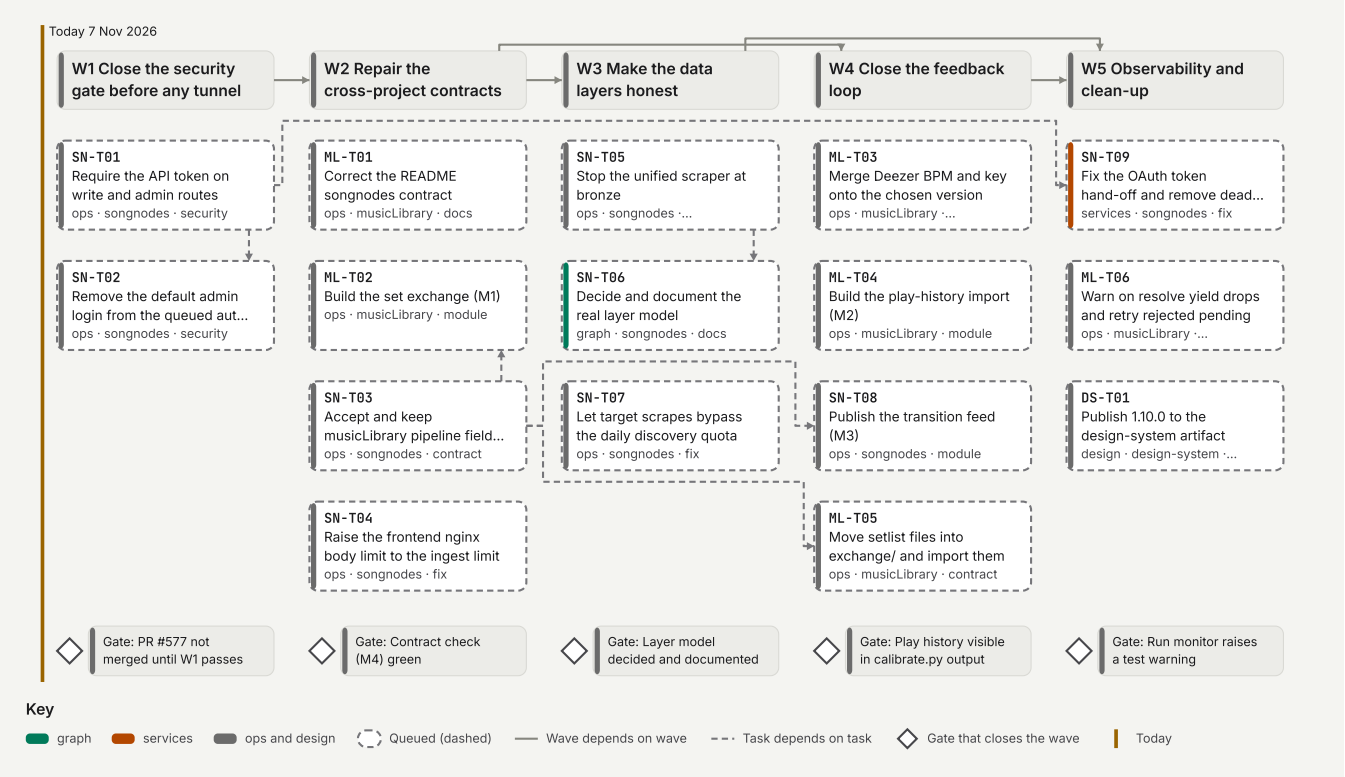
Table: **Roadmap waves** `5 waves`

5 rows

WAVE	GOAL	TASKS	GATE	DEPENDS ON
W1	**Close the security gate before any tunnel**	SN-T01, SN-T02	PR #577 not merged until W1 passes	–
W2	**Repair the cross-project contracts**	ML-T01, ML-T02, SN-T03, SN-T04	Contract check (M4) green	W1
W3	**Make the data layers honest**	SN-T05, SN-T06, SN-T07	Layer model decided and documented	W2
W4	**Close the feedback loop**	ML-T03, ML-T04, SN-T08, ML-T05	Play history visible in calibrate.py output	W2
W5	**Observability and clean-up**	SN-T09, ML-T06, DS-T01	Run monitor raises a test warning	W3, W4

D11: **Roadmap timeline**

5 waves and 16 tasks; the security gate in W1 comes before any tunnel.



D11 · n = 16 · Unit: tasks · ▲ Queued

· Source: roadmap[] (tasks, gate, depends); tasks[] (project, type, depends, gaps); meta.ml_snapshot (Today) · Method: hand-placed columns, dependency arrows routed through row corridors; 1 crossings

Section 8: **Gaps, proposed modules and storage backbone** 28 gaps · 5 modules

Gaps: **Critical data gaps** 28 gaps

Gap: **README contract names a songnodes script that no longer exists**

ml-gap-sync-script-missing · ◐ Medium · ● Built

README and .gitignore say scripts/sync-music-library.sh copies library.json to frontend/public/data/music-library.json. That script and folder are gone; songnodes now loads /api/v1/library/search fed by POST /api/v1/library/ingest (import-music-library.sh, and cmd_sync_songnodes inside cmd_db).

Fix	Update README 'In SongNodes' section to name scripts/import-music-library.sh and the run_weekly db step; drop the .gitignore lines for music-library.json in songnodes.
Data contract	`library.json v1 (build_db.export_artifact_json) → POST /api/v1/library/ingest, Idempotency-Key import-<md5>`
Affects	musicLibrary/README.md, songnodes/.gitignore

Evidence 3

```
musicLibrary/README.md
songnodes/.gitignore:251-253
songnodes/scripts/import-music-library.sh
```

Gap: **Saved-set shape is no longer shared between the two Set Builders**

ml-gap-saved-set-shape · × Critical · ● Built

musicLibrary sets carry tidal, shape, minutes, mix, pool, seed and ledger track ids (h:/bp:). songnodes stores only name, description and uuid track ids server-side and has no tidal flag. A songnodes 'Export sets for the weekly run' file imports into sets/set_tracks with uuid ids that match no ledger track, and sync-sets skips every set (no tidal). /api/v1/library/sets/export returns {sets:[...]} without ids, which load_sets turns into a list of lists and crashes on.

Fix	Version the document: add schema 'naarm.saved-set.v2' {id, name, shape, minutes, mix, pool, seed, tidal, tracks[{ml_id, sn_id, tgt, move}]}; songnodes stores the extra fields in library.sets (jsonb) and maps uuid→source_id via library.track_xref on export; load_sets accepts {sets:[...]} and resolves ids via xref.
Data contract	`naarm.saved-set.v2`
Affects	musicLibrary/pipeline/build_db.py, musicLibrary/pipeline/run_weekly.py, songnodes/frontend/src/features/musicLibrary/data.ts, songnodes/services/library/model/library_db.py

Evidence 4

```
musicLibrary/pipeline/build_db.py:93-106
musicLibrary/pipeline/run_weekly.py:529-547
songnodes/frontend/src/features/musicLibrary/data.ts:140-161
songnodes/services/library/model/library_db.py:50-62
```

Gap: **JohnCourse.json and kiki.json are written but never consumed**

`ml-gap-setlists-orphan` · × Critical · ● Built

Both are songnodes.setlists.v1 DJ tracklists (29 and 5 sets) at the musicLibrary root. cmd_sync_songnodes globs only the parent folder and inbox/, so they are never found; if found, the body {schema, setlists} does not match TracklistImportRequest {tracks, dj_artist, name} and the 422 is swallowed. JohnCourse.json.bak is invalid JSON.

Fix	Move them to inbox/setlists/; post one request per setlist as {name: event, dj_artist: dj, tracks:[{artist,title,position,...}]}; log failures to outputs/<date>/ALERT_songnodes.txt.
Data contract	`songnodes.setlists.v1 → TracklistImportRequest`
Affects	musicLibrary/pipeline/run_weekly.py, musicLibrary root JSON

Evidence 4

```
musicLibrary/pipeline/run_weekly.py:599-616
songnodes/services/rest_api/main.py:96-100
musicLibrary/JohnCourse.json
musicLibrary/kiki.json
```

Gap: **Rekordbox play history and overrides do not feed back automatically**

`ml-gap-rekordbox-feedback` · ◐ Medium · ● Built

Play history reaches only calibrate.py, and only from a hand-dropped overrides/rekordbox_collection.xml (absent); calibrate writes suggestions only and has never run. local_path must be typed into overrides, so the XML export holds 0 tracks and the buy list never shrinks.

Fix	Parse rekordbox_collection.xml in process: match by Location/Artist+Title to set local_path and last_played, add a play appearance (source 'Rekordbox') so rotation sees real play; keep calibrate as the suggestion step.
Data contract	`rekordbox DJ_PLAYLISTS TRACK{Location, Artist, Name, PlayCount, LastPlayed, DateAdded} → ledger local_path,`
Affects	musicLibrary/pipeline/run_weekly.py, musicLibrary/pipeline/calibrate.py

Evidence 2

```
musicLibrary/pipeline/calibrate.py:55-83
musicLibrary/pipeline/djlib/export.py:356-358
```

Gap: **songnodes proven transitions never inform musicLibrary set building**

`ml-gap-transitions-not-used` · ◐ Medium · ● Built

Direction musicLibrary → songnodes exists (library.json ingest; load-library-targets.sql makes library tracks scrape targets). Nothing flows back: library.json and the artifact Set Builder have no transition data, and the songnodes /sets page does not call the graph either.

Fix	Add GET /api/v1/library/transitions?ids=... returning occurrence_count for library-track pairs; have build_db fetch it (optional) and add 'nx' (proven next track ids with counts) to library.json tracks; Set Builder ranks spares by it.
Data contract	`library.json tracks[].nx: [{id, n}]`
Affects	songnodes graph-api, musicLibrary/pipeline/build_db.py, Set Builder pages

Evidence 2

```
songnodes/scripts/load-library-targets.sql
musicLibrary/pipeline/build_db.py:190-229
```

Gap: **Second-source BPM and key are discarded**

`ml-gap-deezer-bpm-not-merged` · × Critical · ● Built

When the chosen version is TIDAL, Deezer's BPM is used only for the cross-check; ref_mb is not queried by the pipeline at all despite the 2026-10-25 decision to cut web calls with it.

Fix	In process, when v.bpm is null take the Deezer twin's bpm (fold to pillar range) and record bpm_source; add an optional ref_mb lookup by ISRC for key/label/release date.
Data contract	`ledger track bpm_source, key_source`
Affects	musicLibrary/pipeline/run_weekly.py

Evidence 2

```
musicLibrary/pipeline/run_weekly.py:226-230
musicLibrary/decisions/2026-10-25-metabrainz-integration-
baseline.md
```

Gap: **Outputs with no reader**

`ml-gap-written-never-read` · × Critical · ● Built

outputs/crates/*.csv, rejected.csv, ledger/history/*, db/dj_library.sqlite (only DB Browser), tests/reports/* (songnodes output) and the review_queue override column have no code reader; ledger beatport_check (243) is read only by the supervised Beatport playbook, which has never run.

Fix	Either delete the step or give it a consumer: have process read the newest review_queue.csv override column into energy_overrides; move tests/reports to songnodes.
Data contract	`review_queue.csv energy_score_override → overrides`
Affects	musicLibrary/outputs, musicLibrary/tests

Evidence 2

```
musicLibrary/pipeline/djlib/export.py:285-305
musicLibrary/pipeline/run_weekly.py:298-308
```

Gap: **TIDAL login collides with songnodes on 127.0.0.1:3006**

`ml-gap-tidal-port` · ◐ Medium · ● Built

Both use the same TIDAL app and redirect; songnodes' /oauth/callback route consumes the single-use code if the frontend is up. Two TIDAL send paths exist (sync-sets 'SET · name' and songnodes /api/v1/tidal/sets/send) with separate token stores and no shared playlist map.

Fix	Register a second redirect (http://127.0.0.1:3016/oauth/callback) on the TIDAL app for musicLibrary, or reuse songnodes' stored token through an endpoint; keep one playlist map keyed by set id.
Data contract	`TIDAL redirect per client; shared set→playlist map`
Affects	musicLibrary/pipeline/djlib/tidal.py, songnodes music_auth / tidal_playlists

Evidence 3

```
musicLibrary/pipeline/djlib/tidal.py:32
songnodes/frontend/src/App.tsx:33
songnodes/services/rest_api/routers/tidal_playlists.py:318
```

Gap: **songnodes drops pipeline fields on ingest**

`ml-gap-songnodes-field-loss` · ◐ Medium · ● Built

library_search returns bp NULL, generated null, empty sources and runs, ids as uuids; rev/ovr/local/sync come back only if mapped. The songnodes Library page therefore cannot show buy links or run history that the artifact shows.

Fix	Persist export extras (bp, sync, rev, ovr, local) from _from_export 'export' into evidence and select them in library_search; return generated and runs from library.runs.
Data contract	`LibraryTrack parity between library.json and /api/v1/library/search`
Affects	songnodes/services/rest_api/routers/library_search.py

Evidence 2

```
songnodes/services/rest_api/routers/library_search.py
songnodes/frontend/src/features/musicLibrary/data.ts:121-129
```

Gap: **Scraper must stop at bronze**

`sn-be-gap-scraper-bronze-only` · ▲ High · ● Built

The unified scraper writes silver directly (layer skip, Unknown Artist, no outbox).

Fix	Remove the tracks/playlists/playlist_tracks/song_adjacency branches from PersistencePipeline._flush_batch (keep bronze_* only) and rely on the ingest_pending follow-up; add a unique index on silver_enriched_tracks (bronze_id) or (lower(artist_name), lower(track_title)) and convert upsert_track to ON CONFLICT.
Data contract	`Scraper output = bronze_scraped_playlists + bronze_scraped_tracks with positions 1..n; silver is written only`
Affects	songnodes/scrapers/pipelines, songnodes/services/common/datalayer

Evidence 2

```
songnodes/scrapers/pipelines/persistence_pipeline.py:910
songnodes/services/common/datalayer/ingest.py:104
```

Gap: **Target scrapes must bypass the daily discovery quota**

`sn-be-gap-target-scrape-quota` · ▲ High · ● Built

1001tracklists targeted searches are blocked by scraped:setlists:1001tracklists:last_run.

Fix	Set self.force_run = True in OneThousandOneTracklistsSpider.__init__ when search_query is given (as MixesDB does at line 116), or have the handler pass force_run=1 via params; and only stamp last_searched / found_nothing when the spider did not close with daily_quota_reached (return a reason from /scrape).
Data contract	`POST /scrape response gains close_reason; handler treats daily_quota_reached as RetryableError.`
Affects	songnodes/scrapers/spiders, songnodes/services/common/datalayer/handlers.py

Evidence 2

```
songnodes/scrapers/spiders/1001tracklists_spider.py:189
songnodes/services/common/datalayer/handlers.py:85
```

Gap: **Run the scrape HTTP call outside the job transaction**

`sn-be-gap-scrape-outside-tx` · ◐ Medium · ● Built

A 16-minute open transaction holds xmin and stalls the outbox.

Fix — In runner.run_one, let handlers declare long_io=True; for scrape, perform the HTTP POST before opening conn.transaction(), then record_playlists_found + follow-ups + complete inside a short transaction; lease (1800s) still guards double-run.

Data contract — `Handler contract: (prepare outside tx) → (commit inside tx).`

Affects — songnodes/services/common/datalayer/runner.py, songnodes/services/common/datalayer/handlers.py

Evidence 1

```
songnodes/services/common/datalayer/runner.py:37
```

Gap: **Decide and document the real layer model**

`sn-be-gap-medallion-doc` · ▲ High · ● Built

Gold/operational exist only in docs and unused Helm cronjobs.

Fix — Either (a) update CLAUDE.md to Bronze → Silver → projection_tracks/graph reads, drop gold_track_graph, or (b) add a gold job kind (REFRESH MATERIALIZED VIEW CONCURRENTLY gold_track_graph after ingest, apply medallion 008) and point graph-api at it. Record the choice in docs/decisions.

Data contract — `Layer contract table: layer → table → sole writer → readers.`

Affects — songnodes/CLAUDE.md, songnodes/services/graph-visualization-api

Evidence 2

```
songnodes/CLAUDE.md:11
songnodes/sql/migrations/medallion/008_gold_track_graph_index_up.sql:3
```

Gap: **Make API token usable by scripts and TIDAL routes**

`sn-be-gap-auth-token-flow` · × Critical · ● Built

Token middleware collides with TIDAL Bearer; import script sends no token; prefix typo. Queued PRs #578/#579 add a separate user login (cookie) but no enforcement on existing routes.

Fix — In security/auth.py prefer X-API-Token over Authorization, fix the prefix to /api/v1/api-keys and add /api/v1/music-auth/spotify/token/status; pass SONGNODES_API_TOKEN and API_KEY_ENCRYPTION_SECRET through compose.dev.yaml; add -H "X-API-Token: $SONGNODES_API_TOKEN" to import-music-library.sh curl calls.

Data contract — `X-API-Token header is the only admin credential; Authorization is reserved for provider tokens.`

Affects — songnodes/services/rest_api/security, songnodes/compose.dev.yaml, songnodes/scripts/import-music-library.sh

Evidence 2

```
songnodes/services/rest_api/security/auth.py:58
songnodes/scripts/import-music-library.sh:45
```

Gap: **Project library_track events into projection_tracks**

`sn-be-gap-library-projection` · ◐ Medium · ● Built

Library search reads stale rows after a library ingest.

Fix	In projection.handle_event treat aggregate_type library_track like track (project_track on aggregate_id), and add a consumer or explicit ignore list for library_set and setlist events.
Data contract	`Outbox event registry: event_type → consumers (projection_tracks, graph_projection).`
Affects	songnodes/services/common/datalayer/projection.py

Evidence 1

```
songnodes/services/common/datalayer/projection.py:75
```

Gap: **Working backup for the compose volume**

`sn-be-gap-backups` · ▲ High · ● Built

No script backs up songnodes-dev_dev-postgres-18.

Fix	Rewrite scripts/backup_database.sh to docker compose -f compose.dev.yaml exec -T postgres pg_dump -U songnodes -Fc songnodes > backups/songnodes-$(date +%Y%m%d).dump with retention, and retire backup_volumes.sh.
Data contract	`Backup file: backups/songnodes-YYYYMMDD.dump (pg custom format).`
Affects	songnodes/scripts

Evidence 1

```
songnodes/scripts/backup_database.sh:52
```

Gap: **Single migration runner with a ledger**

`sn-be-gap-migration-ledger` · ◐ Medium · ● Built

Migrations are a hand list in start-dev.sh.

Fix	Apply 000_schema_migrations_up.sql and extend services/worker/migrate.py to apply every sql/migrations/*_up.sql in numeric order once, recording filename+checksum; start-dev.sh calls docker compose run --rm worker python migrate.py.
Data contract	`schema_migrations(version, checksum, applied_at).`
Affects	songnodes/services/worker/migrate.py, songnodes/scripts/start-dev.sh

Evidence 2

```
songnodes/services/worker/migrate.py:1
songnodes/scripts/start-dev.sh:31
```

Gap: **Enforce the library ingest body contract**

`sn-be-gap-library-ingest-409` · Medium · Built

Same key + different body silently returns the old job.

Fix	Store sha256 of the canonical JSON payload in ingestion_jobs.payload_hash (or compare payload) in jobs.enqueue for library_ingest and raise CatalogError(409) on mismatch.
Data contract	`Idempotency-Key + payload hash.`
Affects	songnodes/services/common/datalayer/jobs.py, songnodes/services/rest_api/routers/library_ingest.py

Evidence 1

`songnodes/contracts/library_ingest.md:16`

Gap: **Music search has no deployed backend**

`sn-be-gap-music-search` · Medium · Built

rest-api proxies to metadata-enrichment:8086, not in compose.

Fix	Move the S1 search router into rest-api using common/providers (tidal, deezer) or add metadata-enrichment to compose; until then return an explicit 503 with cause in the UI.
Data contract	`GET /api/v1/music-search/{service}?q= → [{id, artist, title, isrc}].`
Affects	songnodes/services/rest_api

Evidence 1

`songnodes/services/rest_api/main.py:692`

Gap: **Frontend calls routes that do not exist**

`sn-be-gap-frontend-dead-routes` · Medium · Built

DJInterface posts /api/v1/scrapers/target-tracks/search and /api/v1/scrapers/queue/clear.

Fix	Point DJInterface at POST /api/v1/target-tracks/{id}/search and drop queue/clear (no queue endpoint), or add the routes in a router module.
Data contract	`Use existing target-tracks router.`
Affects	songnodes/frontend/src/components/DJInterface.tsx

Evidence 2

`songnodes/frontend/src/components/DJInterface.tsx:360`
`songnodes/frontend/src/components/DJInterface.tsx:432`

Gap: **OAuth tokens written to sessionStorage have no reader**

`sn-fe-gap-oauth-session-tokens` · ▲ High · ● Built

Settings connects in the same window (window.location.href), so OAuthCallback has no opener: it writes sessionStorage <service>_oauth_tokens and redirects to /. GraphPage and SettingsPanel only read localStorage tidal_oauth_tokens/spotify_oauth_tokens (which OAuthCallback removes), so the browser store never receives the tokens; sendToTidal and useTokenRefresh then lack an access token.

Fix	In OAuthCallback, when window.opener is null, call useStore.getState().credentials.updateCredentials(service, {accessToken,refreshToken,expiresAt,isConnected:true}) before redirecting to /settings/<service>-connection, and delete the sessionStorage write; or have SettingsPage read and clear sessionStorage '<service>_oauth_tokens' on mount.
Data contract	`oauth-handoff/v1: {service, access_token, refresh_token, expires_at} consumed exactly once by the credentials`
Affects	songnodes/frontend/src/components/OAuthCallback.tsx, songnodes/frontend/src/pages/SettingsPage.tsx

Evidence 3

```
songnodes/frontend/src/components/OAuthCallback.tsx:226-270
songnodes/frontend/src/pages/SettingsPage.tsx:362-394
songnodes/frontend/src/pages/GraphPage.tsx:15-33
```

Gap: **Graph page calls scraper endpoints that do not exist**

`sn-fe-gap-scrapers-endpoints` · ◐ Medium · ● Built

DJInterface posts to /api/v1/scrapers/target-tracks/search and /api/v1/scrapers/queue/clear; no rest-api route matches (route scan of services/rest_api).

Fix	Point the actions at POST /api/v1/target-tracks/{id}/search and POST /api/v1/scrape/trigger, or remove the buttons; add a contract test listing every frontend path against the FastAPI OpenAPI document.
Data contract	`frontend-endpoint-manifest.json checked against rest-api /openapi.json in CI`
Affects	songnodes/frontend/src/components/DJInterface.tsx

Evidence 2

```
songnodes/frontend/src/components/DJInterface.tsx:360,395,432
/home/claude/dfa/scratch/sn-frontend/backend_routes.tsv
```

Gap: **No IndexedDB last-good-data cache despite design-system rule**

`sn-fe-gap-indexeddb-cache` · ◐ Medium · ● Built

design-system/16-platform.md:50 says SongNodes keeps the last graph, library and saved sets in IndexedDB and shows them with FreshnessStamp when the API is unreachable. No indexedDB usage exists in frontend/src; library load just errors and Home shows a dash.

Fix — Add frontend/src/lib/offlineCache.ts (idb-keyval style store "sn-cache" with keys graph, library, sets, each {data, fetchedAt}); write on successful loadLibrary/useDataLoader/readSavedSetsAsync, read on failure and pass fetchedAt to FreshnessStamp.

Data contract — `sn-cache/v1 {key, data, fetchedAt ISO, apiVersion}`

Affects — songnodes/frontend/src/features/musicLibrary/data.ts, songnodes/frontend/src/hooks/useDataLoader.ts

Evidence 2

```
songnodes/design-system/16-platform.md:50
songnodes/frontend/src/features/musicLibrary/data.ts:107-135
```

Gap: **artifact_sets.json lacks the tidal flag musicLibrary needs**

`sn-fe-gap-artifact-sets-tidal-flag` · × Critical · ● Built

Set builder exports SavedSet[] without a 'tidal' field; musicLibrary run_weekly.py sync-sets filters sets by s.get('tidal'), so sets exported from songnodes are never pushed to TIDAL by the weekly run (songnodes has its own Send to TIDAL instead). The file is also only a browser download that has to be copied by hand.

Fix — Either add 'tidal: boolean' (set when Send to TIDAL is requested) to SavedSet and the export, or document that songnodes sets go to TIDAL through /api/v1/tidal/sets/send and drop the inbox path; better, have musicLibrary read GET /api/v1/library/sets/export (exists, unused by the frontend).

Data contract — `artifact_sets/v2: SavedSet[] + {tidal:boolean}`

Affects — songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx, musicLibrary/pipeline/run_weekly.py

Evidence 3

```
songnodes/frontend/src/features/musicLibrary/SetBuilderPage.tsx:223-
226
musicLibrary/pipeline/run_weekly.py:529
musicLibrary/pipeline/build_db.py:93-106
```

Gap: **Pending target syncs are never retried and override the server**

`sn-fe-gap-target-pending-sync` · Medium · Built

A failed POST/DELETE leaves pendingSync=true in sn-target-tracks-v1; nothing reads pendingSync, and useUnifiedTracks prefers the local entry over backend state, so the device can disagree with target_tracks forever.

Fix	On useUnifiedTracks.refresh(), replay entries with pendingSync=true through syncTargetTrack, and drop local entries whose backend match agrees; show a count of unsynced targets.
Data contract	`sn-target-tracks-v2 {isTarget, priority, pendingSync, lastAttempt}`
Affects	songnodes/frontend/src/services/trackSyncAdapter.ts, songnodes/frontend/src/hooks/useUnifiedTracks.ts

Evidence 2

```
songnodes/frontend/src/services/trackSyncAdapter.ts:31-100
songnodes/frontend/src/hooks/useUnifiedTracks.ts:93-103
```

Gap: **Library freshness stamp is always unknown**

`sn-fe-gap-library-generated` · Medium · Built

loadLibrary sets generated: null and runs: [] because /api/v1/library/search returns rows only; Home and Library show an en dash for 'Library data from' and the weekly-run launcher summary never appears.

Fix	Add GET /api/v1/library/meta returning {generated, ledger_updated, counts, sources, runs} from the ingest job result and merge it in loadLibrary.
Data contract	`library-meta/v1 = LibraryFile minus tracks`
Affects	songnodes/frontend/src/features/musicLibrary/data.ts, songnodes/services/rest_api/routers/library_search.py

Evidence 2

```
songnodes/frontend/src/features/musicLibrary/data.ts:120-130
songnodes/frontend/src/pages/HomePage.tsx:22-46
```

Gap: **Graph WebSocket has a server but no client**

`sn-fe-gap-graph-ws` · Medium · Built

graph-api exposes WEBSOCKET /api/graph/ws/{room_id} (nginx passes Upgrade headers for /api/graph). The only frontend WebSocket client (LiveDataLoader, /api/ws) is unreachable and targets a path rest-api does not serve.

Fix	Delete LiveDataLoader/useLiveDataLoader, or rewire it to /api/graph/ws/<room> and mount it from GraphPage.
Data contract	`graph-ws/v1 message {type: nodes|edges|delta, payload}`
Affects	songnodes/frontend/src/components/LiveDataLoader.tsx

Evidence 2

```
songnodes/frontend/src/components/LiveDataLoader.tsx:138-140
songnodes/frontend/Dockerfile:81-91
```

Gap: **Library ingest allows 50 MB but frontend nginx accepts 1 MB**

`sn-fe-gap-ingest-body-limit` · ◐ Medium · ● Built

LibraryImportDialog FileDrop limit is 50 MB and the body is the whole export; the frontend image nginx config sets no client_max_body_size, so nginx default 1 MB returns 413 for any larger export. STATUS.md already lists it as a follow-up.

Fix	Add client_max_body_size 64m; inside location /api/ (or a dedicated location = /api/v1/library/ingest) in the frontend Dockerfile default.conf, and map HTTP 413 to an explicit error in LibraryImportDialog.
Data contract	`ingest upload limit = FileDrop maxMb = nginx client_max_body_size = rest-api request limit`
Affects	songnodes/frontend/Dockerfile, songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx

Evidence 3

```
songnodes/frontend/src/features/musicLibrary/components/LibraryImportDialog.tsx:23-
24
songnodes/frontend/Dockerfile:91-99
songnodes/STATUS.md:1899
```

Modules: **Proposed single-purpose modules** `5 modules`

Module: **Set exchange**

Target users	Set planner, weekly-run agent
One verb	**Convert**
Data inputs	songnodes saved sets (/api/v1/library/sets/export), artifact sets (artifact db `sets`)
Core operation	Reads either saved-set shape and writes one versioned `songnodes.saved-set/2` document (name, shape, minutes, tracks[{isrc, tidal_id, ledger_id, tgt}], tidal) to musicLibrary inbox/artifact_sets.json, rejecting rows it cannot map.
Size	about 250 lines (Estimate)
Where it lives	`musicLibrary/pipeline/djlib/set_exchange.py (new)`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Play-history import**

Target users	Library curator
One verb	**Import**
Data inputs	Rekordbox / Lexicon collection XML, play counts and dates
Core operation	Matches plays to ledger tracks by local_path then ISRC, appends `play` events to the ledger and writes rotation evidence that calibrate.py reads.
Size	about 300 lines (Estimate)
Where it lives	`musicLibrary/pipeline/djlib/play_history.py`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Transition feed**

Target users	Set planner, DJ
One verb	**Export**
Data inputs	songnodes silver_track_transitions (occurrence_count) joined to library tracks
Core operation	Publishes `songnodes.transitions/1` (pairs of ISRCs with counts) beside library.json so both Set Builders can rank spares by proven pairings.
Size	about 200 lines (Estimate)
Where it lives	`songnodes/services/rest_api/routers/transitions_export.py`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Contract check**

Target users	Developer, pipeline operator
One verb	**Verify**
Data inputs	contracts/*.schema.json, the newest musicLibrary outputs, the songnodes ingest fixture
Core operation	Validates every cross-project file and request body against its schema in CI and in the weekly run, and fails with the field that broke.
Size	about 180 lines (Estimate)
Where it lives	`songnodes/scripts/check_contracts.py`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Module: **Run monitor**

Target users	Pipeline operator, library curator
One verb	**Alert**
Data inputs	musicLibrary runs table and scrape_log; songnodes ingestion_jobs and web_call_log
Core operation	Compares each run's resolve yield, source counts and job failures with the last four runs and raises one Banner-shaped warning with a source link when a figure drops by half.
Size	about 220 lines (Estimate)
Where it lives	`songnodes/frontend/src/features/pipeline/RunMonitor.tsx`
Design system	Built from the SongNodes design system in full (tokens, sn-* classes, page pattern).

Backbone: **Unified storage backbone**

Keep each project's own data folder and owner. Add one shared contracts folder and one exchange folder with versioned, schema-checked files, so every cross-project hop is a named file or endpoint with one writer.

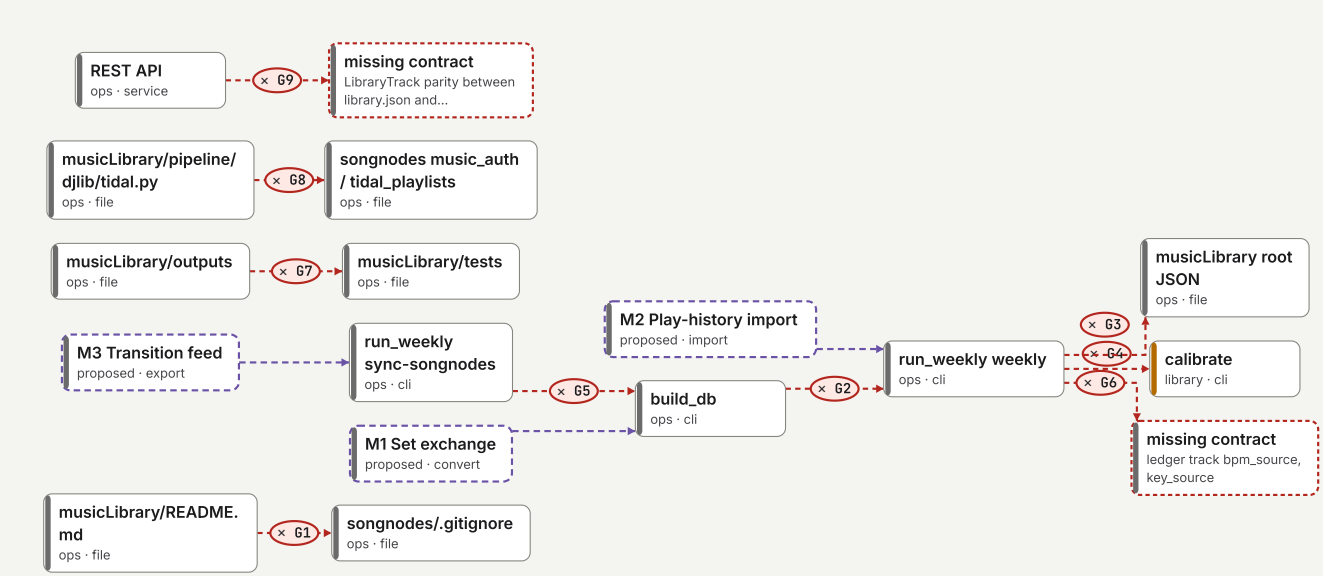
```
songnodes/
  contracts/                      # single source of every cross-project schema (exists; extend)
    library.set.saved.v2.json      # new
    library_ingest.openapi.yaml    # exists
    transitions.v1.json            # new
    setlists.v1.json               # new
  Postgres (compose volume)       # owner: songnodes services only
    bronze_* → silver_* → (gold_* decided in DFA gap) → projection_*
musicLibrary/
  ledger/                         # owner: process step (append-only)
  db/dj_library.sqlite            # rebuilt each run from ledger
  inbox/                          # inputs written by people or other systems
  outputs/<date>/                 # run outputs, read by people
  exchange/                       # NEW: the only files the other project reads or writes
    library.json                  # musicLibrary → songnodes (also POST /api/v1/library/ingest)
    saved-sets.json               # both ways, saved-set/2
    transitions.json              # songnodes → musicLibrary, transitions/1
    setlists/*.json               # tracklists for songnodes import (moved from the root)
```

4 rows

RISK	MITIGATION
Readers still point at old paths (README, import script, artifact)	Keep the old paths as copies for one release and log a warning when they are read.
Saved-set/1 documents in the artifact db	M1 reads both shapes; write only /2; keep /1 readable until no /1 set has been opened for four weeks.
Weekly run writes a file songnodes is reading	Write to a temporary name and rename; songnodes reads only complete files.
Postgres schema change for gold	One migration through the migration ledger gap first; never reset the volume (CLAUDE.md).

D8a: **Gap map – musicLibrary and cross-project**

9 musicLibrary and cross-project gaps, 4 of them critical; dashed brand cards are the modules that close them.



Gap contracts

- **G1** library.json v1 (build_db.export_artifact_json) → POST /api/v1/library/ingest, Idempotency-Key import-<md5> (medium)
- **G2** naarm.saved-set.v2 (critical)
- **G3** songnodes.setlists.v1 → TracklistImportRequest (critical)
- **G4** rekordbox DJ_PLAYLISTS TRACK{Location, Artist, Name, PlayCount, LastPlayed, DateAdded} → ledger local_path, appearances (medium)
- **G5** library.json tracks[].nx: [{id, n}] (medium)
- **G6** ledger track bpm_source, key_source (critical)
- **G7** review_queue.csv energy_score_override → overrides (critical)
- **G8** TIDAL redirect per client; shared set→playlist map (medium)
- **G9** LibraryTrack parity between library.json and /api/v1/library/search (medium)

Key

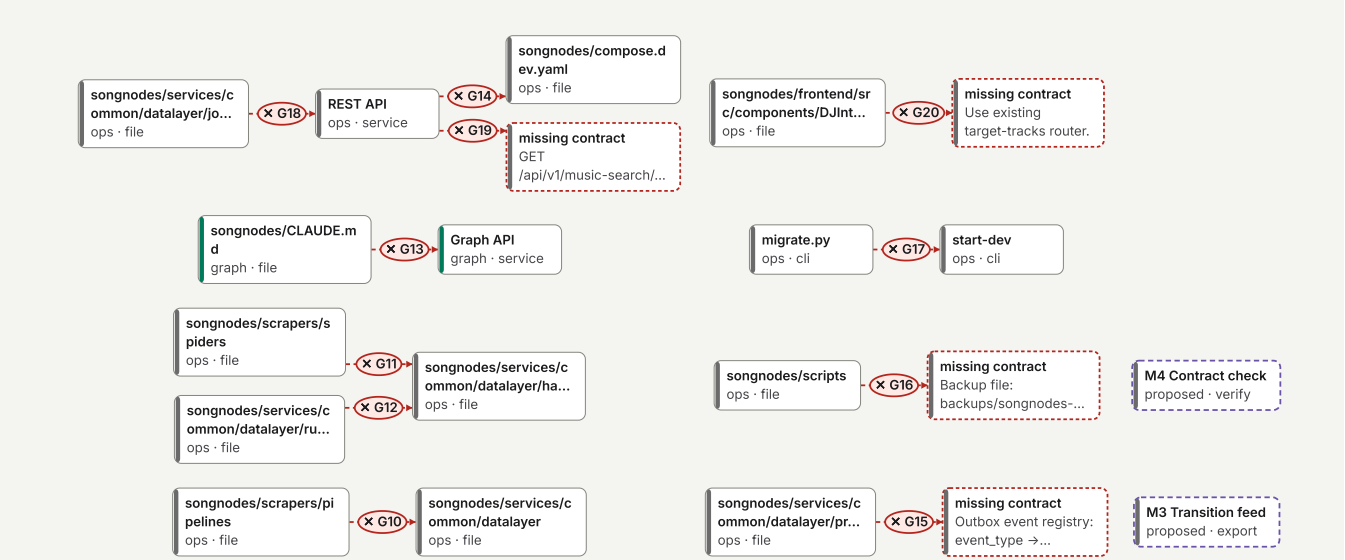
library · ops and design · Production (solid outline) · Proposed (brand, dashed) · Gap (red dashed, ×) · Proposed link (brand dashed)

D8a · n = 9 · Unit: gaps · ● Built

· Source: gaps[] (title, contract, affects, severity, effort); modules[]; tasks[] (gaps, title) · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes; gap endpoints = first two affected paths matched to pages by path prefix

D8b: **Gap map – songnodes backend**

11 songnodes backend gaps, 1 of them critical; dashed brand cards are the modules that close them.



Gap contracts

- **G10** Scraper output = bronze_scraped_playlists + bronze_scraped_tracks with positions 1..n; silver is written only by common/datalayer/ingest.py inside a transaction that appends an...
- **G11** POST /scrape response gains close_reason; handler treats daily_quota_reached as RetryableError. (high)
- **G12** Handler contract: (prepare outside tx) → (commit inside tx). (medium)
- **G13** Layer contract table: layer → table → sole writer → readers. (high)
- **G14** X-API-Token header is the only admin credential; Authorization is reserved for provider tokens. (critical)
- **G15** Outbox event registry: event_type → consumers (projection_tracks, graph_projection). (medium)
- **G16** Backup file: backups/songnodes-YYYYMMDD.dump (pg custom format). (high)
- **G17** schema_migrations(version, checksum, applied_at). (medium)
- **G18** Idempotency-Key + payload hash. (medium)
- **G19** GET /api/v1/music-search/{service}?q= → [{id, artist, title, isrc}]. (medium)
- **G20** Use existing target-tracks router. (medium)

Key

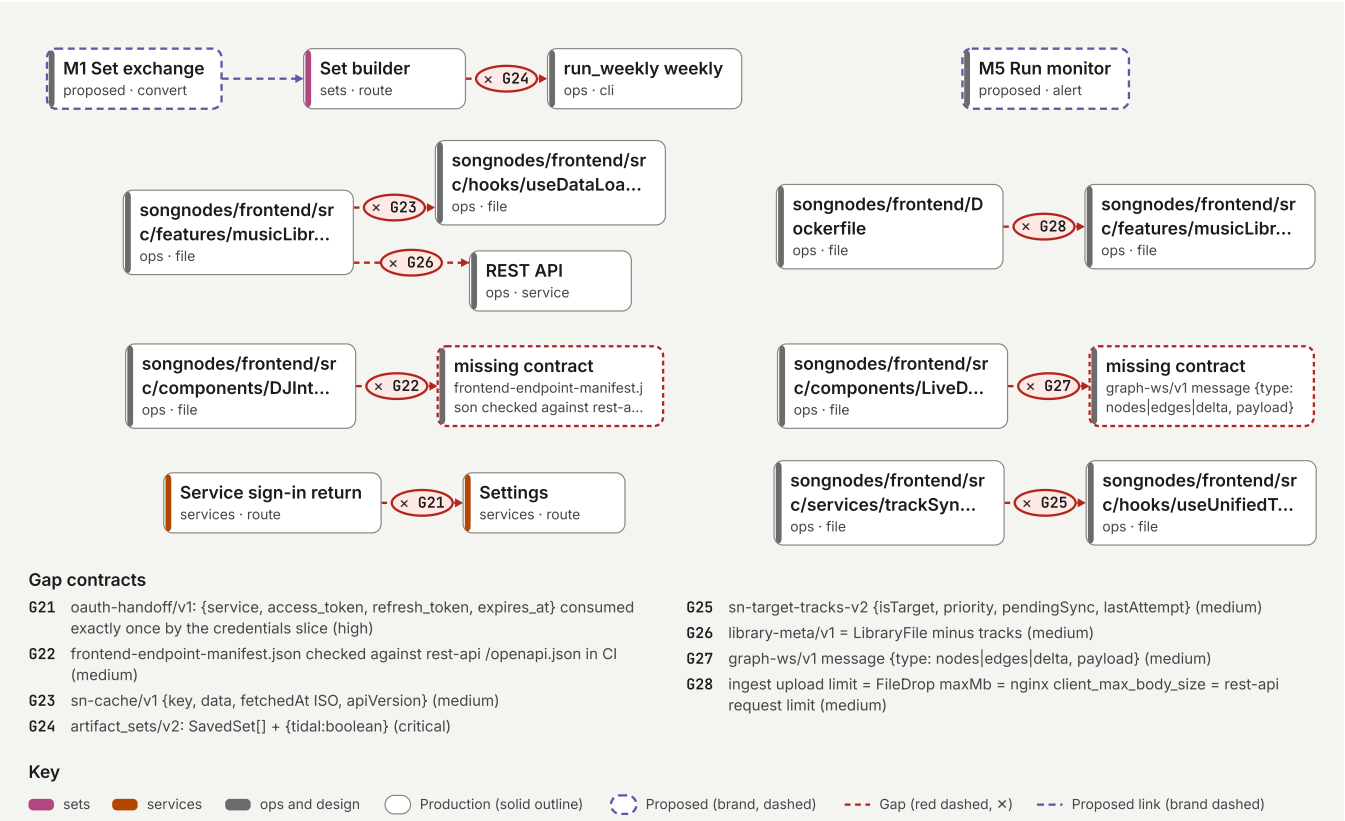
graph · ops and design · Production (solid outline) · Proposed (brand, dashed) · Gap (red dashed, ×) · Proposed link (brand dashed)

D8b · n = 11 · Unit: gaps · ● Built

· Source: gaps[] (title, contract, affects, severity, effort); modules[]; tasks[] (gaps, title) · Method: graphviz dot layered layout (LR, order given, components packed 4 across; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 15 units; 0 crossings, 0 edges through nodes; gap endpoints = first two affected paths matched to pages by path prefix

D8c: **Gap map – songnodes frontend**

8 songnodes frontend gaps, 1 of them critical; dashed brand cards are the modules that close them.



Gap contracts

G21 oauth-handoff/v1: {service, access_token, refresh_token, expires_at} consumed exactly once by the credentials slice (high)

G22 frontend-endpoint-manifest.json checked against rest-api /openapi.json in CI (medium)

G23 sn-cache/v1 {key, data, fetchedAt ISO, apiVersion} (medium)

G24 artifact_sets/v2: SavedSet[] + {tidal:boolean} (critical)

G25 sn-target-tracks-v2 {isTarget, priority, pendingSync, lastAttempt} (medium)

G26 library-meta/v1 = LibraryFile minus tracks (medium)

G27 graph-ws/v1 message {type: nodes|edges|delta, payload} (medium)

G28 ingest upload limit = FileDrop maxMb = nginx client_max_body_size = rest-api request limit (medium)

Key

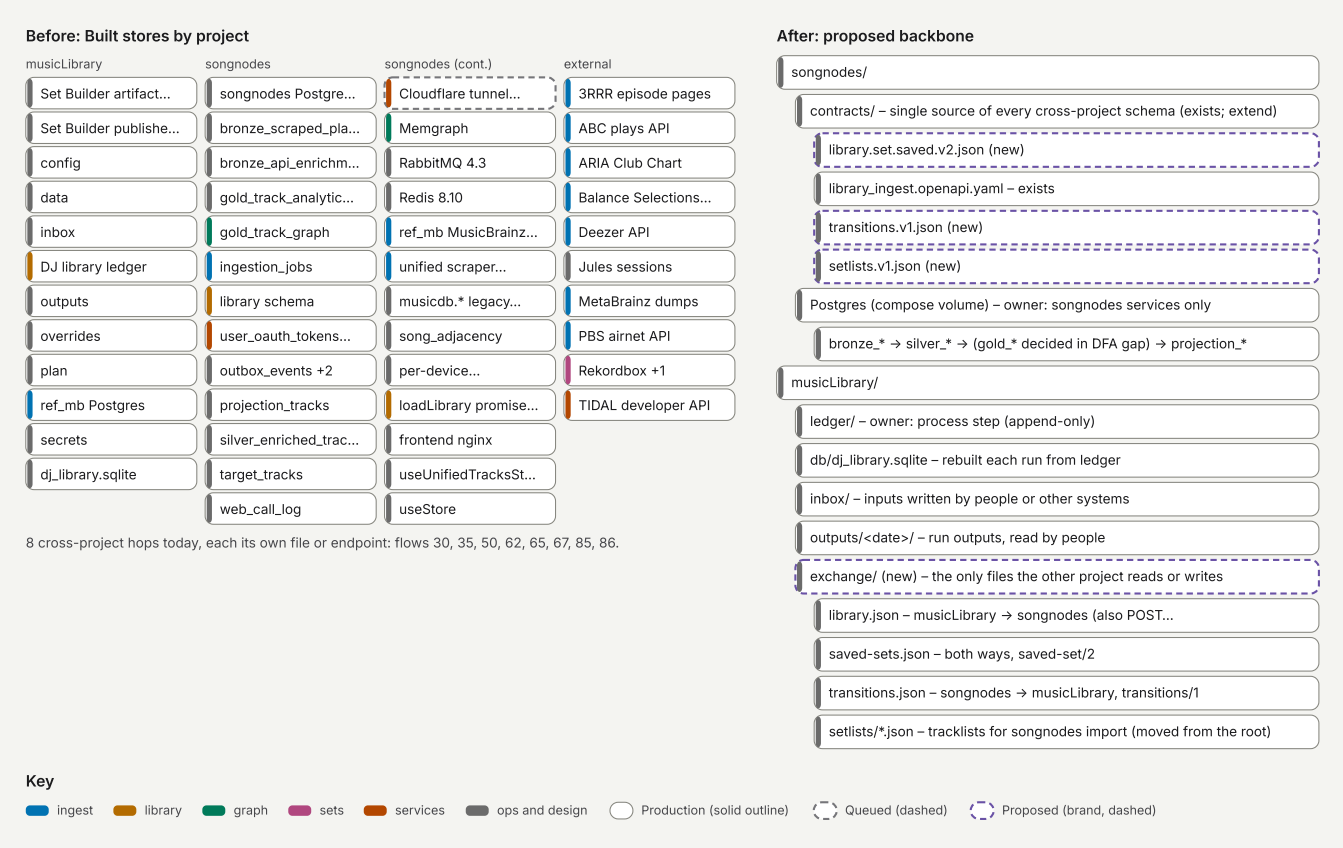
sets · services · ops and design · Production (solid outline) · Proposed (brand, dashed) · Gap (red dashed, ×) · Proposed link (brand dashed)

D8c · n = 8 · Unit: gaps · ● Built

· Source: gaps[] (title, contract, affects, severity, effort); modules[]; tasks[] (gaps, title) · Method: graphviz dot layered layout (LR, order given, components packed 4 across; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes; gap endpoints = first two affected paths matched to pages by path prefix

D10: **Storage backbone before and after**

Each project keeps its own data folder; the change is 4 new contract and exchange entries that carry the 8 cross-project hops.



D10 · n = 66 · Unit: stores and backbone entries · **i built+projected**

· Source: stores[] (repo, kind, domain, state); backbone.tree; flows[] crossing repos · Method: hand-placed two-panel tree; 0 edges

Section 9: **Built and Projected** `37 queued items`

Built = songnodes main @7429821 and the musicLibrary snapshot of 7 Nov 2026. Projected = Built with every queued item applied.

Table: **Queued work** `37 items`

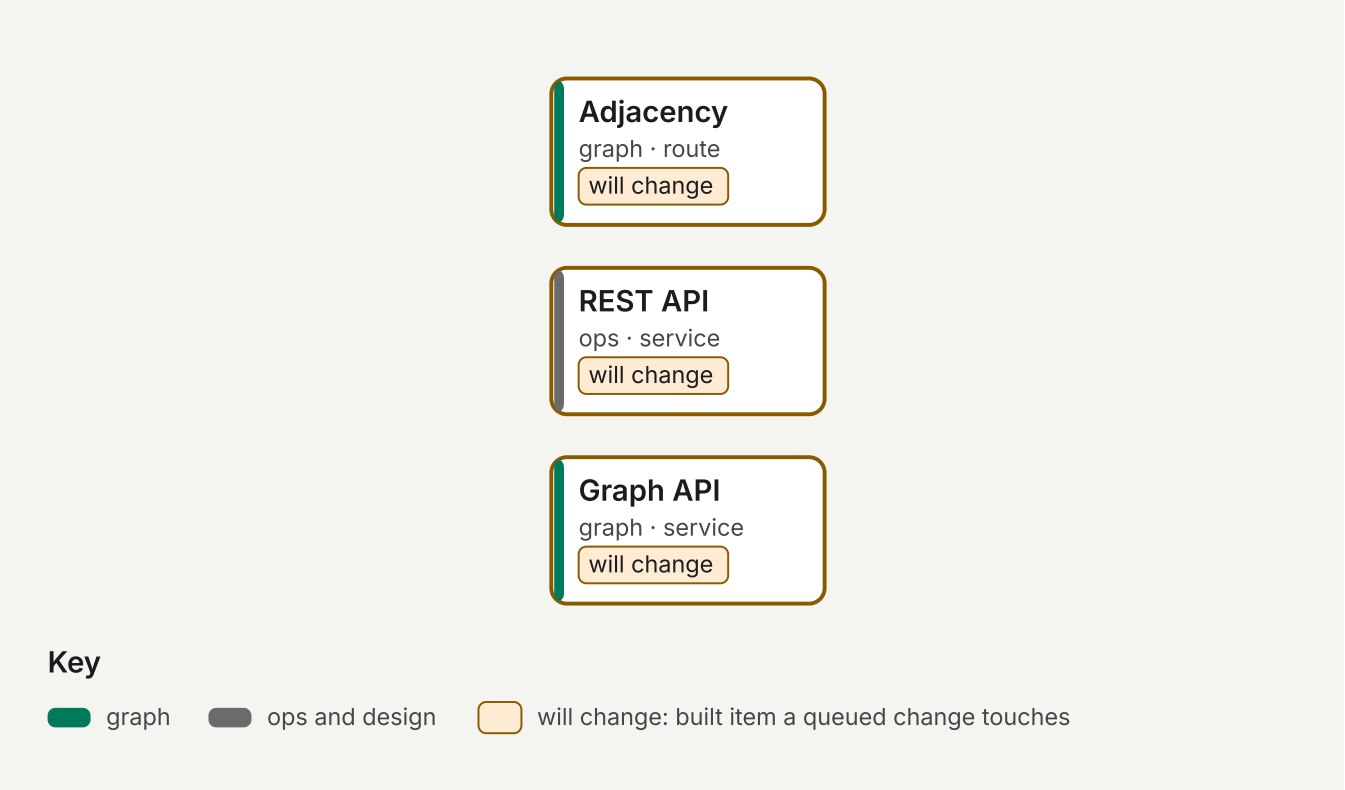
37 rows

ID	WHAT	SOURCE	EFFECT WHEN APPLIED	TOUCHES
`ml-queued-be1`	Real TIDAL and Spotify track search	`musicLibrary/plan/jules_payloads/BE1.json`	–	–
`ml-queued-be2`	Python 3.12 upgrade audit	`musicLibrary/plan/jules_payloads/BE2.json`	–	–
`ml-queued-fe1`	Unified track data bridge	`musicLibrary/plan/jules_payloads/FE1.json`	–	–
`ml-queued-fe10`	AI setlist discovery and import page	`musicLibrary/plan/jules_payloads/FE10.json`	–	–
`ml-queued-fe11`	Navigation audit	`musicLibrary/plan/jules_payloads/FE11.json`	–	–
`ml-queued-fe12`	Pipeline page on real job and outbox data	`musicLibrary/plan/jules_payloads/FE12.json`	–	–
`ml-queued-fe2`	Target Tracks module	`musicLibrary/plan/jules_payloads/FE2.json`	–	–
`ml-queued-fe3`	Route finder page	`musicLibrary/plan/jules_payloads/FE3.json`	–	–
`ml-queued-fe4`	Set Builder: crate export formats and reordering	`musicLibrary/plan/jules_payloads/FE4.json`	–	–
`ml-queued-fe5`	Key and mood Camelot module	`musicLibrary/plan/jules_payloads/FE5.json`	–	–
`ml-queued-fe6`	Pipeline health page	`musicLibrary/plan/jules_payloads/FE6.json`	–	–
`ml-queued-fe7`	Service settings	`musicLibrary/plan/jules_payloads/FE7.json`	–	–
`ml-queued-fe8`	Graph explorer workspace	`musicLibrary/plan/jules_payloads/FE8.json`	–	–
`ml-queued-fe9`	Add library tracks to Target Tracks	`musicLibrary/plan/jules_payloads/FE9.json`	–	–
`ml-queued-few`	Route wiring and cleanup	`musicLibrary/plan/jules_payloads/FEW.json`	–	–
`ml-queued-ref0`	ref_mb as a separate database	`musicLibrary/plan/jules_payloads/REF0.json`	–	–
`ml-queued-sec1`	Remove eval and audit subprocess use	`musicLibrary/plan/jules_payloads/SEC1.json`	–	–
`ml-queued-sec2`	CORS allowlist	`musicLibrary/plan/jules_payloads/SEC2.json`	–	–
`ml-queued-sec3`	Dependency security updates	`musicLibrary/plan/jules_payloads/SEC3.json`	–	–
`ml-queued-sec4`	Secrets and dependency scanning in CI	`musicLibrary/plan/jules_payloads/SEC4.json`	–	–
`ml-queued-sec5`	Opt-in token auth for write and admin routes	`musicLibrary/plan/jules_payloads/SEC5.json`	–	–
`ml-queued-sec6`	Server-side AI provider proxy	`musicLibrary/plan/jules_payloads/SEC6.json`	–	–

`ml-queued-wp0`	Contracts, DDL, fixtures	`musicLibrary/plan/jules_payloads/WP0.json`	–	–
`ml-queued-wp2`	MusicBrainz reference ETL into ref_mb	`musicLibrary/plan/jules_payloads/WP2.json`	–	–
`ml-queued-wp3`	Local MusicBrainz resolver	`musicLibrary/plan/jules_payloads/WP3.json`	–	–
`ml-queued-wp4`	Local MusicBrainz provider in enrichment	`musicLibrary/plan/jules_payloads/WP4.json`	–	–
`ml-queued-wp5`	Library overlay model, importer, ingest job and saved-sets endpoints	`musicLibrary/plan/jules_payloads/WP5.json`	–	–
`ml-queued-wp6`	Ingest API and transactional outbox	`musicLibrary/plan/jules_payloads/WP6.json`	–	–
`ml-queued-wp8`	Graph projection as outbox consumer	`musicLibrary/plan/jules_payloads/WP8.json`	–	–
`ml-queued-wp9`	Server-side search and Library / Set builder pages on the API	`musicLibrary/plan/jules_payloads/WP9.json`	–	–
`ml-queued-wp7-pipeline-client`	MetaBrainz reference data + musicLibrary → songnodes pipeline client	`musicLibrary/plan/agents/WP7_pipeline_client.md`	–	–
`ml-queued-open-prs`	Open songnodes PRs #576-#580 (Duo auth, tunnel, Sentinel)	`gh api repos/zvirb/songnodes/pulls/576..580/files`	–	–
`PR-576`	DUOAUTH: Cisco Duo 2FA and Cloudflare Tunnel architecture guide	`github zvirb/songnodes#576`	Docs only	docs
`PR-577`	DUO3: Cloudflare Tunnel connector integration	`github zvirb/songnodes#577`	Publishes the stack through a public tunnel	compose.dev.yaml, sn-be-q-cloudflared
`PR-578`	DUO1: Cisco Duo Web SDK v4 auth router	`github zvirb/songnodes#578`	Adds /api/v1/auth routes with a default admin login fallback	sn-be-q-user-auth-router
`PR-579`	DUO2: Frontend login page and auth context	`github zvirb/songnodes#579`	Adds /login	sn-fe-login
`PR-580`	Sentinel: sanitise REST API main exception details	`github zvirb/songnodes#580`	Sanitises errors, partly on unreachable routes	sn-be-svc-rest-api

D9: **Built to Projected change map**

Queued work adds 10 cards and 6 edges and changes 3 built items; 32 briefs wait behind them.



D9 · n = 51 · Unit: items · i built+queued

· Source: queued[] (PR touches and effect, brief notes); flows[] and pages[]/stores[] with state queued · Method: graphviz dot layered layout (LR, order given; 12 rank directions and input orders tried, scored by legibility then crossings), splines=ortho, own number badges, type floor 12 units; 0 crossings, 0 edges through nodes

Section 10: **Sister-project tasks** 16 tasks in 3 projects

C12: **Sister tasks by project, type and status**

16 queued tasks over 3 projects; 10 are ready now and 6 wait on another task.

	songnodes (9)	musicLibrary (6)	design-system (1)
fix (3)	**SN-T04** Ready **SN-T07** Ready **SN-T09** Waits on SN-T01		
module (3)	**SN-T08** Waits on SN-T03	**ML-T02** Waits on SN-T03 **ML-T04** Ready	
contract (2)	**SN-T03** Ready	**ML-T05** Waits on SN-T03	
data-integrity (2)	**SN-T05** Ready	**ML-T03** Ready	
docs (2)	**SN-T06** Waits on SN-T05	**ML-T01** Ready	
security (2)	**SN-T01** Ready **SN-T02** Waits on SN-T01		
design-system (1)			**DS-T01** Ready
observability (1)		**ML-T06** Ready	

Every task is Queued (dashed cell). Ready = no dependency; Waits on = blocked by the named task

C12 · n = 16 · Unit: tasks · ▲ Queued

· Source: tasks[] (project, type, depends) · Method: Tasks placed by type (rows, by count) and project (columns, by count). tasks[] has no status field: status is derived from depends (Ready when empty).

Table: **Task matrix** 16 tasks

16 rows

TASK	PROJECT	PACKAGE	TYPE	OWNED PATHS	DEPENDS ON	VERIFICATION	STAT
SN-T01	songnodes	**Require the API token on write and admin routes**	security	services/rest_api/security/auth.py, compose.dev.yaml	–	curl without token gets 401 on every POST/PUT/DELETE; scripts pass with the token	– T
SN-T02	songnodes	**Remove the default admin login from the queued auth router**	security	services/rest_api/routers/user_auth.py (in PR #578)	SN-T01	no login succeeds without a configured user	– T
SN-T03	songnodes	**Accept and keep musicLibrary pipeline fields on library ingest**	contract	services/rest_api/routers/library_ingest.py, contracts/library_ingest.openapi.yaml	–	fixture round-trip keeps energy, rotation, pillar and ISRC	– T
SN-T04	songnodes	**Raise the frontend nginx body limit to the ingest limit**	fix	frontend/Dockerfile	–	a 5 MB library import returns 202, not 413	– T
SN-T05	songnodes	**Stop the unified scraper at bronze**	data-integrity	scrapers/pipelines/persistence_pipeline.py	–	a scrape writes bronze rows only; ingest creates silver once	– T

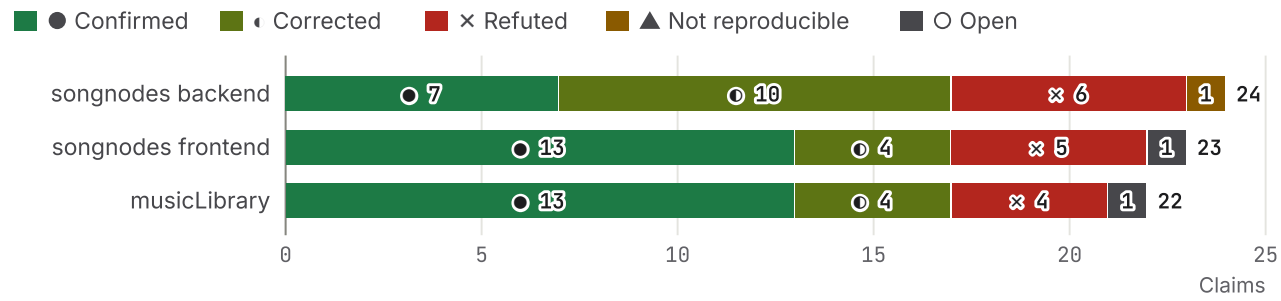
SN-T06	songnodes	**Decide and document the real layer model**	docs	docs/DATA_LAYER.md, CLAUDE.md	SN-T05	docs and compose name the same jobs	– T
SN-T07	songnodes	**Let target scrapes bypass the daily discovery quota**	fix	scrapers/spiders/1001tracklists_spider.py	–	two target scrapes in an hour both run	– T
SN-T08	songnodes	**Publish the transition feed (M3)**	module	services/rest_api/routers/transitions_export.py (new), contracts/transitions.v1.json (new)	SN-T03	GET returns transitions/1 for library ISRCs	– T
SN-T09	songnodes	**Fix the OAuth token hand-off and remove dead scraper calls**	fix	frontend/src/components/OAuthCallback.tsx, frontend/src/pages/SettingsPage.tsx, frontend/src/pages/GraphPage.tsx	SN-T01	Send to TIDAL finds a token after a fresh login; no 404 in the graph page network log	– T
ML-T01	musicLibrary	**Correct the README songnodes contract**	docs	README.md	–	every path in the README exists	– T
ML-T02	musicLibrary	**Build the set exchange (M1)**	module	pipeline/djlib/set_exchange.py (new), pipeline/run_weekly.py	SN-T03	a songnodes set reaches TIDAL through sync-sets	– T
ML-T03	musicLibrary	**Merge Deezer BPM and key onto the chosen version**	data-integrity	pipeline/run_weekly.py, pipeline/djlib/deezer.py	–	more than 0 tracks have BPM and key after one run	– T
ML-T04	musicLibrary	**Build the play-history import (M2)**	module	pipeline/djlib/play_history.py (new)	–	calibrate.py reports rotation timings from plays	– T
ML-T05	musicLibrary	**Move setlist files into exchange/ and import them**	contract	exchange/setlists/ (new), pipeline/run_weekly.py	SN-T03	JohnCourse and kiki appear as songnodes setlists	– T
ML-T06	musicLibrary	**Warn on resolve yield drops and retry rejected pending**	observability	pipeline/run_weekly.py, pipeline/djlib/export.py	–	a run with half the usual yield writes a warning	– T
DS-T01	design-system	**Publish 1.10.0 to the design-system artifact**	design-system	design-system artifact 7f37707d	–	artifact tokens.json version is 1.10.0	– T

Section 11: **Verification register** 69 claims

No earlier data-flow report exists for these repos, so the register re-verifies the repos' own documentation.

C11: **Verification outcome by area**

33 of 69 repo claims hold as written; 18 needed correcting and 15 are refuted.



C11 · n = 69 · Unit: claims · Built · Source: claims[] (outcome, _agent) · Method: Count of claims per scanning area split by outcome; sorted by total.

69 rows

ID	CLAIM	SOURCE	OUTCOME	FINDING
ml-claim-stdlib	Standard-library Python 3.10+. Nothing to install.	musicLibrary/README.md:9	Confirmed	pipeline imports djlib has no third
ml-claim-schedule	Weekly flow runs as a Sunday 7:37am scheduled task	musicLibrary/README.md:13	Refuted	runs dated 24 Oc (Tue), 31 Oct (Fri on a Sunday
ml-claim-resolve-fields	resolve writes TIDAL and Deezer versions with duration, version, BPM, key, ISRC, genre	musicLibrary/README.md (step table)	Corrected	0 of 1007 TIDAL or key; Deezer BI and never copiec version; ledger B
ml-claim-resumable	resolve is resumable	musicLibrary/README.md (step table)	Confirmed	skips keys alreac resolved_<date> 20
ml-claim-db-tables	DB tables tracks ... runs; views v_crates, v_source_yield, v_review_queue, v_buy_list	musicLibrary/README.md (Database)	Confirmed	sqlite_master ma table not listed)
ml-claim-page-source	Page source: artifact/index.html; the weekly run republishes it	musicLibrary/README.md (Set Builder)	Refuted	the published pa artifact/build/ind artifact/index.htm beside it
ml-claim-sync-script	scripts/sync-music-library.sh copies library.json to frontend/public/data/music-library.json	musicLibrary/README.md (In SongNodes)	Refuted	neither exists in s contract is POST /api/v1/library/inc library.sh, cmd_s
ml-claim-same-set-shape	Sets saved in SongNodes use the same document shape as this artifact's saved sets	musicLibrary/README.md (In SongNodes)	Refuted	songnodes serve description and u no tidal, shape o server
ml-claim-review-queue	Tracks go to review_queue.csv when BPM and source methods disagree by 3+ points	musicLibrary/README.md (Energy model)	Corrected	needs_review is than two method tracks are in the
ml-claim-weights	Energy blend BPM 45%, source 40%, version cues 15%	musicLibrary/README.md (Energy model)	Confirmed	rules.energy.weig 0.45/0.40/0.15; ir source weight ap
ml-claim-overrides	Overrides are re-applied every run and the auto score is kept	musicLibrary/README.md (levers)	Confirmed	apply_overrides s once; 0 overrides
ml-claim-manual-stamp	manual_adds rows get stamped processed	musicLibrary/README.md (levers)	Confirmed	stamp written be raw_<date>.json
ml-claim-calibrate	calibrate writes suggestions only	musicLibrary/README.md (levers)	Confirmed	writes outputs/calibratic file absent, never
ml-claim-xml-local	Rekordbox XML only includes tracks with a local_path	musicLibrary/README.md (one-off setup)	Confirmed	0 local paths, XM

ml-claim-tidal-fallback	Until TIDAL setup is done resolve uses Deezer only and TIDAL sync is skipped	musicLibrary/README.md (one-off setup)	● Confirmed	guards on secret tidal_user_token.j files show 403 tr setup was done
ml-claim-db-loud	db fails loudly if the row count or integrity check is wrong	musicLibrary/agent/weekly_run.md:6	● Confirmed	sys.exit on mism
ml-claim-atomic	database is written to a temp file and swapped in atomically	musicLibrary/pipeline/build_db.py:6-7	◐ Corrected	shutil.copyfile ov atomic rename
ml-claim-shared-wins	Shared sets win by id, and sets only in this browser are uploaded	musicLibrary/artifact/app/README.md	● Confirmed	initSavedSets me then local-only u
ml-claim-ds-version	Published Set Builder is built from SongNodes design system 1.1.0	musicLibrary/artifact/app/README.md	i Open	no version string confirm; songnoc and the build car main
ml-claim-isrc-319	Resolver thresholds to be fitted to the 319 ISRC-bearing library tracks	musicLibrary/decisions/2026-10-25-metabrainz-integration-baseline.md	◐ Corrected	all 413 ledger trac ISRC
ml-claim-goodstatic	Good Static is always 0 because it publishes no playlists	musicLibrary/agent/weekly_run.md:9	● Confirmed	0 rows in all four Elektrosophy and fell to 0 on 3 Nov row counts three
ml-claim-only-library-json	The weekly run only replaces library.json	musicLibrary/agent/weekly_run.md:8	● Confirmed	publish step pass and library.json; l equals outputs/a
sn-be-c-unified-only	unified-scraper is the ONLY scraper container	songnodes/CLAUDE.md:58	● Confirmed	Only scraper ser compose.dev.yar
sn-be-c-medallion	Scraper → Bronze → Silver → Gold → Operational → Frontend	songnodes/CLAUDE.md:11	× Refuted	No gold migratio compose; graph- directly; gold_trac reader.
sn-be-c-no-layer-skip	Bronze → Silver → Gold flow, no layer skipping	songnodes/CLAUDE.md:28	× Refuted	Scraper persiste silver tables direc
sn-be-c-etl-sql	Transition ETL inserts (source_track_id, target_track_id, playlist_id)	songnodes/CLAUDE.md:215	◐ Corrected	Columns are from_track_id/to_ playlist_occurren occurrence_cour Python.
sn-be-c-target-searches	Scraper queries target_track_searches	songnodes/CLAUDE.md:44	◐ Corrected	Scheduler reads (migration 012); target_track_sear legacy SQL.
sn-be-c-bronze-schema	Bronze tables carry raw_metadata JSONB	songnodes/CLAUDE.md:199	◐ Corrected	Code writes raw_ scraped_at, scra
sn-be-c-scraper-api	Unified Scraper API body {spider_name, search_params}	songnodes/CLAUDE.md:226	◐ Corrected	ScrapeRequest is search_query, ar max_pages, job_
sn-be-c-enrich-8020	Enrichment delegates to metadata-enrichment:8020	songnodes/CLAUDE.md:174	× Refuted	metadata-enrichi compose; 8020 i enrichment runs common/provide
sn-be-c-ollama	Resilience includes Ollama CAPTCHA solving	songnodes/CLAUDE.md:176	▲ Not reproducible	No Ollama servic the Ollama modu ITEM_PIPELINES
sn-be-c-artist-enforce	Artist validity enforced Database view → API filter → Frontend → ETL	songnodes/CLAUDE.md:37	◐ Corrected	API filter (graph- and ETL (is_place no enforcing data applied migratior writes Unknown
sn-be-c-trigger-scrape	POST /api/v1/scrape/trigger with Idempotency-Key queues a scrape	songnodes/CLAUDE.md:354	● Confirmed	jobs.trigger_scra scrape keyed by

sn-be-c-pools	pool_size=5, max_overflow=10	songnodes/CLAUDE.md:248	◐ Corrected	graph-api uses p max_overflow 15; min 5 max 15.
sn-be-c-resource-limits	Resource limits DB 1-2GB, APIs 512MB, scrapers 1GB, frontend 256MB	songnodes/CLAUDE.md:252	× Refuted	Only audio-analy (1g) in compose.
sn-be-c-pg-unpublished	postgres, redis, rabbitmq not published	songnodes/CLAUDE.md:120	● Confirmed	No ports on thos
sn-be-c-backup	pg_dump -U songnodes -Fc songnodes backup one-liner	songnodes/CLAUDE.md:300	● Confirmed	Matches compos scripts/backup_d alternative target
sn-be-c-scrape-writes-bronze	POST /scrape is synchronous (up to 15 minutes) and writes bronze itself	songnodes/compose.dev.yaml:146	◐ Corrected	It also writes silv song_adjacency.
sn-be-c-audio-no-db	audio-analysis needs no database or RabbitMQ for /analyze-url	songnodes/compose.dev.yaml:187	● Confirmed	Startup tolerates DATABASE_URL a queue consumer
sn-be-c-jobs-idem	A retried request returns the original job	songnodes/services/common/datalayer/jobs.py:7	● Confirmed	ON CONFLICT (kind,idempotenc NOTHING then S
sn-be-c-outbox-skip	A slow writer that commits late can never be skipped	songnodes/services/common/datalayer/outbox.py:4	● Confirmed	Reads only txid < side effect: long consumers.
sn-be-c-ingest-409	Same Idempotency-Key with a different body yields 409	songnodes/contracts/library_ingest.md:16	× Refuted	enqueue returns without comparir
sn-be-c-ingest-rows	Ingest returns per-row results	songnodes/contracts/library_ingest.md:14	× Refuted	Endpoint returns {job_id,status,cre land in the job re
sn-be-c-orchestrator	services/scraper-orchestrator/main_enhanced.py imports automated_scheduler	songnodes/STATUS.md:1567	◐ Corrected	No services/scra directory on mair orchestrator hold enrich_existing_t
sn-be-c-mock-search	REST API still returns Mock Artist until S2	songnodes/plan/RESTRUCTURE.md:14	◐ Corrected	No Mock strings it proxies to meta and returns 503 (always, in comp
sn-be-c-migrate-py	Migrations 012-014 are applied by python migrate.py	songnodes/services/worker/Dockerfile	◐ Corrected	start-dev.sh appl migrate.py is use or a manual com
sn-fe-c-port	frontend at http://127.0.0.1:3006, nginx proxies /api to the APIs	songnodes/CLAUDE.md §2 Services table	● Confirmed	compose publish 127.0.0.1:3006:80 default.conf send /api/(graph|v1/vis setlists) to graph rest-api
sn-fe-c-import-order	Import order: legacy CSS → tokens.css → bundle.css → legacy-aliases.css → app.css	songnodes/CLAUDE.md:66	◐ Corrected	main.tsx imports before tokens.cs bundle, legacy-a
sn-fe-c-g2	feature routes self-register via G2	songnodes/CLAUDE.md §2 Stack	◐ Corrected	true for library, s sources/import; / /sources/targets, and * are still sta routes/registry.ts
sn-fe-c-nav	The NavDrawer is the only site navigation	songnodes/CLAUDE.md §4 Interaction	● Confirmed	AppShell renders (menu + search) App.tsx commen bar
sn-fe-c-stack	React 19, TypeScript 7, PIXI.js v8, Zustand; Node 26 frontend image	songnodes/CLAUDE.md §2 Stack, §1 Prerequisites	● Confirmed	package.json rea ^7.0.2, pixi.js ^8.2 ^5.0.16; Dockerfil alpine

sn-fe-c-ds-sync	frontend design-system copy is synced and --check fails on drift	songnodes/CLAUDE.md §4	● Confirmed	sync-design-sys "design-system in sync"; VERSIO tokens.json 1.10.(
sn-fe-c-no-alert	Never alert/confirm/prompt	songnodes/CLAUDE.md §4 Interaction	● Confirmed	0 calls in fronten
sn-fe-c-one-primary	one primary button per view	songnodes/CLAUDE.md §4 Interaction	× Refuted	Set builder and P primaries at once does not test this
sn-fe-c-en-dash	Missing values are an en dash; no "Unknown"	songnodes/CLAUDE.md §4 Writing, §8	× Refuted	AI setlists show ' OAuth page 'Unk
sn-fe-c-pending-list	Files not yet on the design system are listed in design-system-pending.txt; the conformance job scans every other file	songnodes/frontend/design-system-pending.txt:1-3	● Confirmed	all 37 lines are cc migrated; check_ -check exits 0 wi misses R2 and R grep)
sn-fe-c-pointer-events	Graph uses pointerdown/pointerup, not click	songnodes/CLAUDE.md §5.4	● Confirmed	GraphVisualizatic pointerdown/poir .on('click')
sn-fe-c-shortcuts	Shortcuts D (debug), Space (pause), Escape (clear), Ctrl+A (select all)	songnodes/CLAUDE.md §5.4	i Open	only case 'd' fou GraphVisualizatic useKeyboardSho unreachable; Spa handlers not loca
sn-fe-c-artist-filter	Frontend validation drops NULL/Unknown artists from the graph	songnodes/CLAUDE.md Critical requirement 1	● Confirmed	useDataLoader fi hasValidArtist on fallback loads (fa gap noted in R8)
sn-fe-c-frontend-mem	Resource limits: Frontend 256MB	songnodes/CLAUDE.md §5.3	× Refuted	compose.dev.ya sets no mem_lim
sn-fe-c-indexeddb	SongNodes keeps the last graph, library and saved sets in IndexedDB	songnodes/design-system/16-platform.md:50	× Refuted	no indexedDB us frontend/src
sn-fe-c-frame-deny	nginx/nginx.conf sends X-Frame-Options: DENY and frame-ancestors none	songnodes/design-system/16-platform.md:72	◐ Corrected	nginx/nginx.conf compose; the live default.conf sets
sn-fe-c-error-pages	404/50x static files do not exist yet	songnodes/design-system/13-page-patterns.md:136	● Confirmed	no error_page in default.conf; SPA unknown paths
sn-fe-c-db1-fallback	loadLibrary falls back to /data/music-library.json only when the API fails	songnodes/plan/agents/DB1_one_track_source.md:23	× Refuted	no fallback in dat frontend/public h failure surfaces a
sn-fe-c-stages-poll	Scrape stage cards poll every 5 s while the tab is visible, at most 5 cards	songnodes/STATUS.md:2161	● Confirmed	STAGES_POLL_M 5) on watched ru
sn-fe-c-gv2-next	GV2 next: load the neighbourhood from the API (GV3) instead of the whole graph	songnodes/STATUS.md:2505	● Confirmed	useDataLoader s /api/graph/data?l view param (que
sn-fe-c-body-limit	frontend nginx 1 MB default body limit is a follow-up	songnodes/STATUS.md:1899	● Confirmed	no client_max_bo default.conf whil MB
sn-fe-c-ui9-debug	debug console.log calls in GraphVisualization and useTokenRefresh stay out of production	songnodes/STATUS.md (UI9 slot)	◐ Corrected	true for those tw OAuthCallback st code/state prefix in production
sn-fe-c-banner-day	Banner dismissal should also end at the next Melbourne day (DR-051)	songnodes/STATUS.md:2236	● Confirmed	bannerKey store follow-up still qu

Section 12: **Methods, assumptions and open decisions**

Profile: **Project profile as applied**

Scope	songnodes (E:\songnodes = github zvirb/songnodes main) and musicLibrary (E:\musicLibrary). Sub-projects scanned: songnodes frontend, services, scrapers, design-system; musicLibrary pipeline, artifact, ref_mb, plan.
Page source of truth	songnodes source tree on main @7429821 (no built bundle is kept in the repo); musicLibrary artifact/build/index.html is the live page, artifact/index.html is the legacy page (agent ml, M3 render).
Queued work	Open PRs #576–#580 (Duo 2FA, Cloudflare tunnel, login page, Sentinel sanitisation); musicLibrary plan/jules_payloads (30) and the unmerged WP7 brief.
Baseline report	None found in documentation/, docs/ or project artifacts. The verification register re-checks the repos' own documentation instead.
Standing rules	R1–R8 from songnodes CLAUDE.md §2–§4 and AGENTS.md rule 12 (listed in section 3).
Retired or excluded	deploy/ and k8s/ (kept but unused, CLAUDE.md §1); musicLibrary _to_delete/ and data/ (data/ listed only).
Locale	en-AU, A4, Australia/Melbourne 24-hour time (design system 01-writing).
Personas	Inferred from pages and docs; merged across the three scans into one set (section 6).
Design system	SongNodes design system, artifact 7f37707d and repo mirror songnodes/design-system, 1.9.1 raised to 1.10.0 by this investigation.

Pre-flight: **Tools and access**

songnodes git	main @7429821, clean except untracked docs/DEPLOYMENT_E2E_STATUS_2026_11_06.md on the laptop; push via the session's GitHub app (REST only, GraphQL blocked)
musicLibrary git	Not a git repository: files only, no commits or PRs possible
Tools	node 22.23, python 3.13, graphviz dot 2.43, Chromium 153 (Playwright), poppler 24.02 (pdftoppm, pdffonts, pdftotext), pypdf 6.17; missing: PyMuPDF (not needed)
Laptop shell	tuf-boi Linux VM over the bridge: deletes blocked, no gh; used for the musicLibrary snapshot only
Existing reports	None of this kind. Related: songnodes docs/DATA_LAYER.md, STATUS.md, plan/RESTRUCTURE.md; musicLibrary README.md, docs/mb/*

Methods: **Where the methods agreed and disagreed** `37 notes`

37 rows

ITEM	METHODS	RESULT	NOTE
artifact localStorage keys	M1, M3	● Agree	M1 found sn-music-library-sets, sn-set-draft, sn-filters-library, sn-banner-keyed-*; M3 also saw sn-density (written by SN.prefs, not in feature code). Legacy nsb:* keys found by M1 only, since the legacy page never reaches the draft write without library.json.
window.claude calls	M1, M3	● Agree	Build: use('db') then collection('sets').get; legacy: use('db'), use('user'), user.can('data.write'), onSnapshot('sets'). Without a stub no calls and no errors.
library.json beside the legacy page	M1, M3	▲ Disagree	Code fetches library.json relative to artifact/index.html (M1); file does not exist there, M3 got 404/CORS and 'Library file missing'. Confirms superseded status.
file:// render	M3	● Agree	Both pages fail to load library.json from file:// (CORS, origin null); console errors only from that. Over http the build page renders 413 tracks with 0 console errors (stubbed claude) or one favicon 404.
songnodes sync contract	M1, M2	▲ Disagree	README names scripts/sync-music-library.sh and frontend/public/data/music-library.json; neither exists. Code (M2: run_weekly.cmd_db → cmd_sync_songnodes; songnodes import-music-library.sh) uses POST /api/v1/library/ingest.
artifact page source	M1, M2	▲ Disagree	README says artifact/index.html is the page source; weekly_run.md and app/README say the published page is artifact/build/index.html built from songnodes pages.
BPM and key coverage	M1, M3	▲ Disagree	Code and README describe a BPM-band method and Camelot moves; data (ledger, resolved files, SQLite) and the rendered page show 0 of 413 tracks with BPM or key.
sync-sets tidal_id fallback	M1	▲ Disagree	sync-sets falls back to tr.tidal_id, but both page versions save tr.tidal (a URL), so the fallback never fires.

SQLite schema	M1, M3	● Agree	SCHEMA string matches sqlite_master of db/dj_library.sqlite (13 tables, 4 views); counts recorded from the file.
ref_mb consumers	M1, M2	● Agree	Only songnodes (refmb/settings.py, MB_LOCAL_* in compose.dev.yaml) reads ref_mb; no musicLibrary pipeline module does.
HTTP endpoints	M1, M2, M3	● Agree	AST parse found 215 decorated routes (rest-api 98 incl. main.py, graph-api 18, unified-scraper 5, audio-analysis 4, rest are non-compose services). Registry check: every rest-api router with routes is mounted explicitly or via register(); data_quality is neither (its decorators are commented out, AST found none).
rest-api /api/graph/nodes and /edges	M2, M3	▲ Disagree	AST lists them on rest-api; nginx config routes ^/api/graph to graph-api, so they are unreachable from the frontend.
Silver writers	M1, M2	▲ Disagree	Docs say silver comes from bronze ETL only; grep finds the scraper persistence pipeline, rest-api catalog writes, and one-off scripts also writing silver_enriched_tracks.
Gold layer	M1, M2	▲ Disagree	grep finds gold ETL code and Helm cronjobs; compose/start-dev never run or migrate them; gold_track_graph has zero readers.
RabbitMQ usage	M1, M2	▲ Disagree	grep finds aio_pika in audio-analysis, metadata-enrichment, dlq-manager, websocket-api and a scraper publisher; compose gives RabbitMQ only to worker (Celery) and scraper env; audio-analysis has no RabbitMQ env so its consumer stays off.
Redis graph cache	M1	▲ Disagree	cache_result decorator and CACHE_TTL exist, but no route uses the decorator; only batch:{id} keys are written.
Frontend-to-backend routes	M1, M3	▲ Disagree	Frontend references /api/v1/scrapers/target-tracks/search and /api/v1/scrapers/queue/clear; AST route list has neither.
Memgraph	M1, M2	▲ Disagree	graph_projection code and job kinds exist; compose has no memgraph service or MEMGRAPH_* env.
Migrations applied	M2	● Agree	compose initdb mounts (001,002,006,012,013,014) and start-dev.sh psql lines (101,170,180-185,220,240,260,270,280, init/07) cross-checked with worker/migrate.py REQUIRED list.
Open PRs #576-#580	M1, M2	● Agree	Read via gh REST pulls/{n}/files and patches; items marked state queued. #578 adds no route-gating middleware; #577 adds public ingress.
R6 router registration	M2, M3	▲ Disagree	AST check for top-level register() (router_registry._defines_register) vs main.py include_router list: 11 routers lack register() and rely on explicit wiring.
routes	M1, M2	▲ Disagree	Grep finds PlannedModulePage for /paths in STATIC_ROUTES; registry replacement logic (M2) shows features/pathfinder/route.tsx overrides it, so the planned page is dead.
LiveDataLoader /api/ws	M1, M2	▲ Disagree	Grep finds a WebSocket client; import graph from main.tsx shows LiveDataLoader and useLiveDataLoader are never imported.
services/api.ts clients	M1, M2	▲ Disagree	Grep lists ~20 endpoints; only api.get/post/put/delete (target-tracks client) and trackApi (from unreachable NowPlayingDeck) are imported; backend route scan shows most paths do not exist.
songnodes_ai_prompt_history	M1, M2	▲ Disagree	Key found by grep; its only consumer AIAssistantModal is unreachable.
OAuth token storage	M1, M2	▲ Disagree	Writers use sessionStorage <service>_oauth_tokens; readers use localStorage tidal_oauth_tokens; same-window redirect means no postMessage path.
display preference keys	M1, M3	● Agree	bundle.js and index.html both use sn-theme/sn-text-size/sn-density/sn-motion; Playwright render of reference.html wrote sn-density only.
design-system previews	M1, M3	▲ Disagree	Grep shows previews write sn-disclosure:* and sn-skip-moved-targets; Playwright standalone render throws before writing because window.SongNodes is injected by the card host.
nginx proxy target	M2	● Agree	compose.dev.yaml builds frontend/Dockerfile whose inline default.conf is the only live proxy; nginx/ and nginx-local-proxy.conf are not referenced by compose (only Helm values mount a default.conf).

frontend endpoints vs backend	M1, M2	▲ Disagree	Static route scan of services/rest_api and graph-visualization-api (120 decorators) found no match for /api/v1/scrapers/*, /api/v1/search/*, /api/v1/pathfinding/*, /api/graph/tracks|positions, /api/ws; prefixes defined via include_router in main.py were not resolved, so a few may exist (see unverified).
public/data/music-library.json	M1, M2	▲ Disagree	Plan briefs (DB1, L8, QA7) describe the static file and its fallback; the file is absent from frontend/public and no code fetches /data/*.json. Only tests/e2e/data/library.fixture.json remains (test-only).
R2/R5 conformance	M1, repo checker	▲ Disagree	check_design_system.py --check and design-system/linters/check.py report 0 findings on all route page files, but grep finds two simultaneous primaries (Set builder, Pathfinder), Title Case labels and "Unknown" fallbacks.
static HTML conformance	design-system linter, M3 Playwright	● Agree	linter flags tracker.html hex (12) and reference.html emoji/caps; Playwright shows tracker loads only Google Fonts and needs the artifact runtime.
claim ml-claim-ds-version	scanner: ml, verifier: GitHub REST history of design-system/tokens.json vs build file mtimes; token-name set diff of build style.css against tokens.css at 29619fe5b	▲ Disagree	Disagree with open: the build was made 11 minutes after 1.1.0 landed and before 1.1.1, and its token set matches 1.1.x (1.1.0-1.2.0 tokens.css are identical, so tokens alone cannot pin the patch). Also: main @7429821 is 1.9.1, not 1.10.0; 1.10.0 is the unpushed local commit 78d1d4f.
claim sn-be-c-enrich-8020	scanner: sn-backend, verifier: Followed the registered scraper pipeline to its default URL	▲ Disagree	Disagree with refuted: the scraper does still delegate to metadata-enrichment:8020 as documented; the host just does not exist in compose, so every call lands in service_unavailable and real enrichment happens in the worker.
claim sn-be-c-ollama	scanner: sn-backend, verifier: Checked the downloader-middleware layer (where CAPTCHA solving lives), spider custom_settings and compose	▲ Disagree	Disagree with not-reproducible: it is checkable and false. Code exists (middlewares/captcha_middleware.py, ollama_captcha_solver.py) but is never registered. The scanner looked at ITEM_PIPELINES, the wrong layer.
claim sn-fe-c-shortcuts	scanner: sn-frontend, verifier: Read the window keydown handler in full	▲ Disagree	Disagree with open: all four handlers exist in the same switch the scanner cited. Caveat: Ctrl+A calls toggleNodeSelection, so already-selected nodes get deselected.

Secrets: **Secrets scan (location and type only)** `32 findings`

32 rows

LOCATION	TYPE	RISK
`musicLibrary/secrets/tidal_app.json (not in snapshot)`	TIDAL client id and client secret copied from songnodes .env TIDAL_CLIENT_ID/TIDAL_CLIENT_SECRET	▲ Real-looking
`musicLibrary/secrets/tidal_user_token.json (not in snapshot)`	TIDAL OAuth access and refresh token	▲ Real-looking
`musicLibrary/secrets/tidal_pkce_pending.json (not in snapshot)`	PKCE verifier and state (blanked to {} after use)	▲ Real-looking
`musicLibrary/secrets/tidal_playlists.json (not in snapshot)`	crate and set to TIDAL playlist id map (not a credential)	● Placeholder
`musicLibrary/ref_mb/secrets/ref_mb_admin_password.txt`	Postgres admin password file (32 bytes, Docker secret)	▲ Real-looking
`musicLibrary/ref_mb/secrets/ref_mb_reader_password.txt`	Postgres read-only role password file (32 bytes)	▲ Real-looking
`musicLibrary/ref_mb/.env`	REF_MB_PORT only, no secret	● Placeholder
`env JULES_API_KEY (shell only)`	Google Jules API key	● Placeholder
`songnodes/.env.example:40`	TIDAL client id/secret placeholders	● Placeholder
`songnodes/.env.example:34`	Spotify client id/secret placeholders	● Placeholder
`songnodes/.env.example:59`	API_KEY_ENCRYPTION_SECRET placeholder (not wired into compose)	● Placeholder
`songnodes/compose.dev.yaml:60`	provider credentials passed from host env with empty defaults	● Placeholder
`songnodes/compose.dev.yaml:101`	RabbitMQ guest/guest default broker credentials in CELERY_BROKER_URL	× Committed

songnodes/compose.dev.yaml:11	Postgres POSTGRES_HOST_AUTH_METHOD trust (no password) inside compose network	× Committed
songnodes/sql/init/07-api-keys.sql:61	hard-coded fallback pgp_sym encryption key used when no secret is supplied	× Committed
songnodes/sql/init/07-api-keys.sql:308	POSTGRES_ENCRYPTION_SECRET example in comment	● Placeholder
songnodes/sql/migrations/006_user_oauth_tokens_up.sql:126	OAuth access/refresh tokens stored plain text in user_oauth_tokens (runtime data, not committed)	▲ Real-looking
songnodes/k8s/base/secret.yaml:20	Kubernetes Secret with CHANGE_ME placeholders (DB, Redis, Spotify, TIDAL, Last.fm, YouTube, Discogs, Anthropic)	● Placeholder
songnodes/deploy/helm/songnodes/values.yaml:777	Helm secretKeyRef names only	● Placeholder
songnodes/services/api-gateway/.env.example:6	JWT secret and Redis password examples	● Placeholder
songnodes/services/db-connection-pool/start.sh:25	default DB password fallback in shell parameter expansion	▲ Real-looking
songnodes/scripts/run_integration_tests.sh:138	default DB password fallback	▲ Real-looking
songnodes/scrapers/middlewares/captcha_middleware.py:58	CAPTCHA API key placeholder literal	● Placeholder
PR #578 user_auth.py	DUO_CLIENT_ID/SECRET, JWT_SECRET_KEY, AUTH_USERS from env; AUTH_USERNAME/AUTH_PASSWORD default admin/admin in code	× Committed
PR #577 compose.dev.yaml	CLOUDFLARE_TUNNEL_TOKEN from host env	● Placeholder
browser localStorage songnodes-store .state.musicCredentials (runtime, not in repo)	TIDAL/Spotify access and refresh tokens, client id and client secret	▲ Real-looking
browser sessionStorage <service>_oauth_tokens (runtime)	OAuth access/refresh tokens	▲ Real-looking
songnodes/frontend/src/store/useStore.ts:1577-1581	debugZustand.injectTestCredentials placeholder client id/secret	● Placeholder
songnodes/frontend/src/components/APIKeyManager.tsx:529,569	AI provider API key input placeholders (prefix hints only); keys POSTed to /api/v1/api-keys, legacy raw keys under songnodes_key_*/ai_key_*/apiKey_* removed on migration	● Placeholder
songnodes/frontend/src/features/setBuilder/sendToTidal.ts:38-44	browser-held TIDAL access token sent as Authorization Bearer to rest-api	▲ Real-looking
songnodes/nginx/ssl/ (ca.crt, musicdb.crt, musicdb-chain.crt, ca.srl)	TLS certificates only; musicdb.key referenced by nginx/conf.d/blue-production.conf but not committed	● Placeholder
PR #579 frontend/src/features/auth/AuthContext.tsx	session cookie (credentials: include) for Duo login	● Placeholder

Unverified: **Unverified items** 19 items

- musicLibrary/secrets/ is not in this copy: tidal_app.json, tidal_user_token.json, tidal_playlists.json contents and whether TIDAL login is live are inferred from sync_result files (403 tracks TIDAL_ONLY) and code only
- Artifact db capability config and the 'only you write' rule are from app/README.md, not read from the live artifact
- Whether TIDAL honours a reused Idempotency-Key by replaying the earlier response
- Why TIDAL v2 returns no bpm/key (scope, country or catalogue coverage)
- data/ contents beyond _data_manifest.txt (58 GB not copied); ref-mb running state inferred from postmaster.pid date
- Scheduled task configuration itself (cron, device, folders) was not inspected, only agent/weekly_run.md
- songnodes Send to TIDAL backend behaviour beyond the route signature
- Runtime behaviour: nothing was executed against a running stack (no Docker here); daily-quota and duplicate-silver effects are inferred from code.
- scripts/sync-music-library.sh named in the brief does not exist; the music library import script is scripts/import-music-library.sh.

- External API host list is from URL literals; per-provider rate limits and auth were not traced line by line.
- Legacy service endpoints are listed from decorators only; their table use was not traced.
- common/datalayer/audio.py exact call line to /analyze-url not pinned.
- Backend route existence was checked by decorator scan only; routers mounted with include_router(prefix=...) in services/rest_api/main.py were not resolved, so /api/v1/scrapers/* could exist under a different mount.
- Whether sendToTidal reads the token from songnodes-store musicCredentials.tidal.accessToken was inferred from the Authorization header code path, not traced to its caller.
- Behaviour of POST /api/v1/library/ingest (writes library.tracks) belongs to the backend agent.
- design-system/source/previews files were grouped by folder count; individual titles not extracted.
- Pathfinder export JSON branch content (pathfinder.machine.ts:80-90) not read in full.
- PR #579 login page reviewed by grep of its files at the PR head only; not rendered.
- sn-fe-sources-import R2 third-primary visibility inferred from markup, not rendered.

Decisions: **Assumptions and open decisions**

- The laptop copy of songnodes equals origin/main @7429821 (git status showed only one untracked doc).
- musicLibrary data/ (58 GB) holds raw dumps and caches; it was listed, not read.
- Effort sizes (S, M, L) are Estimates from evidence spread against the AGENTS.md package budget.
- Gap severity is the highest severity of debt sharing evidence with the gap, otherwise medium (Estimate).
- Open decision: keep a gold layer or document silver-plus-projection as the model (SN-T06).
- Open decision: which Set Builder is primary once both read the same saved-set shape.

Section 13: **Handover**

Handover: **Card**

Branch docs/data-flow-investigation

Gates: **Verification gates** `0 gates`

Next: **Recommended next steps**